

EXECUTIVE SUMMARY

NASA Launch Services II (NLS II)

RFP10-99-0021 – Revision T

The NLS II contract “On-Ramp” clause authorizes NASA to invite domestic launch service providers that meet minimum contract requirements to submit proposals to furnish launch services for NASA and NASA-sponsored payloads on an indefinite delivery / indefinite quantity (IDIQ) basis. Proposals are solicited and evaluated during specified “open seasons” conducted each year during the life of the NLS II contract. Resulting contract(s) will also require performance of non-standard services and special task assignments in support of launch services. Generally, the payloads to be launched will support the goals and objectives of various NASA programs.

During this open season, NASA invites proposals for risk category 2 and 3 launch services capable of delivering, at a minimum, a 250 kg payload to orbit at an altitude of 200 km and a launch inclination of 28.5 degrees. Although there are no firm missions included in this solicitation, offerors are requested to identify and price, on a not-to-exceed (NTE) basis, standard and non-standard launch services that meet NLS II contract requirements. Standard launch services shall be offered for all of the order years remaining on the NLS II contract, *i.e.*, order years 2026 through June 30, 2030 (Non-Standard and Special Task Assignment Labor Rates through December 31, 2032).

Selection for award of an NLS II contract will be determined by NASA in accordance with (IAW) the evaluation criteria specified in Attachment E3 of the RFP which includes, among other things, a requirement that the launch service provider be a domestic source who is AS9100 certified by the date of contract award. Other factors that will be evaluated include the offeror’s ability to successfully deliver to orbit at least one payload using a launch service that is capable of placing, at a minimum, a 250 kg payload into a 200 km circular orbit at a launch inclination of 28.5 degrees; ability of the offered launch service to meet NASA’s performance and launch vehicle certification requirements; the reasonableness of the offeror’s NTE prices and firm fixed-prices, when applicable; and the offeror’s record of past performance. The performance capabilities of proposed launch services will be incorporated into the resultant NLS II contract(s) for those services meeting NASA’s defined performance requirements.

As future NASA and NASA-sponsored missions become firm, each NLS II contractor will be given a fair opportunity to be considered for launch service task orders for those missions in accordance with the task ordering provisions of the NLS II contract and subpart 16.505 of the Federal Acquisition Regulations, provided that the common launch vehicle configuration to be proposed has achieved one successful flight.

NASA’s intent is to utilize existing commercial providers’ capabilities and processes to the maximum extent possible. NASA has attempted to accommodate commercial business practices and methods employed by industry in the development of this solicitation. The following provisions are incorporated:

- ◆ The Statement of Work (SOW) defines NASA's performance requirements in lieu of a detailed design specification and requests the Offerors propose their launch service capabilities for incorporation into the resulting contracts.
- ◆ Compliance documents are limited to those items mandatory from an Agency policy perspective.
- ◆ No certified cost or pricing data is required.
- ◆ Government insight and approval is limited to those items mandatory in accordance with Agency policy.

Due to the anticipated number of submissions and available personnel, the Government may evaluate proposals sequentially to optimize available resources. If so, the Contracting Officer will determine the sequence of evaluation to ensure a thorough and efficient review process that best serves the interests of the program (e.g., increasing competition for specific, upcoming task orders). The sequence of evaluation is an administrative matter and will not impact the substantive criteria used for making contract award determinations. Regardless of the evaluation sequence, the Government will evaluate all proposals against the criteria set forth in Section M. The order of evaluation does not imply a preference for any specific offeror, nor does it guarantee a sooner award date. Awards may be made on a rolling basis.

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SECTION A
STANDARD FORM 1449

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SECTION B

STANDARD FORM 1449 CONTINUATION

1.0 SCHEDULE OF SUPPLIES AND/OR SERVICES TO BE PROVIDED

The Contractor shall provide all services, facilities, and resources (except as may be expressly stated in this contract as furnished by the Government) necessary to furnish the Contract Line Item Number (CLIN) items (Tables B-4 through B-10) in accordance with (IAW) the Statement of Work (SOW), Exhibits, and Documents attached hereto.

1.1 Reserved

Table B-1: Reserved

1.2 Reserved

Table B-2: Reserved

1.3 Reserved

Table B-3: Reserved

1.4 Reserved

Table B-4: Reserved

1.5 Special Task Assignments (Studies and Analyses)

CLIN	Description of Supplies and Service	Qty	Unit	Unit Price (\$)	Total Amount (\$)
10	Special Task Assignments – Firm Fixed-Price Composite Labor Hours	A/R	Labor Hour	See Table B-6	See Table B-6

Table B-5: Special Task Assignments (Studies and Analyses)

HOURLY LABOR RATE (COMPOSITE SKILL MIX AND BURDENED)	FIRM FIXED-PRICE (\$) IN CALENDAR YEAR ORDERED*						
	2026	2027	2028	2029	2030	2031	2032
Labor Rate							

* The proposed labor rates shall be negotiated using the applicable procedures of FAR Part 15 and incorporated herein.

Table B-6: Hourly Labor Rate - FFP List

Table B-7: Reserved

1.6 IDIQ Launch Service Task Order (LSTO)

The Government reserves the unilateral right to order launch services at any time from contract award through June 30, 2030, in accordance with contract Section C, Clause 14.0, Launch Service Task Ordering Procedures. These launch services are identified in Tables B-9 and B-10 and described in Exhibits 2 and 3 to the Statement of Work. Launch dates will be mutually agreed upon prior to issuance of an LSTO.

NTE prices for launch services for IDIQ missions will be part of the price evaluation and best value selection for award of an IDIQ task order contract.

In no event shall the Government be obligated to pay for any launch service prior to Authority To Proceed and obligation of funds for that launch service. The minimum order amount for each NLS IDIQ contract awarded in accordance with Section C, Clause 2.0, On-Ramp and Technology Insertion, is \$20,000. The maximum potential quantity for this IDIQ contract portion for all NLS contracts combined is one hundred (100) Launch Service Task Orders and associated non-standard services/mission unique services.

Once payment begins on a qualified standard launch service and NASA insight commences, the Government reserves the right to approve any Contractor initiated substitution of the launch vehicle or reallocation of launch vehicle hardware designated for a NASA mission under this contract, in accordance with Section C, Clause 25.0, Government Insight and Approval.

CLIN	Description of Supplies and Services	Contract Mod ⁴	Qty	Unit	Unit Price (\$)	Total Amount (\$)
X¹	IDIQ Launch Service Task Order (LSTO) Launch Vehicle Configuration ____ TBD ____ for Mission ____ TBD ____ Launch Date ____ TBD ____ Launch Period, if applicable ____ TBD ____	TBD	TBD²	Service	TBD	TBD
SubCLIN						
X¹A	Standard Launch Services and Standard Mission Integration in accordance with the SOW and Exhibit 2 Launch Vehicle Configuration ____ TBD ____		TBD³	Lot	See Table B-9	
X¹B	Non-Standard Services in accordance with the SOW and Exhibit 3 Nonstandard Services ____ TBD ____		TBD	Lot	See Table B-10	
X¹C	Known Mission Unique Hardware and Software Non- recurring Items in accordance with SOW Exhibit 6 Mission Unique Services ____ TBD ____		TBD	Lot	TBD	

¹ The Contracting Officer shall determine the CLIN number to be assigned.

² The actual number of LSTOs to be awarded is TBD.

³ The actual number of missions to be included in each LSTO is TBD.

⁴ As various Contract Modifications are issued for services ordered, their respective number will be listed in this table.

Table B-8: IDIQ Launch Service Task Order

1.7 NTE Prices - Standard Launch Services for IDIQ Missions

The Government shall have the right to order the standard launch services at the prices and for the launch vehicle configurations identified in Table B-9. Nothing in this clause precludes the Contractor from proposing more favorable prices or discount terms in response to specific requests for launch service proposals in accordance with contract Section C, Clause 14.0.

The Offeror shall add as many rows (in Table B-9 format) necessary to identify all standard launch service NTE prices for each launch vehicle configuration proposed. The Offeror shall also propose a table(s) (Table B-9a) of quantity discounts that shall correspond and apply to the proposed NTE standard launch service prices in Table B-9 for each launch vehicle configuration. The proposed quantity discount shall apply in the event the Government orders more than one launch service in the same year and/or under an individual LSTO. The quantity discount may be in terms of a percentage or dollar amount reduction from the stated NTE price. The quantity discount may apply to the number of launch services generally or may be applied to the number of launch vehicle configurations specifically, or both. The Offeror's proposal in response to a Request for Launch Service Proposal (RLSP) shall identify any conditions that the discount is contingent upon.

LAUNCH VEHICLE CONFIGURATION	NOT-TO-EXCEED PRICE (\$) IN CALENDAR YEAR ORDERED*					
	2025	2026	2027	2028	2029	2030

* Pricing is based on the standard integration period of L-30 plus or minus 3 months.

Table B-9: NTE Price List - Standard Launch Services for IDIQ Missions

1.8 NTE Prices – Non-Standard Services for IDIQ Missions

The Government shall have the right to order the following non-standard services at the prices and for the launch vehicle configurations identified in Table B-10. Nothing in this clause precludes the Contractor from proposing more favorable prices or discount terms in response to specific requests for launch service proposals in accordance with contract Section C, Clause 14.0. For non-standard services which have non-recurring costs the offeror shall also propose “Subsequent purchase” pricing. “Subsequent purchase” pricing shall apply in the event the Government orders a non-standard service more than once from a provider, regardless of the mission specific implementation and/or repetitive nature of the mission.

NON-STANDARD SERVICE ITEM	Launch Vehicle Configuration TBP From Table B-9							
	NOT TO EXCEED (NTE) PRICE (\$) IN CALENDAR YEAR ORDERED*							
	2026	2027	2028	2029	2030	2031	2032	

* Pricing is based on the standard integration period of L-30 plus or minus 3 months.

Table B-10: NTE Price List - Non-Standard Services for IDIQ Missions

2.0 TOTAL AWARD AMOUNT

The guaranteed minimum value of this contract is \$20,000. The maximum potential value of contract number TBD is \$15,000,000,000. The total maximum potential value of all NLS contracts combined is \$15,000,000,000.

3.0 LAUNCH SERVICE FLEXIBILITY

3.1 General

The Government reserves the unilateral right, at any time, to substitute payloads, as well as bilaterally substitute standard launch services and/or add/delete non-standard services thereto, for any mission awarded to the Contractor under this contract. Substitution of those standard launch services and non-standard services shall be subject to the NTE caps identified in Tables B-9 and B-10. In addition, cost impacts resulting from payload substitutions and substitution of standard launch services shall be subject to the changes clause.

3.2 Reserved

Table B-11: Reserved

3.3 Reserved

Table B-12: Reserved

4.0 MOST FAVORED CUSTOMER

The Contractor hereby certifies the CLIN prices for standard launch services (including standard mission integration) under this contract are no higher than the lowest price charged to any other commercial or U.S. Government customer for an equivalent launch service during the twelve (12) months both preceding and following the placement of a launch service task order. The Government shall be entitled to a one-time reduction in contract price for each standard launch service failing to meet this certification. The price credit will be equal to the difference between the standard launch service price under this contract and the lower price awarded for an equivalent launch service.

5.0 IDIQ LAUNCH SERVICE AND PRICE UPDATES

Successful Offerors who receive NLS II IDIQ contract awards may propose additional IDIQ NTE launch services and prices via the On-Ramp and Technology Insertion clause, Section C, Clause 2.0. IDIQ contract holders will not be entitled to unilaterally adjust NTE prices or delete launch services already in the contract. Contractor initiated changes to the launch vehicle configuration, vendor, or design shall be subject to the launch vehicle certification requirements of Section C, Clause 3.0. Contractor initiated changes to the launch vehicle configuration, vendor, or design, after mission ATP, shall be at no increase to the contract price and subject to the written approval of the Contracting Officer. New launch vehicle certification plans and revisions to existing certification plans may be proposed for inclusion into the contract when submitting Launch Service proposals pursuant to Section C, Clause 14.0, and shall be subject to the written approval of the Contracting Officer.

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CONTRACT TERMS AND CONDITIONS

1.0 FAR 52.212-4 CONTRACT TERMS AND CONDITIONS--COMMERCIAL PRODUCTS AND COMMERCIAL SERVICES (DEVIATION) Nov 2025(TAILORED)

- (a) (a) *Definitions.* The clause at Federal Acquisition Regulation (FAR) 52.202-1, Definitions, is incorporated by reference.
- (b) *Inspection/Acceptance.* Pursuant to FAR Subpart 12.402(b), FAR clause 52.246-11 Higher-Level Contract Quality Requirement (Dec 2014) shall govern and apply to this contract and is provided in full text in Section C, Clause 16.0. For launch service Contract Line Item Numbers (CLINs) prior to launch, the provisions of FAR clause 52.246-4 Inspection of Services—Fixed-Price (Aug 1996) shall govern and apply to this contract and is incorporated by reference under contract Section C, Clause 36.0. The Government reserves the right to perform in-process inspection or testing of any supplies or launch services tendered for acceptance prior to launch.

The Government will accept only those launch services that successfully deliver a separated and undamaged payload to the proper orbit conditions and insertion accuracies and do not exceed the environmental parameters stated in the Interface Control Document (ICD) except as otherwise provided in this contract. Acceptance of launch service CLINs, and remedies for partial or failed missions shall be in accordance with contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions. The Contractor shall only tender for acceptance those items conforming to the contract requirements. For services other than launch services, the Government will require re-performance of nonconforming services at no increase in contract price. For launch services, the Government must exercise its post-acceptance rights within the time specified in contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions. For supplies or services other than launch services, the Government must exercise its post-acceptance rights:

- (1) Within a reasonable time after the defect was discovered or should have been discovered; and
 - (2) Before any substantial change occurs in the condition of the item, unless the change is due to the defect in the item.
- (b) *Assignment.* The Contractor or its assignee may assign its rights to receive payment due as a result of performance of this contract to a bank, trust company, or other financing institution, including any Federal lending agency in accordance with the Assignment of Claims Act (31 U.S.C.3727). However, when a third party makes

payment (e.g., use of the Government-wide commercial purchase card), the Contractor may not assign its rights to receive payment under this contract.

- (c) *Changes.* FAR clause 52.243-1 Changes - Fixed-Price (Jun 2025) Alt I (Jun 2025) is hereby incorporated by reference under contract Section C, Clause 36.0. The time requirement for the Contractor to assert its rights for Equitable Adjustment (EA) is increased from thirty (30) days to (60) sixty days.
- (d) *Disputes.* This contract is subject to 41 U.S.C. chapter 71, Contract Disputes. Failure of the parties to this contract to reach agreement on any request for equitable adjustment, claim, appeal or action arising under or relating to this contract shall be a dispute to be resolved in accordance with the clause at FAR 52.233-1, Disputes, which is incorporated herein by reference. The Contractor shall proceed diligently with performance of this contract, pending final resolution of any dispute arising under the contract.
- (e) *Definitions.* The clause at FAR 52.202-1, Definitions, is incorporated herein by reference under contract Section C, Clause 36.0.
- (f) *Excusable delays.* The Contractor shall be liable for default unless nonperformance is caused by an occurrence beyond the reasonable control of the Contractor and without its fault or negligence such as, acts of God or the public enemy, acts of the Government in either its sovereign or contractual capacity, fires, floods, epidemics, quarantine restrictions, strikes, unusually severe weather, and delays of common carriers. The Contractor shall notify the Contracting Officer in writing as soon as it is reasonably possible after the commencement of any excusable delay, setting forth the full particulars in connection therewith, shall remedy such occurrence with all reasonable dispatch, and shall promptly give written notice to the Contracting Officer of the cessation of such occurrence. Excusable delays for launch delays by either party are identified and shall be resolved in accordance with contract Section C, Clause 19.0, Adjustments to Launch Schedule.
- (g) *Invoice.* The Government will handle invoices according to the Prompt Payment Act (31 U.S.C. 3903) and 5 CFR part 1315. The Contractor shall submit invoices to the address designated in the contract to receive invoices. An invoice must include the information required by 5 CFR part 1315.9(b).
- (h) *Patent indemnity.* The Contractor shall indemnify the Government and its officers, employees and agents against liability, including costs, for actual or alleged direct or contributory infringement of, or inducement to infringe, any United States or foreign patent, trademark or copyright, arising out of the performance of this contract, provided the Contractor is reasonably notified of such claims and proceedings.
- (i) *Payment.*

- (1) *Items accepted.* Payment shall be made for items accepted by the Government that have been delivered to the delivery destinations set forth in this contract.
- (2) *Prompt Payment.* The Government will make payment in accordance with the Prompt Payment Act (31 U.S.C. 3903) and prompt payment regulations at 5 CFR Part 1315.
- (3) *Discount.* In connection with any discount offered for early payment, time shall be computed from the date of the invoice. For the purpose of computing the discount earned, payment shall be considered to have been made on the date which appears on the payment check or the specified payment date if an electronic funds transfer payment is made.
- (4) *Overpayments.* If the Contractor becomes aware of a duplicate contract financing or invoice payment or that the Government has otherwise overpaid on a contract financing or invoice payment, the Contractor shall—
 - (i) Remit the overpayment amount to the payment office cited in the contract along with a description of the overpayment including the—
 - (A) Circumstances of the overpayment (*e.g.*, duplicate payment, erroneous payment, liquidation errors, date(s) of overpayment);
 - (B) Affected contract number and delivery order number, if applicable;
 - (C) Affected contract line item or subline item, if applicable; and
 - (D) Contractor point of contact.
 - (ii) Provide a copy of the remittance and supporting documentation to the Contracting Officer.
- (5) *Interest.*
 - (i) All amounts that become payable by the Contractor to the Government under this contract shall bear simple interest from the date due until paid unless paid within thirty (30) days of becoming due. The interest rate shall be the interest rate established by the Secretary of the Treasury as provided in 41 U.S.C. 7109, which is applicable to the period in which the amount becomes due, as provided in (i)(6)(v) of this clause, and then at the rate applicable for each six-month period as fixed by the Secretary until the amount is paid.
 - (ii) The Government may issue a demand for payment to the Contractor upon finding a debt is due under the contract.

- (iii) Final decisions. The Contracting Officer will issue a final decision as required by FAR Part 33 if—
 - (A) The Contracting Officer and the Contractor are unable to reach agreement on the existence or amount of a debt within thirty (30) days;
 - (B) The Contractor fails to liquidate a debt previously demanded by the Contracting Officer within the timeline specified in the demand for payment unless the amounts were not repaid because the Contractor has requested an installment payment agreement; or
 - (C) The Contractor requests a deferment of collection on a debt previously demanded by the Contracting Officer (see FAR Part 32).
- (iv) If a demand for payment was previously issued for the debt, the demand for payment included in the final decision shall identify the same due date as the original demand for payment.
- (v) Amounts shall be due at the earliest of the following dates:
 - (A) The date fixed under this contract.
 - (B) The date of the first written demand for payment, including any demand for payment resulting from a termination for cause.
- (vi) The interest charge shall be computed for the actual number of calendar days involved beginning on the due date and ending on—
 - (A) The date on which the designated office receives payment from the Contractor;
 - (B) The date of issuance of a Government check to the Contractor from which an amount otherwise payable has been withheld as a credit against the contract debt; or
 - (C) The date on which an amount withheld and applied to the contract debt would otherwise have become payable to the Contractor.
- (vii) The interest charge made under this clause may be reduced under the procedures for interest credits prescribed in FAR Part 32 in effect on the date of this contract.

- (j) *Risk of loss.* Risk of loss of or damage to Government property under this contract will be governed by FAR clause 52.246-25 Limitation of Liability--Services (Feb 1997) and is hereby incorporated by reference under contract Section C, Clause 36.0.
- (k) *Taxes.* The contract price includes all applicable Federal, State, and local taxes and duties.
- (l) *Termination for the Government's convenience.* The Government reserves the right to terminate this contract, or any part hereof, for its sole convenience subject to the terms of this contract specified in contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement under FAR Clause 52.212-4(l).
- (m) *Termination for Cause.* The Government may terminate this contract, or any part hereof, for cause in the event of any default by the Contractor, including failure to meet launch dates, certify or maintain certified launch vehicles, or if the Contractor fails to comply with any contract terms and conditions, or fails to provide the Government, upon request, with adequate assurances of future performance. In the event of Termination for Cause, the Government shall not be liable to the Contractor for any amount for supplies or services not accepted, including all launch service payments previously paid and all launch service payments remaining to be paid, and the Contractor shall be liable to the Government for any and all rights and remedies provided by law. If it is determined that the Government improperly terminated this contract for default, such termination shall be deemed a termination for convenience.
- (n) *Title.* Unless otherwise specified in a Task Assignment, title to items furnished in conjunction with services under this contract shall remain with the Contractor. In accordance with 42 U.S.C. 2465e, NASA shall not take title to launch vehicles under contracts for launch services.
- (o) *Warranty.* Unless otherwise specified in a Task Assignment, the Contractor makes no warranty, express or implied, with respect to the services delivered or performed hereunder.
- (p) *Limitation of liability.* Except as provided for in paragraphs (a) and (j) above, the Contractor will not be liable to the Government for consequential damages resulting from any defect or deficiencies in accepted items.
- (q) *Other compliances.* The Contractor shall comply with all applicable Federal, State and local laws, executive orders, rules and regulations applicable to its performance under this contract.
- (r) *Compliance with laws unique to Government contracts.* The Contractor agrees to comply with 31 U.S.C. 1352 relating to limitations on the use of appropriated funds to influence certain Federal contracts; 18 U.S.C. 431 relating to officials not to

benefit; 40 U.S.C. chapter 37, Contract Work Hours and Safety Standards Act; 41 U.S.C. chapter 87, Kickbacks; 49 U.S.C. 40118, Government-financed air transportation; and 41 U.S.C. chapter 21 relating to procurement integrity.

(s) *Order of precedence.* Any inconsistencies in this solicitation or contract shall be resolved by giving precedence in the following order:

- (1) The schedule and description of supplies/services (Section B, C, and Statement of Work, excluding contract clauses incorporated by reference). The ICD (Exhibit 8) shall take precedence over the Mission Solution (Exhibit 9) once the ICD is incorporated by reference into the contract.
- (2) The Assignments, Disputes, Payments, Invoice, Other Compliances, and Compliance with Laws Unique to Government Contracts paragraphs of this clause.
- (3) Other contract clauses incorporated in the solicitation or contract;
- (4) Addenda to this solicitation or contract, including any license agreements for computer software.
- (5) Solicitation provisions if this is a solicitation.
- (6) Other paragraphs of this clause.
- (7) The Standard Form 1449.
- (8) Other documents, exhibits, and attachments.
- (9) The specification.

(t) Unauthorized Obligations.

- (1) Except as stated in paragraph (t)(2) of this clause, when any supply or service acquired under this contract is subject to any End Use License Agreement (EULA), Terms of Service (TOS), or similar legal instrument or agreement, that includes any clause requiring the Government to indemnify the Contractor or any person or entity for damages, costs, fees, or any other loss or liability that would create an Anti-Deficiency Act violation (31 U.S.C. 1341), the following shall govern:

- (i) Any such clause is unenforceable against the Government.
- (ii) Neither the Government nor any Government authorized end user shall be deemed to have agreed to such clause by virtue of it appearing in the EULA, TOS, or similar legal instrument or

agreement. If the EULA, TOS, or similar legal instrument or agreement is invoked through an “I agree” click box or other comparable mechanism (e.g., “click-wrap” or “browse-wrap” agreements), execution does not bind the Government or any Government authorized end user to such clause.

(iii) Any such clause is deemed to be stricken from the EULA, TOS, or similar legal instrument or agreement.

(2) Paragraph (u)(1) of this clause does not apply to indemnification by the Government that is expressly authorized by statute and specifically authorized under applicable agency regulations and procedures.

(u) *Comptroller General examination of record.* This paragraph applies if this contract was awarded using other than sealed bid procedures and is in excess of the simplified acquisition threshold on the date of award of this contract.

(1) The Comptroller General of the United States, or an authorized representative of the Comptroller General, shall have access to and right to examine any of the Contractor’s directly pertinent records involving transactions related to this contract.

(2) The Contractor shall make available at its offices, at all reasonable times, the records, materials, and other evidence for examination, audit, or reproduction, until 3 years after final payment under this contract or for any shorter period specified in FAR part 4, longer period required by statute, or periods specified in other clauses of this contract. If this contract is completely or partially terminated, the records relating to the work terminated shall be made available for 3 years after any resulting final termination settlement. Records relating to appeals under the disputes clause or to litigation or the settlement of claims arising under or relating to this contract shall be made available until such appeals, litigation, or claims are finally resolved.

(3) As used in this clause, records include books, documents, accounting procedures and practices, and other data, regardless of type and regardless of form. This clause does not require the Contractor to create or maintain any record that the Contractor does not maintain in the ordinary course of business or pursuant to a provision of law.

(v) Incorporation by reference. The Contractor’s representations and certifications, including those completed electronically via the System for Award Management (SAM), are incorporated by reference into the contract.

2.0 ON-RAMP AND TECHNOLOGY INSERTION

2.1 The purpose of the IDIQ On-Ramp is to create an opportunity for qualified new, emerging launch service providers and for incumbent launch service providers to

introduce launch vehicles not available at the time of the award of the initial contract and to compete for additional requirements not identified as firm or option requirements under the basic contract. The intent of the On-Ramp is to foster competition for future requirements for launch services.

The minimum IDIQ On-Ramp qualification criteria are:

- (A) The Launch Service Contractor (LSC) shall be a domestic LSC pursuant to Section 201 of Public Law 105-303, Commercial Space Act of 1998.
- (B) The LSC must obtain an AS9100 third party certification from a Registrar that is accredited by a body recognized by either the ANSI-ASQ National Accreditation Board (ANAB) or the International Aerospace Quality Group (IAQG), for any corporation, corporate divisions, subsidiaries, joint ventures, partner(s) and/or other business entity actually performing launch vehicle manufacturing, management, payload/launch vehicle integration, testing and launch.

- 2.2 The parties mutually agree that the original solicitation, as revised, shall remain open during the life of this contract and that at any time the Government may award additional contracts for IDIQ requirements. Each August during the life of this contract, or at any other time established via synopsis, the Government, will accept proposals from new launch service providers for IDIQ contracts and proposals from existing IDIQ contractors for additional launch services. If the Government issues a solicitation notice, new launch service providers and current IDIQ contractors will be allowed to submit proposals, within the notice's stated response time.

The minimum contract requirements (as revised), the technical acceptability standards, evaluation factors, solicitation terms and conditions, price reasonableness, and basis for award shall remain in full force and effect for each new proposal. Upon award of each additional contract, the Government shall notify all present Contractors of the award, and the new Contractor shall thenceforth be eligible to compete with all present Contractors for the award of IDIQ task orders. A new launch vehicle orientation may be requested by the Government when appropriate as referenced in Attachment D1, Paragraph 2.2.5.

2.3 Expansion of Performance Capabilities

NASA reserves the right to consider, in the future, expansion of the performance capabilities covered by the IDIQ portion of the NLS II contract, by properly soliciting offers from all interested sources capable of meeting the requirements in the expanded performance capabilities.

2.4 Additional Launch Services Requirements

- (A) Prior to proposal submittal for a Launch Services Task Order (LSTO), the proposed launch vehicle must have one successful flight, as established in contract Section C, Clause 14.0, Launch Service Task Ordering Procedures.

- (B) Launch vehicles will be certified based on the requirements established by NPD 8610.7, ELV Risk Mitigation Policy for NASA-Owned and NASA-Sponsored Payloads, and contract Section C, Clause 3.0, Launch Vehicle Certification, prior to launch of a NASA mission.
 - (C) Launch services must comply with all the conditions and requirements of contract Attachment D1, Statement of Work (SOW).
- 2.5 The minimum order amount for IDIQ contracts awarded in accordance with this On-Ramp Clause is \$20,000.
- 2.6 In preparation for eventual on-ramp and certification of new common launch vehicle configurations (CLVC) in active development by NLS II IDIQ LSC's, NASA is available to review, comment, provide advice and/or advance independent assessment on various launch vehicle systems and performance as follows:
 - A) Each CLVC governed by this clause will be listed via bi-lateral contract modification in clause 2.6 F. There is no requirement for the LSC to offer NASA information under this clause for any new CLVC's in development unless it is listed in clause 2.6 F or has been on ramped. Clause 2.6 F will include specific guidelines governing the interaction for each CLVC. The LSC may unilaterally remove a vehicle from clause 2.6 F at any time.
 - B) The LSC will determine the extent of the data to be provided to NASA. However, for listed CLVC's, NASA shall be invited to major reviews (e.g. System Requirements Reviews, Preliminary Design Reviews, Critical Design Reviews, Qualification Test Reviews, Technical Interchange Meetings) and NASA will use its best efforts to support them. The LSC should provide NASA the presentation materials in advance in either electronic and/or hardcopy. LSP may request, and the LSC may provide at their discretion, any additional supporting information or analyses.
 - C) NASA will use reasonable efforts to participate in person or via telecon in reviews and TIMs with LSP personnel it deems appropriate and consistent with its available resources. NASA will consider on a case-by-case basis any LSC requests for independent assessment, or analyses to be performed by NASA. NASA will use its best efforts to provide comments, advice and results of any assessments or analyses it performs with the provided data back to the LSC who requested the assessment.
 - D) NASA LSP may, with the LSC's consent, share LSC data received by NASA under this clause with NASA employees and/or NASA's support contractors with executed nondisclosure agreements (directly between the contractor and the LSC) working on the NASA Commercial Crew Program, the Commercial Resupply Services Contract, NASA Spacecraft Programs considering the potential use of the new CLVC, and/or NASA HQ.

NASA LSP assessments/opinions that have been previously provided to the LSC under Clause 2.6 C with reasonable time for comment, and do not contain LSC proprietary data, may be provided to one or more of the noted groups concurrent with notification to the LSC.

The transfer of data to above named agencies/personnel outside of NASA LSP is for the direct recipient(s) only, and will be marked as such to avoid being transferred from that recipient to others. NASA LSP recognizes that data received under this agreement does not permit the rendering of opinions/decisions on mission success and flight worthiness of missions currently on the manifest, other than to the involved LSC, NASA HQ and the NASA Spacecraft customer Program/Project to which it applies.

- E) Nothing in Clause 2.6 shall be construed as modifying any of the parties' rights and responsibilities with respect to Clause 25 (Insight and Approval) and any provision of Data thereunder. Clause 25 (Insight and Approval) is not applicable to new CLVCs until they have been on ramped.
- F) CLVC's and associated guidelines governing the interaction for each CLVC shall be identified below as applicable.

3.0 LAUNCH VEHICLE CERTIFICATION

3.1 Any common launch vehicle configuration utilized by the Contractor to provide launch services under this contract must be certified by NASA in accordance with NPD 8610.7, ELV Risk Mitigation Policy for NASA-Owned or NASA-Sponsored Payloads. Common launch vehicle configuration certification will be performed in accordance with LSP-PLN-324.01, Expendable Launch Vehicle Certification.

3.2 Certification Criteria:

- (A) Reserved
- (B) Reserved
- (C) For contract award of IDIQ launch services, the Contractor must provide a detailed, viable plan, subject to Government assessment and acceptance, to achieve all certification requirements prior to launch. If a certification plan (Attachment D1, Exhibit 7) is submitted, it shall be exempt from disclosure as provided under the FOIA.
- (D) Services under this contract will only include launch vehicles certified to risk mitigation Category 2 and 3 in accordance with NPD 8610.7. The Contractor shall submit all required documentation for NASA evaluation and determination of certification category. NASA shall not bear any cost associated with the

development of any LSC documentation required for the certification of a common launch vehicle configuration.

- (E) The proposed common launch vehicle configuration shall achieve one successful flight prior to submittal of a proposal for a Launch Service Task Order (LSTO).
- (F) Prior to launch of the NASA payload, the proposed common launch vehicle configuration shall be certified to the required payload risk category.
- (G) NASA reserves the right to require a specific launch vehicle certification category and alternative for selected payloads, regardless of the payload risk classification. This determination will be made prior to the Launch Services Task Order (LSTO) competition.
- (H) A launch vehicle certified to a higher category is inherently qualified to launch payloads designated as a lower risk classification (i.e., launch vehicles certified to launch vehicle risk category 3 are also certified to launch vehicle risk category 1 and 2).
- (I) If no certified launch vehicle configurations exist to meet specific mission requirements solicited under a NASA request for contract proposal, the Launch Service Program (LSP) may elect to award a mission contingent on the launch vehicle configuration achieving certification requirements tailored by NASA.

3.3 For new launch vehicles and major modifications to certified launch vehicles that do not require a new certification, NASA may require additional technical insight into the design, manufacturing, testing, integration, and first launch of the effected systems and launch vehicle.

3.4 In the event of a launch failure of a certified common launch vehicle configuration, LSP may participate in or perform a failure investigation/return-to-flight board to approve the cause and corrective action for the launch vehicle configuration to maintain a particular risk mitigation certification category.

4.0 OFF-RAMP CLAUSE

4.1 Notification of Discontinued Launch Service Availability

The Contractor may determine, during the life of this contract, that launch services contracted hereunder are no longer economically viable in the marketplace. In such event, the Contractor shall provide written notification to the Government at least 12 months prior to the proposed discontinuation of any launch service available under this contract. The Contractor's notification shall include a brief explanation for the discontinuation and a request for a contract modification to remove the launch service. If the launch service is the only launch service available under the contract, the Contractor shall submit a request for a contract modification to terminate the contract at no cost to

either party after the completion of all current contracted orders for the discontinued launch service. In any event, Launch Service Task Orders existing at the time of notification shall not be affected.

4.2 Reserved

4.3 Reserved

4.4 Reserved

(A) Reserved

(B) Reserved

(C) Reserved

(D) Reserved

5.0 LAUNCH SERVICE PAYMENTS, MILESTONE EVENTS AND COMPLETION CRITERIA

5.1 The Contracting Officer will notify the Contractor of ATP of the launch service via award of the task order.

5.2 Upon successful completion of a milestone event and submission of a properly certified invoice, the Contractor may request commercial interim payments. The commercial interim payments will be paid in accordance with the payment schedule shown in Table C-1 and based on the milestone events identified in the contract Work Plan (Attachment D8).

5.3 Commercial interim payments are contract financing payments that are not payment for accepted items. Commercial interim payments are fully recoverable, in the same manner as progress payments, in the event of default. Commercial interim payments are contract financing payments and, therefore, are not subject to the interest-penalty provisions of prompt payment. However, these payments shall be made in accordance with the Agency's policy for prompt payment of contract financing payments. In accordance with 42 U.S.C. 2465d, NASA shall not take title to launch vehicles under contract for launch services.

5.4 In the event an LSTO is issued at twenty-four (24) months or less prior to launch, the initial payment for such launch service will be the cumulative total of all payments that would have been paid if the LSTO had been issued at L-30 months. If a non-standard service is ordered when an LSTO is issued, the initial payment shall include the catalog price adjustment.

- 5.5 Scheduled dates in Table C-1 are tentative and represent invoice submission dates. Payment schedules may be deferred or canceled by the Government if the Contractor fails to make substantial progress in accomplishing the major launch service milestone events in the attached contract Work Plan. Payments falling due in the first quarter of each fiscal year (October - December) shall be paid promptly to the maximum extent practicable, but shall not be considered late until January 31 of the following calendar year. In the event the contractor completes a milestone ahead of the milestone completion date, the contractor may submit a proper invoice and the Government will consider on a case-by-case basis, the early payment of the milestone.
- 5.6 In the event launch service SubCLINs are terminated independently of standard launch service CLINs, the parties mutually agree the same termination refund schedule percentage(s) shall be used and applied against the price for the terminated SubCLINs. Table C-1 milestone events shall apply to each IDIQ LSTO issued under this contract.
- 5.7 This paragraph will apply to any postponement declared by the Government or the Contractor for any reason, including launch schedule adjustments and Contractor failure to make substantial progress in accordance with launch service milestone events (as determined by the Government under this clause). In the event of a launch schedule adjustment by the Government or Contractor in accordance with contract Section C, Clause 19.0, Adjustments to Launch Schedule, the payment schedule for the applicable launch service CLINs and SubCLINs shall be postponed for the length of the delay, if necessary, by the Government to correspond with the new launch date and the milestone events in the attached Work Plan. The requirement to make substantial progress in general conformance with the attached Work Plan, however, is not waived for any postponed launch service. In the event of a launch schedule adjustment by the Government or Contractor, the Government or Contractor shall be entitled to an equitable adjustment as determined by the contract Section C, Clause 19.0, Adjustments to Launch Schedule.
- 5.8 The Contractor agrees in the event of a termination of this contract pursuant to contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement Under FAR Clause 52.212-4(l), the Government shall not be obligated in any event to pay or reimburse the Contractor any amount in excess of the amount already obligated to the contract. The Contractor shall not be obligated to continue performance of the work beyond such point. The Government shall not be obligated in any event to pay or reimburse the Contractor in excess of the amount obligated to the contract on a quarterly basis, notwithstanding anything to the contrary in contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement Under FAR 52.212-4(l) and NFS 1852.232-77 Limitation of Funds (Fixed-Price Contract) (Mar 1989).

Launch Service Payment Schedule				
Milestone/ Commercial Interim Payment No.	Payment(s) Months Before / After Launch ¹ <i>Offerors to insert additional column to the left containing invoice submission dates</i>	Amount (% of Launch Service CLIN Price)	Cumulative Amount of Launch Service Payments (\$) <i>Offerors to insert additional column to the left containing individual dollar amounts</i>	Termination for Convenience of the Government Repayment Schedule – Percentage (%) of Cumulative Payments Made to Date to be Returned to Government
1	L-“X” ² / L-30	10	TBD	TBD (50-100) / 50
2	L-27	5	TBD	50
3	L-24	5	TBD	50
4	L-21	10	TBD	50
5	L-18	10	TBD	50
6	L-15	10	TBD	50
7	L-12	10	TBD	50
8	L-09	3	TBD	50
9	L-05	5	TBD	50
10	L-03	2	TBD	50
11	L-02	10	TBD	50
12	Launch ³	10 ³	TBD	--
13	Launch + 03 ³	10 ³	TBD	--

Table C-1: Launch Service Payment Schedule

¹ The Government reserves the right to extend or shorten the above payment schedule by plus or minus three months (i.e., L-33M or L-27M) at no increase in contract value to accommodate mission specific requirements.

² In the event that a mission requires an integration period greater than L-30M plus 3 months, the Government shall have the right to order the Extended Integration non-standard services at the prices and for the launch vehicle configurations identified in Table B-10. The payment numbers for L-“X”M through launch shall be adjusted as specified in the RLSP. The cumulative amount of the payments for the extended

integration periods (L-“X”M) combined with the L-30M payment shall not exceed 10% of the total launch service price unless specified in the RLSP.

- ³ The Contracting Officer will either approve the final payment (Launch+ 03M) within thirty (30) days after receipt of the Final Flight Report, (CDRL C4-13) or withhold the final payment in accordance with the provisions of contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions.

The final payment (Launch + 03M) shall not be made in event of partial mission success or failed mission determination by the Government in accordance with contract Section C, Clause 24.0. In the event of a partial mission success or failed mission, the final payment shall be forfeited by the Contractor and is not recoupable. In the event of a failed mission determination, an additional 15% of the launch service price shall be applied as a credit to another launch service or be returned to the Government if it cannot be applied to a subsequent launch service at the Government’s discretion. If funds are not credited or returned within thirty (30) days of a failed mission determination, the funds shall be subject to interest penalties at the prevailing U.S. Treasury interest rate established for Prompt Payment.

- 5.9 The Contracting Officer will unilaterally determine the Contractor’s accomplishment and successful completion of each milestone event. The Contracting Officer’s determination of milestone event completion will include, but is not limited to, the accomplishment criteria listed for the major milestone events set forth in Attachment D8, Work Plan. In addition, the Contracting Officer will determine if the following are complete for each payment requested: all Contract Data Requirements List (CDRL) data item deliverables for which delivery is required prior to the requested payment and all previous events have been met. Approval of the final payment will be made in accordance with contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions.

**6.0 NFS 1852.232-77 LIMITATION OF FUNDS (FIXED-PRICE CONTRACT)
(MAR 1989)**

- (a) Of the total price of CLINs [TBD] through [TBD], the sum of \$TBD is presently available for payment and allotted to this contract. It is anticipated that from time to time additional funds will be allocated to the contract as required by the payment schedule in contract Section C, Clause 5.0, Launch Service Payments, Milestone Events and Completion Criteria, until the total price of said CLINs is allotted.
- (b) The Contractor agrees to perform or have performed work on the items specified in paragraph (a) of this clause up to the point at which, if this contract is terminated pursuant to contract Section C, Clause 1.0(l) “Termination for the Government’s convenience” of this contract, the total amount payable by the Government pursuant to contract Section C, Clause 1.0(l) would equal the amount retained by the Contractor pursuant to Section C, Clause 5.0 and Table C-1. The Contractor is not obligated to continue performance of the work beyond that point. The Government is not obligated in any event to pay or reimburse the Contractor more

than the amount from time to time allotted to the contract, anything to the contrary in contract Section C, Clause 1.0(l) "Termination for the Government's convenience" notwithstanding.

- (c) (1) It is contemplated that funds presently allotted to this contract will cover the work to be performed until [TBD].
- (2) If funds allotted are considered by the Contractor to be inadequate to cover the work to be performed until that date, or an agreed date substituted for it, the Contractor shall notify the Contracting Officer in writing when within the next sixty (60) days the work will reach a point at which, if the contract is terminated pursuant to contract Section C, Clause 1.0(l) "Termination for the Government's convenience" of this contract, the total amount payable by the Government pursuant to contract Section C, Clause 1.0(l) would equal the amount retained by the Contractor pursuant to contract Section C, Clause 5.0 and Table C-1.
- (3) (i) The notice shall state the estimate when the point referred to in paragraph (c)(2) of this clause will be reached and the estimated amount of additional funds required to continue performance to the date specified in paragraph (c)(1) of this clause, or an agreed date substituted for it.
- (ii) The Contractor shall, sixty (60) days in advance of the date specified in paragraph (c)(1) of this clause, or an agreed date substituted for it, advise the Contracting Officer in writing as to the estimated amount of additional funds required for the timely performance of the contract for a further period as may be specified in the contract or otherwise agreed to by the parties.
- (4) If, after the notification referred to in paragraph (c)(3)(ii) of this clause, additional funds are not allotted by the date specified in paragraph (c)(1) of this clause, or an agreed date substituted for it, the Contracting Officer shall, upon the Contractor's written request, terminate this contract on that date or on the date set forth in the request, whichever is later, pursuant to contract Section C, Clause 1.0(l) "Termination for the Government's convenience".
- (d) When additional funds are allotted from time to time for continued performance of the work under this contract, the parties shall agree on the applicable period of contract performance to be covered by these funds. The provisions of paragraphs (b) and (c) of this clause shall apply to these additional allotted funds and the substituted date pertaining to them, and the contract shall be modified accordingly.
- (e) If, solely by reason of the Government's failure to allot additional funds in amounts sufficient for the timely performance of this contract, the Contractor incurs additional costs or is delayed in the performance of the work under this contract,

and if additional funds are allotted, an equitable adjustment shall be made in the price(s) (including appropriate target, billing, and ceiling prices where applicable) of the items to be delivered, or in the time of delivery, or both.

- (f) The Government may at any time before termination, and, with the consent of the Contractor, after notice of termination, allot additional funds for this contract.
- (g) The provisions of this clause with respect to termination shall in no way be deemed to limit the rights of the Government under the Termination for Cause clause of this contract. The provisions of this Limitation of Funds clause are limited to the work on and allotment of funds for the items set forth in paragraph (a) of this clause. This clause shall become inoperative upon the allotment of funds for the total price of said work except for rights and obligations then existing under this clause.
- (h) Nothing in this clause shall affect the right of the Government to terminate this contract pursuant to contract Section C, Clause 1.0(l) "Termination for the Government's convenience" of this contract.

7.0 SECURITY FOR LAUNCH SERVICE PAYMENT FINANCING

- 7.1 *Requirements for payment.* Payments will be made under this contract upon submission of properly certified invoices or vouchers by the Contractor, and approval by the administering office, NASA John F. Kennedy Space Center. The amount of all invoices or vouchers submitted shall not exceed the total contract price for all CLINs, authorized LSTOs issued under the IDIQ contract, and special task assignments (studies and analyses).
- 7.2 *Security.* Pursuant to FAR Subpart 32.202-4 Security for Government Financing and 10 U.S.C. 2307(f) and 41 U.S.C. 255(f), the Government is required to obtain adequate security for Government financing. Adequate security for payments made under this contract shall be required in the form of a preferred creditor's lien. The Contractor shall grant the Government a preferred creditor's lien i.e., a first lien paramount to all other liens against all work in process sufficient to recompense the Government for all monies advanced under this contract should the Contractor's performance prove to be materially unsatisfactory.
- 7.3 *Insurance.* The Contractor represents and warrants that it maintains with responsible insurance carriers (1) insurance on plant and equipment against fire and other hazards to the extent similar properties are usually insured by others operating plants and properties of similar character in the same general locality; (2) adequate insurance against liability on account of damage to persons or property; and (3) adequate insurance under all applicable workers' compensation laws. The Contractor agrees that, until work under this contract has been completed and all payments made under this contract have been liquidated, it will maintain this insurance and furnish any certificates with respect to its insurance that the administering office may require.

7.4 *Representations and Warranties.* The Contractor represents and warrants the following:

- (A) The balance sheet, the profit and loss statement, and any other supporting financial statements furnished to the administering office fairly reflect the financial condition of the Contractor at the date shown or the period covered, and there has been no subsequent materially adverse change in the financial condition of the Contractor.
- (B) No litigation or criminal or civil proceedings are presently pending or threatened against the Contractor, which would jeopardize performance under this contract, except as shown in the financial statements.
- (C) The Contractor has disclosed all contingent liabilities in the financial statements furnished to the administering office.
- (D) None of the terms in this clause conflict with the authority under which the Contractor is doing business or with the provision of any existing indenture or agreement of the Contractor.
- (E) The Contractor has the power to enter into this contract and accept payments, and has taken all necessary actions to authorize the acceptance under the terms of this contract.
- (F) The assets of the Contractor are not subject to any lien or encumbrance of any character, which would jeopardize performance under this contract, except for current taxes not delinquent and except as shown in the financial statements. There is no current assignment of claims under any contract affected by these payment provisions.
- (G) All information furnished by the Contractor to the administering office in connection with each request for payment is true and correct.
- (H) These representations and warranties shall be continuing and shall be considered to have been repeated by the submission of each invoice for payments.

7.5 *Work Plan Submission.* The Contractor shall comply with the Work Plan that identifies the major milestone events and a corresponding narrative of the work activity necessary to accomplish the major milestone events.

8.0 SCOPE OF WORK – SPECIAL TASK ASSIGNMENTS (STUDIES AND ANALYSES) (CLIN 10)

8.1 The Contractor shall perform specific task assignments that will be defined through the issuance of task orders in accordance with contract Section C, Clause 9.0, Ordering Procedure and Payment for Special Task Assignments (Studies and Analyses) (CLIN 10) and paragraphs 8.3 and 8.4 herein.

8.2 Reserved

8.3 Only hours expended against specific written task orders authorized by the Contracting Officer shall be paid under this contract. Task orders shall be performed on an FFP, completion-effort basis. The Government is under no obligation to reimburse the Contractor for any costs, which exceed the Firm-Fixed-Price authorized on each task order.

8.4 The Government may require Special Task Assignments (Studies and Analyses) in any calendar year from 2026 through 2032 at the FFP composite labor rates identified in Section B, paragraph 1.5. The proposed labor rates shall be negotiated using the applicable procedures of FAR Part 15 and incorporated herein.

8.5 Reserved

9.0 ORDERING PROCEDURE AND PAYMENT FOR SPECIAL TASK ASSIGNMENTS (STUDIES AND ANALYSES) (CLIN 10)

9.1 The Contractor shall incur hours under CLIN 10 of this contract only in the performance of task orders and revisions to task orders issued in accordance with this ordering procedure. No other costs or hours under CLIN 10 are authorized. Special Task Assignments shall be performed on a Firm-Fixed-Price (FFP) completion basis unless specifically authorized otherwise by the Contracting Officer.

9.2 Special Task Assignments shall be subject to the following ordering procedure:

(A) The Contracting Officer shall request a Task Plan from the Contractor. The request shall include:

- (i) Statement of Work
- (ii) Required Completion Date
- (iii) Government Point of Contact
- (iv) Due date for Contractor's Task Plan

(B) The Contractor's Task Plan shall include:

- (i) Discussion of the technical approach for performing the work and rationale for any proposed changes to the Statement of Work and required completion date.
- (ii) Required authorization to proceed date.
- (iii) Basis of Estimate (BOE) - The total price broken down by task with the following elements identified:
 - (a) The direct labor hours broken down by work element subject to the Hourly Labor Rates in Section B, Clause 1.5, Table B-6

- (b) Other Direct Costs identified by element (Travel, Material, Subcontracts) if applicable.
 - (C) The Contracting Officer shall accept the Contractor's Task Plan within ten (10) working days of receipt or enter into negotiations with the Contractor. The parties shall execute a bi-lateral Special Task Assignment.
 - (D) The Special Task Assignment will contain, as a minimum, the following information:
 - (i) Signature of the Contracting Officer's Technical Representative and the Government's Contracting Officer
 - (ii) Contract number, Special Task Assignment number, effective date and completion date
 - (iii) Title and Statement of Work
 - (iv) Total firm fixed price, including the number of labor hours
 - (v) Quality assurance standards, as appropriate
 - (vi) Any other necessary information
 - (E) The Special Task Assignments may be revised in the same manner as they are executed via formal revisions.
- 9.3 Executed Special Task Assignments shall be incorporated by reference into the contract via contract modification to include the following: Special Task Assignment number, title, number of labor hours, and firm fixed price. The contract modification will reflect the firm fixed price of the Special Task Assignments with respect to the adjustment of the contract value and any other terms deemed pertinent.
- 9.4 Costs and/or labor hours incurred in excess of those authorized by the Contracting Officer shall be at the Contractor's own expense. Failure to agree to any adjustment shall be handled IAW the Contract's Disputes clause. However, nothing in this clause shall excuse the Contractor from proceeding with the Special Task Assignment as written.
- 9.5 The price for completion of special task assignments under CLIN 10 will be paid upon satisfactory completion of each individual Special Task Assignment.
- 10.0 RESERVED**
- 11.0 FAR 52.216-18 ORDERING (AUG 2020)**
[Note: Applicable to IDIQ contract portion only]
- (a) Any supplies and services to be furnished under this contract shall be ordered by issuance of delivery orders or task orders by the individuals or activities designated in the Schedule. Such orders may be issued from the effective date of the contract through June 30, 2030. Special task assignments, Mission Unique Services, and

non-standard service task orders may be issued from the effective date of the contract through the completion of all LSTOs.

- (b) All delivery orders or task orders are subject to the terms and conditions of this contract. In the event of conflict between a delivery order or task order and this contract, the contract shall control.
- (c)
 - (1) If sent by mail (includes transmittal by U.S. mail or private delivery service), the Government deposits the order in the mail;
 - (2) If sent by fax, the Government transmits the order to the Contractor's fax number; or
 - (3) If sent electronically, the Government either—
 - (i) Posts a copy of the delivery order or task order to a Government document access system, and notice is sent to the Contractor; or
 - (ii) Distributes the delivery order or task order via email to the Contractor's email address.
 - (d) Orders may be issued by methods other than those enumerated in this clause only if authorized in the contract.

12.0 FAR 52.216-19 ORDER LIMITATIONS (OCT 1995)

[Note: Applicable to IDIQ contract portion only]

- (a) *Minimum order.* When the Government requires supplies or services covered by this contract in an amount of less than:
 - (1) One Launch Service, or
 - (2) One Special Task Assignment, or
 - (3) One Non-Standard Service

The Government is not obligated to purchase, nor is the Contractor obligated to furnish, those supplies or services under the contract.

- (b) *Maximum order.* The Contractor is not obligated to honor—
 - (1) During any calendar year, any order for a single item in excess of
 - TBP** *[To be inserted by Offeror]* Launch Service or
 - TBP** *[To be inserted by Offeror]* Special Task Assignment or
 - TBP** *[To be inserted by Offeror]* Non-Standard Service
 - (2) Any order for a combination of items in excess of **TBP** *[To be inserted by Offeror]*; or

- (3) A series of orders from the same ordering office within thirty (30) days that together call for quantities exceeding the limitation in paragraph (b) (1) or (2) of this clause.
- (c) Reserved
- (d) Notwithstanding paragraphs (b) and (c) of this clause, the Contractor shall honor any order exceeding the maximum order limitations in paragraph (b), unless that order(s) is returned to the ordering office within thirty (30) days after issuance, with written notice stating the Contractor's intent not to ship the item(s) called for and the reasons. Upon receiving this notice, the Government may acquire the supplies or services from another source.

13.0 52.216-22 INDEFINITE QUANTITY (DEVIATION NOV 2025)

[Note: Applicable to IDIQ contract portion only]

- (a) This is an indefinite-quantity contract for the supplies or services specified and effective for the period stated, in Section B and C of the Schedule. The quantities of supplies and services specified in the Schedule are estimates only and are not purchased by this contract.
- (b) Delivery or performance shall be made only as authorized by orders issued in accordance with contract Section C, Clause 11.0, FAR 52.216-18, Ordering (Oct 1995). The Contractor shall furnish to the Government, when and if ordered, the supplies or services specified in the Schedule up to and including the quantity designated in the Schedule as the "maximum." The Government shall order at least the quantity of supplies or services designated in the Schedule as the "minimum."
- (c) Except for any limitations on quantities in contract Section C, Clause 12.0, FAR 52.216-19 Order Limitations (Oct 1995) or in the Schedule, there is no limit on the number of orders that may be issued. The Government may issue orders requiring delivery to multiple destinations or performance at multiple locations.
- (d) Any order issued during the ordering period of this contract and not completed within that period shall be completed by the Contractor within the time specified in the order, which may include order options to be exercised after the ordering period of this contract but before the end of the period of performance of the order. The contract shall govern the Contractor's and Government's rights and obligations with respect to that order to the same extent as if the order were completed during the contract's ordering period; provided, the Contractor shall not be required to make any deliveries under this contract after December 31, 2032.

14.0 LAUNCH SERVICE TASK ORDERING PROCEDURES

[Note: Applicable to IDIQ contract portion only]

14.1 *Requirements for Competition.* The intent of IDIQ is to support missions in addition to the awarded missions. The Government will provide all multiple award Contractors a fair opportunity to be considered for task orders issued under this contract based upon the specific task order requirements, unless the Contracting Officer determines that one of the following apply:

- (A) The Agency need is of such urgency that competing the requirements among Contractors would result in unacceptable delays
- (B) Only one Contractor is capable of providing the service requested at the level of quality required because the service ordered is unique or highly specialized
- (C) The order should be issued on a sole-source basis in the interest of economy and efficiency as a logical follow-on to an order issued under the contract, provided that all Contractors were given a fair opportunity to be considered for the original order; or
- (D) It is necessary to place an order to satisfy the minimum guarantee.

14.2 Types of Task Orders

There are two types of task orders that may be issued under this contract. The first type is an LSTO. Any required services related to a specific launch service will be included in the LSTO. All items other than launch services and items not related to a specific launch service will be obtained by task orders referred to herein as Special Task Assignments (Studies and Analyses). However, secondary payloads shall be obtained as mission unique services in accordance with Statement of Work paragraph 4.3.

14.3 Launch Service Task Orders

- (A) *Requests for Launch Service Proposals (RLSPs).* Prior to the issuance of an RLSP, exchanges and fact-finding may take place with multiple award Contractors. The RLSP will include specific information unique to the mission such as Mission Success risk posture (Section C, Clause 24.4, Table C-8); payload mass and volume; orbital requirements (such as altitude and/or inclination); target launch date and science window, if appropriate; and any unique payload design considerations.

The RLSP will provide any special instructions regarding the level of detail required in the proposal. The RLSP will include a date and time for submission of the proposal. Proposals will be due within thirty (30) calendar days from the date of the RLSP unless stated otherwise.

- (B) *LSTO Proposal.* The proposed common launch vehicle configuration shall achieve one successful flight prior to submission of a proposal in order to be considered for LSTO award. The Contractor, when submitting its firm-fixed price proposal, shall

indicate the proposal is compliant with the contract terms, statement of work, and the specific requirements contained in the RLSP.

Firm-fixed pricing for all proposals shall not exceed the prices contained in the Schedule. The prices contained in the Schedule are prices, which can be adjusted downward by the Contractor for the specific launch service being proposed. Any Contractor proposed reduction will be applicable to the current proposal only and will not be deemed as a permanent reduction of the prices contained in the Schedule. Any mission unique modifications not priced in the non-standard services must be separately identified and individually priced, and are not limited by the unit price in the Schedule.

The proposed launch service price, including applicable priced non-standard services and mission uniques shall be totaled for a single firm-fixed price for all efforts required under the order for that launch service. The total firm fixed-price shall be applied to the percentages in contract Section C, Clause 5.0, Launch Service Payments, Milestone Events and Completion Criteria, for conversion to dollar amounts for each payment event; and shall identify any consideration being provided as a result of changes to the Mission Success risk posture, as applicable. In addition, the Work Plan milestones shall be appropriately modified to add any non-standard and/or mission unique services. The Launch Service Task Order Mission Solutions shall be incorporated in Exhibit 9.

- (C) *Mandatory LSTO Proposal Submission.* Unless otherwise agreed to by the Contracting Officer, it is mandatory for contract holders under multiple award NLSII contracts to respond via a Proposal or Request for Waiver to each order for launch service requirements provided these requirements are identified in the schedule and do not conflict with the contract ordering limitations. The response shall include a Proposal or Request for Waiver for all common launch vehicle configurations that meet the requirements for a particular launch service task order. In the event there arises legitimate reasons for an awardee not to submit a proposal for a particular launch service task order (e.g., limited capacity to perform, excessive performance capability, lack of one successful flight, or launch service is no longer viable), the Contracting Officer may waive the requirement for proposal submission.
- (D) *LSTO Proposal Evaluation/LSTO Selection Criteria.* All LSTO proposals shall be submitted by the date and time specified in the RLSP, or it will be treated as a late proposal in accordance with FAR 52.212-1, Instructions to Offerors – Commercial Products and Commercial Services. If this requirement is met, the Contracting Officer will consider the following three factors prior to award of an LSTO:
 - (i) Technical capability/risk, including the following:
 - (a) Contractor's ability to meet SOW and specific payload requirements, including launch period and launch date

- (b) Launch vehicle certification or reasonableness of the Offeror's plan to obtain certification prior to mission launch
- (c) Information gained through Government insight and approval activities
- (d) Status of significant baseline vehicle changes and resolution of anomalies
- (e) Unique terms and conditions contained in a contract, which may impact price, performance or risk
- (f) Proposed launch vehicle's demonstrated flight experience
- (ii) Reasonableness of proposed price, including any proposed quantity discounts.
- (iii) Past performance, with emphasis given to the most recent and more relevant experience, including small business achievements.

The Requests for Launch Service Proposals shall state, whether all evaluation factors other than cost or price, when combined, are—

- (i) Significantly more important than cost or price;
- (ii) Approximately equal to cost or price; or
- (iii) Significantly less important than cost or price.

The Government reserves the right to modify the evaluation criteria for individual LSTOs. The final evaluation criteria will be defined in the request for LSTO proposal. In any event, the Government intends to award an LSTO to the Contractor who provides the best value in launch services that meet the Government's requirements.

(E) *Award of an LSTO.*

- (i) The Contracting Officer will notify the Contractor of ATP of the launch service via award of the task order.
- (ii) Each of the IDIQ Contractors will be notified of the Government's award of an LSTO.

1. Pursuant to FAR 16.505(a)(10), No protest under Subpart 33.1 is authorized in connection with the issuance or proposed issuance of an

order under a task-order contract or delivery-order contract, except for—

- (A) A protest on the grounds that the order increases the scope, period, or maximum value of the contract; or
- (B) A protest of an order valued in excess of \$25 million. Protests of orders in excess of \$25 million may only be filed with the Government Accountability Office, in accordance with the procedures at 33.104.

2. The authority to protest the placement of an order is governed by FAR 16.505(a)(10). (10 U.S.C. 2304a(d) and 2304c(d), and 41 U.S.C. 253h(d) and 253j(d)).

3. Pursuant to FAR 16.505(b)(1)(iv), for task or delivery orders in excess of \$5.5 million, the requirement to provide all awardees a fair opportunity to be considered for each order shall include an opportunity for a post-award debriefing in accordance with FAR 16.505(b)(6).

- (iii) Launch service task order mission solutions will be attached to the contract (Exhibit 9) upon award of task order(s)

(F) *Modifications to LSTOs.* After an LSTO is issued, it may be necessary to add priced non-standard services to the LSTO. These additions will be accomplished via modifications to the original LSTO. In addition, mission uniques that are not included in the priced non-standard services may be added to, or deleted from LSTOs via modification to the original LSTO. In this instance, the terms of the existing LSTO, such as price, milestone events, and Work Plan completion criteria, may be modified to reflect the change. The remainder of the price of the modification will be applied to the remaining payment milestone event amounts, as performance dictates.

(G) *Task Order Authorization and Content.* The only persons authorized to issue task orders under this contract are the KSC Contracting Officers. Task orders will be issued in writing. However, any facsimile, or electronic, task orders issued by the Contracting Officer will be confirmed in writing within five (5) business days. The Contractor will acknowledge receipt and acceptance of the task order by signing the task order and returning it to the Contracting Officer. Each task order will include the following information:

- (i) Date of the task order
- (ii) Contract number and task order number
- (iii) Statement of Work and any other documentation on which the price is based

- (iv) Product or service to be delivered
- (v) Task order price
- (vi) Completion/Delivery date
- (vii) Reserved
- (viii) Any other necessary information

15.0 PRESERVATION, PACKING, PACKAGING, AND MARKING FOR DOCUMENTATION

Preservation, packing, packaging and marking for shipment of all items ordered hereunder shall be in accordance with commercial practice and adequate to insure safe transportation, acceptable by common carrier, and transportation at the most economical rate(s). The Contractor shall place identical requirements on all subcontracts for items delivered to NASA.

16.0 FAR 52.246-11 HIGHER LEVEL CONTRACT QUALITY REQUIREMENT (DEC 2014)

(a) The Contractor shall comply with the higher-level quality standard selected below.

Title	Number	Date	Tailoring
☒ Aerospace QMS	AS9100	Rev D	

(b) The Contractor shall include applicable requirements of the higher-level quality standard(s) listed in paragraph (a) of this clause and the requirement to flow down such standards, as applicable, to lower-tier subcontracts, in—

- (1) Any subcontract for critical and complex items (see 46.203(b) and (c)); or
- (2) When the technical requirements of a subcontract require—
 - (i) Control of such things as design, work operations, in-process control, testing, and inspection; or
 - (ii) Attention to such factors as organization, planning, work instructions, documentation control, and advanced metrology.

17.0 INSPECTION SYSTEM RECORDS

The Contractor shall maintain records evidencing inspections in accordance with the Inspection clauses of this contract for one year after delivery of all items and/or completion of all services called for by the contract.

18.0 REQUIRED TIME OF DELIVERY FOR LAUNCH SERVICES

- 18.1 The Government requires launch service delivery to be made according to the schedule provided in Table C-4.

Table C-2: Reserved

Table C-3: Reserved

IDIQ LSTOs				
CLIN	Mission	Quantity	Limited Launch Period*	Launch Date/ Launch Period
TBD	TBD	TBD	TBD	TBD

* Indicates whether or not the mission has a limited launch period due to specific Planetary, Lunar, Scientific, and/or Lunar Exclusion requirements

Table C-4: IDIQ LSTO Delivery Schedule

- 18.2 Reserved

19.0 ADJUSTMENTS TO LAUNCH SCHEDULE

- 19.1 NASA/Contractor will give written notice of any desired change in the launch schedule as soon as possible. In the case of a request for postponement of the launch date by NASA/Contractor, NASA/Contractor will propose a new launch date. Within two weeks of receipt of the written request of a launch schedule adjustment, the Contractor/NASA will inform NASA/Contractor whether a launch opportunity exists as requested or will propose an alternatively available launch date. NASA/Contractor will work to provide written agreement within thirty (30) days following receipt of the Contractor's/NASA's proposition, and will work together to deconflict and resolve any possible Range conflicts. If mutual agreement on the revised launch date cannot be reached due to launch vehicle or payload readiness, resolution between the parties shall be handled IAW the Contracts Disputes Clause.

- 19.2 The postponement periods for both NASA and the Contractor shall include the applicable grace period plus any excusable delay (e.g. a "No Fault" delay as described in section 19.9) plus an additional eighteen (18) months each. Postponements during this period shall be subject to the postponement fees pursuant to sections 19.3 through 19.6 and will be paid by the postponing party. In the event of a single postponement, or cumulative postponements for each launch service by NASA/Contractor exceeds eighteen (18) months in addition to any applicable grace period or excusable delay, the contract shall be subject to equitable adjustment for that portion of delay exceeding eighteen (18)

months. Any amounts owed by the postponing party as equitable adjustment shall be in addition to the maximum postponement fees for the eighteen (18) month delay, and/or other available remedies provided for under contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement Under FAR 52.212-4(l), and contract Section C, Clause 1.0(m), Termination for Cause. Alternatively, any delay declared by NASA, which results in a launch date later than the contract period of performance shall be subject to an equitable adjustment.

- 19.3 If the Contractor requests a postponement of the launch date and NASA agrees to the postponement, the Parties agree that, in lieu of Termination for Cause and in place of actual damages, and as fixed, agreed, and liquidated damages, the price of each applicable launch service shall be reduced, dependent upon when receipt of the notification is received, for each calendar day of delay in excess of the grace period by the amount as shown in Table C-5. In this event, NASA shall have the right of approval of the revised launch date prior to its implementation. NASA shall also have the right of approval of the reallocation of launch vehicles if milestone payments have already been made towards a designated launch vehicle. Finally, NASA shall have the right of approval prior to any revision in its position (i.e., the order in the Contractor's launch manifest queue sequence, if it results in the postponement of the NASA launch date).
- 19.4 If the Contractor fails to request a postponement and the major program milestone event designated as the launch is delayed (i.e., the launch service is not completed by the contractual launch date), for reasons other than those excusable delays described in paragraph 19.9, then the Contractor shall pay NASA for each day of delay in accordance with Table C-5. Alternatively, if delivery or performance is so delayed by the Contractor, NASA may terminate this contract in whole or in part under contract Section C, Clause 1.0(m), Termination for Cause, of this contract and in that event, the Contractor shall be liable for fixed, agreed, and liquidated damages accruing until the time NASA may reasonably obtain delivery or performance of similar supplies or services, up to the maximum specified in Clause 19.7. The liquidated damages shall be in addition to any other costs under contract Section C, Clause 1.0(m), Termination for Cause.
- 19.5 Each postponement request by the Contractor that is not the result of paragraph 19.9 herein is subject to the liquidated damages shown in Table C-5. The delay damages are calculated on a per mission basis.

Offerors shall propose, in Table C-5, Liquidated Damages paid to NASA for each day of launch delay. The Time Periods and Grace Days shall not be adjusted.

Contractor Liquidated Damages and Grace Periods		
During Months Prior to Launch Date – First Day to Last Day (Months)	Liquidated Damages Paid to NASA for Each Day of Launch Delay (\$)	Grace Period (Days)
L-“X” (ATP) through L-24	TBP	150
L-24 through L-12	TBP	120

Contractor Liquidated Damages and Grace Periods		
During Months Prior to Launch Date – First Day to Last Day (Months)	Liquidated Damages Paid to NASA for Each Day of Launch Delay (\$)	Grace Period (Days)
L-12 through L-6	TBP	90
L-6 through L-3	TBP	60
L-3 through L-11 Days	TBP	30
L-10 Days through Launch	TBP	7

Table C-5: Contractor Liquidated Damages and Grace Periods

- 19.6 If NASA requests a postponement of the launch date, the Parties agree that, in place of actual damages, and as fixed, agreed, and liquidated damages, the price of each applicable launch service shall be increased, dependent upon when receipt of the notification by the Contracting Officer is received, for each calendar day of delay in excess of the grace period by the amount as shown in Table C-6. Each postponement request by NASA that is not the result of paragraph 19.9 is subject to the postponement fees shown in Table C-6.

In Table C-6, Offerors shall propose amounts or grace days equal to that proposed in Table C-5, decreases only to the dollar amounts proposed in Table C-5 or increases only to the number of days specified in the Grace Periods in Table C-5. The Time Periods shall not be adjusted. See additional instructions in Attachment E2.

NASA Postponement Fees and Grace Periods		
During Months Prior to Launch Date – First Day to Last Day (Months)	Postponement Fees Paid to Contractor for Each Day of Launch Delay (\$)	Grace Period (Days)
L-“X” (ATP) through L-24	TBP	TBP
L-24 through L-12	TBP	TBP
L-12 through L-6	TBP	TBP
L-6 through L-3	TBP	TBP
L-3 through L-11 Days	TBP	TBP
L-10 Days through Launch	TBP	TBP

Table C-6: NASA Postponement Fees and Grace Periods

The Liquidated Damages/Postponement fees (Tables C-5 and C-6) cover all costs related to the delay except vehicle de-stack/re-stack (de-mate/re-mate) costs, necessary repeated performance of studies and analyses, and new scope. The party causing the delay will be responsible for the aforementioned costs.

- 19.7 The maximum amount paid for delays (in excess of the grace period) by the Contractor/NASA is \$TBP, and shall be the lesser amount of either 5% of the highest priced launch vehicle configuration at year of contract award or a total of \$9,400,000. The maximum amount applies only to the Liquidated damages/Postponement fees shown in Tables C-5 and C-6. Both parties are entitled to recover additional monies for Equitable Adjustment as referenced in 19.2.

A grace period is defined as the number of days the Contractor/NASA may delay the launch date, without incurring liquidated damages/postponement fees. The amount of grace available for any period is the smaller of the number of days in the grace period column in Table C-5 compared to the remaining available grace days from the previous period.

For example, if 100 days are used during the period L-X through L-24, there would be 50 days available during the period L-24 through L-12. If no further days are used through the period L-6 through L-3, there would be 30 days available during the period L-3 through L-11 days; with 20 days having expired.

- 19.8 In the event that NASA/Contractor postpones the launch date, the payment schedule shall be suspended for the length of the delay and then resumed with all remaining payments shifted by the amount (length) of the delay. For NASA delays only, the Contractor may elect to receive payments as major program milestone events are completed in lieu of the payment of postponement fees or the Contracting Officer may approve the addition of a new milestone payment event at the time of postponement, provided funding is available and obligated to the contract.

- 19.9 NASA/Contractor will not be charged with postponement fees/liquidated damages when the delay in delivery or performance arises solely out of causes beyond the control of NASA/Contractor and not due to the fault or negligence of NASA/Contractor. Such causes include, but are not limited to the following:

Delays resulting from Range Mission Rules and Range Launch Requirements (Mandatory and Required Assets), acts of God, acts (including delay or failure to act) of any Governmental authority (De Jure or De Facto), wars (declared or undeclared), riots, revolution, hijacking, fires, freight embargoes, sabotage, epidemics, strikes and interruptions of essential services such as electricity, natural gases, fuels and water, or any condition which jeopardizes the safety of the employees of the Contractor, NASA, or its subcontractors; or a launch vehicle failure investigation, provided NASA retains its original position in the order of the queue sequence and that all data related to the failure investigation is made available to NASA without restriction.

- 19.10 Once a specific launch date and time have been established, they may be changed by the cumulative amount of the grace period established above by NASA or the Contractor

without consideration to the other party if the mission requirements can be met. The cumulative total of grace periods exercised by either NASA or the Contractor shall not exceed 150 days for either party.

- 19.11 The amount of postponement fees shall be added to/subtracted from (as applicable) the next milestone payment made pursuant to the contract Section C, Clause 5.0, Launch Service Payments, Milestone Events and Completion Criteria. In addition, the delivery schedule shall be adjusted to reflect the revised launch date.
- 19.12 In the event of an anomaly on any sub-orbital or orbital vehicle provided by the Contractor that involves hardware or software directly applicable to this launch service, NASA reserves the right to delay the launch service, until the next available launch opportunity, without accruing damages or providing an equitable adjustment to the Contractor until acceptable resolution of the anomaly.
- 19.13 This clause, including grace periods, shall not apply to missions with limited launch periods. Requests for launch adjustments for missions with limited launch periods shall be in accordance with the “Changes” clause of this contract. Contractor requests for launch adjustments for missions with limited launch periods are subject to equitable adjustment and require approval of NASA prior to implementation. If NASA approval is granted, the Contractor shall be obligated to launch at the next opportunity capable of meeting the mission’s scientific requirements, irrespective of other launch commitments or customer’s priorities.

20.0 PLACE OF PERFORMANCE

The place of performance and launch site will be identified in each launch service or task order. The delivery schedule and/or period of performance of this contract are based upon the launch dates specified in the schedule or each individual task order.

21.0 RESERVED

Table C-7: Reserved

22.0 RESERVED

23.0 USE OF GOVERNMENT PROPERTY, FACILITIES, AND ASSETS

- 23.1 The Contractor shall obtain and maintain any necessary agreements between the Contractor and any Government Agency authorizing the use of Government property, facilities, and assets accountable to those agencies, or services provided by those agencies the Contractor may wish to utilize in performance of this contract.
- 23.2 The Government makes no warranty whatsoever as to the suitability for use of Government property, facilities and other assets made available under the terms and conditions of any Government use agreements or contracts. Any costs necessary to

maintain, restore, refurbish, and/or replace any assets, for use under this contract, shall result in no increase in the price of this contract.

- 23.3 The Contractor is responsible for determining the suitability for use of all materials, property, and facilities acquired or made available to the Contractor by the Government under any contract agreement. Any use of Government-Furnished Property (GFP), materials, or facilities and services shall not relieve the Contractor of full launch service performance responsibility.

24.0 MISSION SUCCESS DETERMINATION, INVESTIGATION, AND CORRECTIVE ACTIONS

24.1 Mission Success Criteria

- (A) A mission will be determined a *Mission Success*, if:
 - (i) The payload is placed into the required orbit by the launch vehicle, and
 - (ii) Received telemetry data shows the Interface Control Document (ICD) environments and parameters were not exceeded, and
 - (iii) The launch vehicle causes no damage to the payload during deployment or thereafter from collision or contamination products, or
 - (iv) There is a violation of the criteria documented in Mission Success Criteria & Determination Methodology (CDRL C2-5) and reported in Final Flight Report (CDRL C4-13); however, there is no launch vehicle/service caused degradation of the payload's ability to perform its intended function as defined in the Payload Program Level Requirements.
- (B) A mission will be determined a *Partial Mission Success* when there is a violation of the criteria documented in Mission Success Criteria & Determination Methodology (CDRL C2-5) and reported in Final Flight Report (CDRL C4-13) during launch/flight which results in degradation of the payload's ability to perform its intended function as defined in the Payload Program Level Requirements.
- (C) A mission will be determined a *Failed Mission* when the payload:
 - (i) Is destroyed during launch/flight, or
 - (ii) Cannot be separated from the launch vehicle, or
 - (iii) Experiences a violation of the criteria documented in Mission Success Criteria & Determination Methodology (CDRL C2-5) and reported in Final Flight Report (CDRL C4-13) during flight which results in degradation of the payload's ability to meet minimum success as defined in the Payload Program Level Requirements.

24.2 Mission Success Determination

- (A) NASA will provide the Contractor the Payload Program Level Requirements that establish payload mission success criteria.
- (B) The Contractor shall be responsible for providing telemetry data confirming the required orbit conditions and payload environments were met as documented in

Mission Success Criteria & Determination Methodology (CDRL C2-5) and reported in Final Flight Report (CDRL C4-13). The Contractor shall measure and provide telemetry data to the extent required by the SOW.

- (C) The Contracting Officer has authority to determine whether a launched mission is a Mission Success, Partial Mission Success, or a Failed Mission in accordance with Section C, paragraph 24.1.

24.3 Procedures

- (A) Not less than two (2) months prior to launch of a payload, NASA will provide the Contractor the Payload Program Level Requirements that establish payload mission success criteria.
- (B) Within thirty (30) calendar days from submittal of the Final Flight Report (CDRL C4-13), the Contracting Officer will either determine the launch a Mission Success or inform the Contractor of the Government's intent to withhold final mission success payment(s) until a final determination of mission success is made.
- (C) In the event of a partial mission success or mission failure under paragraphs 24.1(B) or 24.1(C), the Contracting Officer will utilize all available data for the mission, in particular the focus will be on the Final Flight Report (CDRL C4-13), findings from the Contractor's investigation board, if activated, and findings from the NASA review board(s) if activated. The Contracting Officer will provide a final mission success determination within thirty (30) calendar days of the delivery of final reports for all applicable review board(s).

24.4 Performance-Based Payment for Final Mission Success Determination

- (A) In the event the Contracting Officer determines the launch service a Mission Success, the Contractor shall receive full payment of the final payment event for the launch service.
- (B) In the event the Contracting Officer determines the launch service a Partial Mission Success, the Government will pay no portion of the final payment in accordance with Table C-8.

Mission Success Determination	% of Cumulative Payments	% of Total Launch Service (CLIN) Price Reallocated / Refunded to the Government
Full Success	100	0
Partial Mission Success	90	0
Failed Mission	90	15

Table C-8: Mission Success Payment Schedule¹

¹The Government reserves the right to tailor the combined amount for mission failure from 0% to 25% of the total launch service price in a Request for Launch Service Proposal (RLSP).

- (C) In the event the Contracting Officer determines the launch service a *Failed Mission*, the Government will pay no portion of the final payment (reference Table C-8) and the Contractor shall reallocate 15% of the total launch service (CLIN) price to another NASA mission or refund the 15% if a subsequent launch service is unavailable at the Government's discretion. Withholding of the final payment and an additional 15% reallocation/refund to the Government are the sole remedies of the Government for a *Failed Mission* determination.

24.5 Investigation and Corrective Action

- (A) In the event of an anomaly or failed mission, a NASA review board(s) will determine the cause of anomaly or failure, if activated. The review board(s) will evaluate all available data from the launch vehicle, payload, Range, and other sources in order to determine if the mission failure was attributable to the launch vehicle or conditions for which the Contractor would normally be expected to control or avoid. Based on all available pertinent data, including the findings and recommendations of the review board(s) the Contracting Officer shall make the final determination as to partial mission success or failed mission. If the Contractor disagrees with the determination, the decision shall be subject to the Disputes clause of this contract.
- (B) If one or more of the environmental requirements are exceeded or orbit requirements are not achieved, as documented in CDRL C2-5 and reported in CDRL C4-13 on a particular mission, the Contractor shall investigate the anomaly or failure at its own expense. The Contractor shall determine the scope of the investigation and shall conduct and control the investigation. The Government may designate representatives to observe and participate in the Contractor's failure investigation board. If the Contractor changes the launch vehicle design, the Contractor shall provide NASA insight into the change. The Government may establish an independent assessment team to assess the Contractor's investigative and corrective actions.

- (C) The Contractor shall present to the Government its findings resulting from the investigation and the proposed corrective actions (return to flight activities), if any. The Contractor shall be responsible for proving the corrective action is sufficient to return to flight. The Contracting Officer may either accept or reject any finding, root cause, or corrective action. If the Contracting Officer accepts a finding and the related corrective action, the Contractor shall be responsible for the cost of the corrective action including Contractor's cost for re-acceptance for NASA missions.

The Government may, at its option and its expense, conduct its own investigation of the anomaly or failure. The Contractor shall cooperate with and fully support the Government investigation, at the Contractor's expense. In the event the Government requires additional analyses or tests beyond those planned by the Contractor, the Contractor shall implement the Contracting Officer's written direction to perform the additional tests or analyses. The costs of implementing these additional tests or analyses may be the basis for an adjustment to this contract.

- (D) The Contractor shall report to NASA any flight anomalies from non-NASA missions. For non-NASA missions, the requirements of paragraphs 24.5(B) and (C) apply to these anomalies as the Contracting Officer finds them to be related to NASA missions. Rights under the Default clause and the right to require corrective action before return to flight shall also apply.

24.6 Finality of Contracting Officer's Determination

- (A) Each of the Contracting Officer's Determinations, under paragraph 24.2, are subject to FAR 52.233-1, Disputes.
- (B) Reserved
- (C) In the event the Contractor appeals the Contracting Officer's determination under the Disputes clause in accordance with FAR subpart 33.2 and submits a claim under the Contract Disputes Act, the parties hereby agree the burden of proof shall rest on the Contractor to prove the failure was not due to the LV. The Contractor assumes the responsibility for providing confirming data. The Government will be responsible for providing proof of payload degradation or reduction in operational capacity or longevity.

24.7 Acceptance

Final acceptance of the launch service will be accomplished following the Contracting Officer's mission success determination.

25.0 GOVERNMENT INSIGHT AND APPROVAL

- 25.1 The Contractor shall provide NASA an adequate level of insight into and/or approval of certain Contractor tasks and milestones in order to ensure all reasonable steps have been taken that result in the highest probability of mission success. This includes insight into any corporation, corporate divisions, subsidiaries, joint ventures, partner(s), and/or any other business entity actually performing launch vehicle manufacturing, management, payload/launch vehicle integration, testing and launch. This also includes insight into certain major sub-contractor tasks and milestones (i.e., those sub-contractors that perform major portions of manufacturing or integration of the launch vehicle system).
- 25.2 The Government's monitoring of launch services provided by the private sector has two elements: approval and insight. Government approval is defined as providing authority to proceed and/or formal acceptance of requirements, plans, tests, or success criteria in specified areas. Where Government approval is required, the Contractor shall submit the necessary documentation to the Contracting Officer and copies to the Contracting Officer's Technical Representative (COTR).
- 25.3 Government insight is defined as gaining an understanding necessary to knowledgeably concur/non-concur with the Contractor's actions through interaction, watchful observation, documentation review, meeting attendance, reviews, tests and compliance evaluations. Where Government insight is required, the Contractor shall notify the Contracting Officer, the Government Resident Office or the appropriate Government operations organization of meetings, reviews, or tests in sufficient time to permit meaningful Government participation.
- 25.4 Should approval or insight identify non-compliance with the terms and conditions of the contract, a difference in interpretation of test results, or disagreement with the Contractor technical directions, the Government will take appropriate action within the terms of the contract to ensure compliance via written direction to the Contractor.
- 25.5 NASA shall have insight into any Contractor initiated fleet changes or any changes that may affect NASA missions. This insight shall be accommodated with no increase in contract price.
- 25.6 Specific areas where the Government requires approval and/or insight are listed in paragraphs 25.7 and 25.8 respectively. The paragraphs document requirements specified in NPD 8610.23 Launch Vehicle Technical Oversight Policy.
- 25.7 Specific areas requiring Government approval are:
- (A) Spacecraft-to-launch vehicle interface control documents/drawings.
 - (B) Decisions/resolutions of action items as determined by joint NASA/Contractor mission integration teams.
 - (C) Mission unique hardware design, analysis, manufacture, and test.

- (D) Mission unique software design, analysis, and test.
- (E) Baseline and changes to the Contractor's Risk Management, Reliability, Safety and Health, and Quality Management Plans, consistent with the intent of NPD 1280.1, NPR 7120.5, and NPR 8715.3.
- (F) Top-level test plans, requirements, and success criteria for integrated vehicle systems and for tests that verify the integrated vehicle interfaces.
- (G) Launch commit criteria.
- (H) Closeout of actions from NASA-Chaired Mission and Flight Readiness Reviews.
- (I) Spacecraft handling procedures and deviations.
- (J) Integrated spacecraft/vehicle mate, test, and closeout procedures and deviations.
- (K) Integrated spacecraft/vehicle mate, test, and closeout as performed procedures and deviations.
- (L) Launch countdown procedures and deviations that affect spacecraft/vehicle integrated assembly.
- (M) Anomaly resolutions that affect the integrated assembly.
- (N) Launch Go/No-Go.
- (O) Reserved

25.8 Specific areas to be open to Government insight are:

- (A) Baseline vehicle design, analyses, models, and configuration management.
- (B) Production program reviews, plans, and schedules.
- (C) Production and systems test, and Material Review Boards.
- (D) Critical flight hardware pedigree.
- (E) Safety and Mission Assurance compliance evaluations (prime and subcontractors).
- (F) Pre-ship reviews.
- (G) Design and qualification reviews.

- (H) Major/critical problems.
- (I) Major system and integrated systems tests.
- (J) Post-test data.
- (K) Anomaly resolutions.
- (L) Failure analysis.
- (M) Vehicle/ground support equipment procedures.
- (N) Launch site support work schedules and plans.
- (O) Launch site vehicle preparations and closeout data.
- (P) Vehicle walkdown inspections.
- (Q) Operations and procedure discipline.
- (R) Work practices and documentation.
- (S) Conduct of Contractor-chaired Mission, Launch, and Flight Readiness Reviews.
- (T) Post-flight vehicle, tracking, and Range data.
- (U) Post-flight anomaly investigations/close-outs.
- (V) Flight Software

25.9 Notwithstanding the insight and approvals set forth in Clause 25.0 herein, the Contractor assumes full performance responsibility as set forth in this contract, and neither the Government's insight nor its approval under this paragraph 25 shall be construed as a defense to any finding of mission success or final acceptance / rejection of the launch service.

26.0 GOVERNMENT LAUNCH READINESS ASSESSMENT

26.1 The Contractor shall participate in a NASA-chaired Launch Readiness Review (LRR) (reference 2.2.1.3(E) of the SOW). At the LRR, the Contractor shall summarize the status of its launch vehicle and all supporting elements and attest to readiness to launch the mission. If, after due consideration of the status of the launch vehicle, payload and other launch support systems, NASA does not agree that the total mission is ready for launch, NASA has the right to delay the launch.

- 26.2 The Contractor shall poll NASA in the final launch countdown and any re-cycle procedure during status checks for their approval of the final launch readiness assessment. NASA may declare a "HOLD" and delay the launch at any time during the final launch countdown.
- 26.3 In the event NASA delays the launch, as a result of exercising its rights under this clause, and the causes of the delay were within the control of or due to the fault or negligence of the Contractor or its Subcontractors at any tier, then the provisions of Section C, Clause 19.0, Adjustments to Launch Schedule shall govern and the Contractor shall be deemed to have caused the delay. For the purpose of this clause, the burden of proof for showing that the causes of delay were within the control or fault and/or negligence on the part of the Contractor or its Subcontractors at any tier rests with NASA. If NASA delays the launch beyond the grace period established in Section C, Clause 19.0 and the causes of the delay were not within the control or due to the fault or negligence of the Contractor or its Subcontractors at any tier, then the provisions of Section C, Clause 19.0, Adjustments to Launch Schedule shall apply.

27.0 LICENSES AND PERMITS FOR A LAUNCH SERVICE OPERATOR

The Contractor shall obtain and maintain the necessary licenses, permits and clearances that may be required by the Department of Transportation, Department of Commerce, Department of Defense, NASA, or other Governmental agencies in order to provide launch services under this contract. No Federal Aviation Administration commercial launch license is required under this contract. All costs and fees associated with obtaining licenses, permits and clearances are included in the standard launch service price. Approvals required by the payload are the responsibility of NASA.

28.0 ADVANCE UNDERSTANDING REGARDING TERMINATION SETTLEMENT UNDER FAR 52.212-4(I)

- 28.1 In the event the Government decides to exercise its right to terminate all or part of this contract under Section C, Clause 1.0(I), Termination for the Government's convenience, it is agreed in advance that the Contractor, after receipt of a written notice of termination, will have satisfied all obligations and discharged all duties required by Section C, Clause 1.0(I), Termination for the Government's convenience, when the Contractor has refunded that portion of the milestone-based payment(s) for each launch service affected by the termination, in accordance with Table C-1.
- 28.2 The parties agree that by virtue of the refund specified in Table C-1, any and all claims for equitable adjustment as a result of the termination are fully satisfied and discharged. The parties agree that this settlement represents fair compensation for Contractor effort accomplished for the terminated portions of the contract and that the terms as stated herein represent full and final settlement between the parties. The parties agree that the Contractor shall retain title to all hardware associated with the terminated launch service. The parties agree that the above settlement shall represent the total amount to be paid to

the Contractor without agreeing on or segregating the particular elements of costs or profits comprising this amount.

- 28.3 The refund amount shall be payable in full no later than thirty (30) days after receipt of the written notice of termination. Delinquent payment(s) shall be subject to interest at the applicable rate as determined by the Secretary of the Treasury.

Table C-9: Reserved

- 28.4 The provisions of this Contract clause shall only apply from Award until the point of intentional ignition of the launch vehicle. The provision of this Contract clause shall in no way be deemed to limit the rights of the Government under Section C, Clause 1.0(m), Termination for Cause. In the event the Government exercises its rights under Section C, Clause 1.0(m), Termination for Cause, the provisions of this contract clause will not apply.

29.0 CO-MANIFESTED PAYLOADS

29.1 Definitions

- (A) Primary Payload: the payload which serves to determine the launch day and time, and without which the vehicle would not launch.
- (B) Secondary Payload: Any payload carried by the launch vehicle in space not required by the primary payload. The secondary payload requires the approval of the primary payload customer prior to manifesting. If the secondary payload is not ready to support scheduled launch day, the secondary payload will be demanifested and replaced with a mass simulator.

29.2 NASA Secondary Payload with NASA Primary Payload

(A) Procedures

NASA may request to manifest a NASA secondary payload on a NASA primary mission utilizing the excess performance available to NASA under a launch service previously acquired under the contract. The addition of a secondary payload service will be acquired as a Mission Unique Service in accordance with Statement of Work paragraph 4.3. The Contracting Officer will provide as part of the mission unique service request for proposal, a description of the proposed secondary payload including, as a minimum: SOW including volume, mass, attachment requirements, and data requested by the Contractor.

Upon receipt of the request for proposal, the Contractor shall develop a proposal to accommodate the secondary payload or provide a response that the Contractor cannot accommodate the secondary payload on the requested mission or on any existing planned missions.

If the Government is unable to provide the secondary payload for launch vehicle integration or there is insufficient time to complete a new mission analysis before the launch date, the Government will bear the cost and the Contractor shall be responsible for designing, fabricating, and installing a secondary payload mass simulator.

In the event of termination for convenience of the NASA secondary payload services, all costs associated with terminating such service shall be determined in accordance with contract Section C, Clause 1.0(1), "Termination for the Government's convenience."

(B) Delays

Any delays shall be subject to contract Section C, Clause 19.0, Adjustments to Launch Schedule.

(C) Mission Success

Mission success determination for the NASA primary and secondary payload shall be made in accordance with contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions with the following exception:

In the event the secondary payload causes primary mission failure, the primary mission shall be determined a full success for purposes of contract Section C, paragraph 24.2.

29.3 Non-NASA Payload with NASA Primary Payload

(A) Procedures

The Contractor may propose to manifest a non-NASA payload on a NASA primary mission. The Contractor shall propose the mission requirements and consideration to be received by NASA. The Contractor shall submit, at no cost to NASA, a detailed payload description, a dual payload compatibility assessment, and any additional documentation or analyses requested by NASA. NASA maintains the right to approve or disapprove the non-NASA payload proposed for manifesting on a NASA primary mission. NASA will respond to the Contractor's request within sixty (60) calendar days after receipt of all NASA requested documentation and analyses.

If the Contractor is unable to provide the non-NASA payload for launch vehicle integration or there is insufficient time to complete a new mission analysis before the launch date, the Contractor shall bear the cost and be responsible for designing, fabricating, and installing a secondary payload mass simulator.

(B) Delays

NASA shall incur no costs or damages associated with delaying the non-NASA mission under any circumstance. The non-NASA payload shall not cause the NASA primary mission launch date to be delayed without NASA approval. If NASA approves a launch delay caused by the non-NASA payload, the NASA Contracting Officer will unilaterally determine whether the Government will be entitled to equitable adjustment under contract Section C, Clause 19.0, Adjustments to Launch Schedule.

(C) Mission Success

Mission success determination for the NASA primary payload shall be made in accordance with contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions with the following exception:

In the event the non-NASA payload causes primary mission failure, the primary mission shall be determined a full success for purposes of contract Section C, paragraph 24.2.

29.4 NASA Secondary Payload on a Non-NASA Primary Payload

(A) Procedures

NASA may request to manifest a NASA secondary payload on a non-NASA primary mission via a task order requesting a proposal to accommodate a secondary payload. The Contracting Officer will provide a description of the proposed secondary payload including, as a minimum: volume, mass, attachment requirements, and data requested by the Contractor. The task order will also identify any additional non-standard services or mission unique requirements. If the Government is unable to provide the secondary payload for launch vehicle integration or there is insufficient time to complete a new mission analysis before the launch date, the Government will bear the cost and the Contractor shall be responsible for designing, fabricating, and installing a secondary payload mass simulator.

Upon receipt of the task order, the Contractor shall develop a proposal to accommodate the secondary payload or provide a response that the Contractor cannot accommodate the secondary payload on any existing planned missions. The Contractor's proposal shall include a firm-fixed price for the secondary payload, any additional non-standard services required, any mission unique services, and a proposed payment schedule with accomplishment criteria. The Contractor shall identify the primary mission and the target launch date for the mission. The Contractor shall make all arrangements for and coordinate the concurrence of manifesting the secondary payload with the primary mission customer.

The terms and conditions of this contract shall apply to NASA secondary payload efforts, however the level of NASA insight and approval will be limited to those activities directly affecting the secondary payload. The Contracting Officer shall identify any additional reductions to the terms and conditions of the contract in each task order for NASA secondary payload services.

In the event of termination for convenience of the NASA secondary payload services, all costs associated with terminating such service shall be determined in accordance with contract Section C, Clause 1.0(l), "Termination for the Government's convenience."

(B) Delays

The Contractor shall incur no costs or damages associated with delaying the secondary mission. The secondary mission shall not cause the primary mission launch date to be delayed without Contractor approval. If the Contractor approves a launch delay caused by the secondary payload, delay damages shall not apply.

(C) Mission Success

Mission success determination for the NASA secondary payload shall be made in accordance with contract Section C, Clause 24.0, Mission Success Determination, Investigation, and Corrective Actions with the following exception:

In the event the primary payload causes secondary mission failure, the secondary mission shall be determined a full success for purposes of contract Section C, paragraph 24.2.

29.5 Contractor Risk Determination for NASA Secondary Missions

The Contractor shall, as part of NASA secondary payload integration activities, perform a payload compatibility assessment for all manifested NASA secondary payloads. The Contractor shall notify the Contracting Officer, in writing if, in the Contractor's opinion, the NASA secondary payload would pose unacceptable risk to the success of the primary mission. The notification shall provide detailed substantiation of said risks.

In the event the Contractor, or the primary mission Customer, determines the risk is unacceptable, the Contractor and the Contracting Officer will mutually agree to terminate the task order/mission unique service or re-manifest the NASA secondary payload on another mission. In the event the task order/mission unique service is terminated, NASA will only be liable for the cost of the compatibility analysis and any other payments made to the Contractor shall be repaid to the Government or reallocated to another mission.

30.0 EXPORT CONTROL AND FOREIGN NATIONALS

- 30.1 The Contractor shall comply with all U.S. export control laws and regulations, including the International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120-130, and the Export Administration Regulations (EAR), 15 CFR Parts 730-744, in the performance of this contract. The Contractor shall be responsible for obtaining export licenses, where required.
- 30.2 The Contractor shall be responsible for obtaining the required export licenses before utilizing foreign nationals in the performance of this contract, including instances where the work is to be performed at launch sites. NASA will be responsible for all ITAR requirements for their foreign national customers, as applicable.
- 30.3 The Contracting Officer, or designated representative, may authorize the Contractor to export ITAR-controlled technical data pursuant to the exemption set forth in 22 CFR 125.4(b)(3) where an international agreement provides for the export of such data and the data does not disclose the details of the design, development, production or manufacture of any defense article.

31.0 DOMESTIC SOURCE CRITERIA

- 31.1 In addition to the certification regarding United States commercial provider of space transportation services, 51 U.S.C. §50131(a), the Contractor shall continue to comply with domestic source criteria. Failure to comply with the criteria may be grounds for “Termination for Cause” in accordance with contract Section C, Clause 1.0(m).
- 31.2 Participation in this procurement is restricted to prime Contractors from the United States launch vehicle/services industry. “United States industry” means any corporation, partnership, joint venture, association, or other entity which is organized or existing under the laws of the United States or any State, and whose controlling interest is held by United States citizens. “Launch services” means all services required in the performance of this contract, excluding those necessary to produce or manufacture launch vehicles, its components and other equipment and facilities required in the performance of the contract. “Controlling interest” means ownership of an amount of equity in such entity sufficient to direct management or to void transactions entered into by management. Ownership of at least fifty-one (51) percent of the equity creates a rebuttable presumption that such interest is controlling.
- 31.3 The Contractor shall provide in the performance of this contract launch vehicles that are domestic end products. The launch vehicle shall be a domestic end product only if the cost of its components, mined, produced or manufactured in the United States exceeds 50 percent of the cost of all its components. The cost of each component includes transportation costs to the place of incorporation into the launch vehicle and any applicable duty (whether or not a duty-free entry certificate is issued). “Components,” as used in this contract clause, means those materials and supplies directly incorporated into the end product.
- 31.4 The Contractor shall provide, in the performance of this contract, domestic launch services. Launch services shall be considered to be domestic if the cost for launch

services performed by United States industry sources exceeds 50 percent of the cost of the total required launch services.

32.0 LIABILITY FOR THIRD PARTY CLAIMS

32.1 This contract clause applies to Third Party claims that arise out of the conduct of hazardous launch activities during the provision of launch services under this contract. More specifically, this contract clause allocates between the Government and the Contractor the risk of Third Party claims for damage to or loss of property or personal injury or death arising from the burning, explosion, detonation, combustion or impact of a launch vehicle, its payload, or a component thereof, whether or not the payload is separated from the launch vehicle, from the time of launch until thirty (30) days after launch.

32.2 Definitions:

Covered Launch Activities: Any and all activities involved in the preparation of a launch vehicle and payload for launch, and conduct of the launch, when those activities take place at a launch site in the United States.

Launch: The intentional ignition of the first-stage motor(s) of the launch vehicle that has been integrated with the payload.

Launch Vehicle: The baseline LVS consisting of a common core booster section and any strap on motors attached, one (1) interstage, an orbital adjust module, the payload fairing and the payload adapter.

Party or Parties: The Contractor or NASA, or both.

Payload: All NASA or NASA-sponsored equipment that has been or will be integrated with the launch vehicle for transportation into earth orbit or escape trajectories.

Related Party:

- (i) Any of the Parties' directors, officers, agents, employees or customers
- (ii) Any of the Parties' contractors, subcontractors, or suppliers at any tier involved directly or indirectly in the performance of this Contract
- (iii) Any entity having any right, title or interest, whether through sale, lease or service arrangement or otherwise, directly or indirectly, in the payload, the launch vehicle, or the launch service.

Third Party: Any person or entity other than NASA, the Contractor and Related Parties.

32.3 Required Insurance for Liability to Third Parties

- (A) The Contractor shall continue in effect or acquire insurance to protect the Parties and the Related Parties from liability for claims from Third Parties for damage to or loss of property or personal injury or death arising in connection with the covered launch activities under this contract. The amount of the required insurance shall be the maximum amount available in the commercial marketplace at reasonable cost, but shall not exceed \$500 million for each launch. The policy or policies shall name NASA and the related parties as additional insured parties. Required insurance coverage shall attach no later than the arrival of the launch vehicle at the launch site and shall remain in force for at least thirty (30) days following launch.
- (B) The Contractor shall provide acceptable evidence to the Contracting Officer of required insurance no later than thirty (30) days prior to the beginning of the covered launch activities. The amount of required insurance and the terms and conditions for the policy or policies shall be subject to review by the Contracting Officer. Once reviewed, the policy or policies may not be modified or canceled without the prior, written approval of the Contracting Officer.
- (C) The foregoing insurance requirement does not preclude the Contractor from acquiring or continuing in effect any additional insurance to protect the interests of the Contractor or its Related Parties.

32.4 Third Party Claims in Excess of Required Insurance

- (A) NASA has determined that launches, under this contract, are conducted by NASA in performance of its functions, as specified in 42 U.S.C. § 2473(a). As a result, once the Contractor or its insurers have paid out for Third Party claims the amount of required insurance under paragraph 32.3(B), NASA will consider any additional Third Party claims for damage to or loss of property or personal injury or death arising from the launches as claims against the United States under the authority of 42 U.S.C. § 2473(c)(13).
- (B) The Contractor (once it or its insurers have paid to Third Party claimants, from their own funds, an amount equal to the amount of required insurance for a Launch) shall adjust, settle and pay meritorious and reasonable additional Third Party claims in excess of the amount of required insurance. To the extent NASA determines that such costs exceed \$25,000, it will forward such claim to the Secretary of Treasury for certification and payment pursuant to 31 U.S.C. § 1304(a). Such costs are subject to the availability of funds and the usual tests for allowability and the total of such costs shall be paid up to a limit of \$1.5 billion above the insurance obtained by the Contractor for each launch.
- (C) In evaluating Third Party claims against the United States paid by the Contractor, NASA will consider such a claim to be meritorious unless the claim represents:

- (i) Liabilities for which the Contractor is otherwise responsible under the express terms or conditions of the contract or a task order issued under this contract
- (ii) Liabilities for which the Contractor has failed to insure or to maintain insurance as required by the Contracting Officer
- (iii) Liabilities for which the Contractor has not reasonably adjusted, settled, or paid on a meritorious and reasonable basis.
- (iv) Liabilities that result from willful misconduct or lack of good faith on the part of any of the Contractor's directors, officers, managers, superintendents, or other representatives who have supervision or direction of:
 - (a) All or substantially all of the Contractor's business
 - (b) All or substantially all of the Contractor's operations at any one plant or separate location in which this contract is being performed
 - (c) A separate and complete major industrial operation in connection with the performance of this contract
- (v) Liabilities that arise from the willful misconduct or gross negligence of the Claimant or, in the case of a claim based on death, the claimant's descendant.

32.5 Third Party Liability for NASA Secondary Payloads on Non-NASA Primaries

The requirements of this clause 32.0 shall apply to all launch services provided under this contract except for those services involving NASA secondary payloads which are manifested on a launch service for non-NASA (commercial) primary payloads. In the event that a NASA secondary payload is manifested on a launch service for a non-NASA (commercial) primary payload, the contractor shall obtain third party liability insurance and indemnification for third party claims in excess of insurance pursuant to the Commercial Space Launch Act, Title 51.

33.0 FAR 52.204-19 INCORPORATION BY REFERENCE OF REPRESENTATIONS AND CERTIFICATIONS (DEC 2014)

The Contractor's representations and certifications, including those completed electronically via the System for Award Management (SAM), are incorporated by reference into the contract

34.0 RESERVED

35.0 RESERVED

36.0 LISTING OF CLAUSES INCORPORATED BY REFERENCE

NOTICE: The following contract clauses pertinent to this section are hereby incorporated by reference:

36.1 FEDERAL ACQUISITION REGULATION (48 CFR CHAPTER 1) CLAUSES

FAR 52.202-1	Definitions (JUN 2020)
FAR 52.211-15	Defense Priority and Allocation Requirements (Apr 2008)
FAR 52.222-1	Notice to the Government of Labor Disputes (DEVIATION Nov 2025)
FAR 52.227-14	Rights in Data—General (May 2014)
FAR 52.227-14	Rights in Data—General (May 2014) Alternate II (Dec 2007)

(g)(3) Notwithstanding paragraph (g)(1) of this clause, the contract may identify and specify the delivery of limited rights data, or the Contracting Officer may require by written request the delivery of limited rights data that has been withheld or would otherwise be entitled to be withheld. If delivery of that data is required, the Contractor shall affix the following “Limited Rights Notice” to the data and the Government will treat the data, subject to the provisions of paragraphs (e) and (f) of this clause, in accordance with the notice:

Limited Rights Notice (Dec 2007)

(a) These data are submitted with limited rights under Government Contract No. TBD (and subcontract TBD, if appropriate). These data may be reproduced and used by the Government with the express limitation that they will not, without written permission of the Contractor, be used for purposes of manufacture nor disclosed outside the Government; except that the Government may disclose these data outside the Government for the following purposes, if any; provided that the Government makes such disclosure subject to prohibition against further use and disclosure:

- (i) Use (except for manufacture) by support service contractors.
- (ii) Evaluation by nongovernment evaluators.

(b) This Notice shall be marked on any reproduction of these data, in whole or in part.

(End of notice)

FAR 52.227-14	Rights in Data—General (May 2014) Alternate III (Dec 2007)
FAR 52.232-18	Availability of Funds (Apr 1984)
FAR 52.233-1	Disputes (DEVIATION Nov 2025)

FAR 52.242-15	Stop-Work Order (Aug 1989)
FAR 52.243-1	Changes—Fixed-Price (DEVIATION Jun 2025) Alternate I (DEVIATION Jun 2025)
FAR 52.246-4	Inspection of Services—Fixed-Price (Aug 1996)
FAR 52.246-25	Limitation of Liability--Services (Feb 1997)

36.2 NASA FAR SUPPLEMENT (48 CFR CHAPTER 18) CLAUSES:

NFS 1852.215-84	Ombudsman (Nov 2023) Edwin Martinez at (321-867-1699)
NFS 1852.219-75	Individual Subcontract Reports (Apr 2015)
NFS 1852.226-71	Safety and Health Measures and Mishap Reporting (Feb 2026)
NFS 1852.226-73	Major Breach of Safety or Security (Feb 2026)
NFS 1852.225-70	Export Licenses (Feb 2000) Alternate I (Feb 2000)
NFS 1852.228-75	Minimum Insurance Coverage (DEVIATION Sept 2025)

36.3 Reserved

37.0 FAR 52.252-2 CLAUSES INCORPORATED BY REFERENCE (FEB 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address(es):

<http://www.acquisition.gov/far/>

GSA FAR Homepage

<https://www.hq.nasa.gov/office/procurement/regs/NFS.pdf>

NASA FAR Supplement
Homepage

38.0 RESERVED

39.0 SPECIAL UNDERSTANDING REGARDING DAMAGE TO GOVERNMENT PAYLOADS

Prior to intentional ignition of the launch vehicle on the launch pad, the Contractor shall not be responsible for damage to the spacecraft while the spacecraft is under the control of the Contractor, except when such damage is caused by the gross negligence, willful misconduct, or lack of good faith by the Contractor. In the event the Contractor is determined to be responsible for such damage, the Contractor shall reimburse the Government for the cost of spacecraft repairs as well as any costs associated with launch delays as set forth in Section C, Clause 19.0, entitled "Adjustments to Launch Schedule." After intentional ignition, the provisions of Contract Section C, clause 24.0 shall apply.

40.0 SPECIAL UNDERSTANDING REGARDING LIABILITY FOR THIRD PARTY CLAIMS FOR NUCLEAR INCIDENTS

In the event that a launch service acquired under this contract carries a payload containing a Radioisotope Thermoelectric Generator (RTG) or other nuclear materials for which third party liability insurance is not commercially available, NASA will obtain from the Department of Energy (DOE) the necessary agreement to provide to the Contractor indemnification against third party claims pursuant to the Price Anderson Act, 42 U.S.C. §2210. Such agreement shall be obtained, and the Launch Service Task Order (LSTO) for the subject launch service shall be modified to reflect the Price Anderson Act indemnification prior to integration of the RTG or other nuclear materials into the launch vehicle. The Contractor shall not be required to integrate the RTG or other nuclear materials into the subject mission's launch vehicle prior to the time at which Price Anderson Act or other indemnification for the nuclear risk has been obtained. Any delay by NASA in timely obtaining Price Anderson Act indemnification prior to scheduled integration which results in a launch delay, shall constitute a Government-caused delay under clause 19.0 of the Contract. Such a Government-caused delay to the subject mission shall be subject to the contract's changes clause.

41.0 DEPARTMENT OF ENERGY (DOE) 952.250-70 NUCLEAR HAZARDS INDEMNITY AGREEMENT (Dec 2024)

(a) *Definitions.* Except as otherwise specified within this clause, all definitions set forth in the Atomic Energy Act of 1954, as amended (hereinafter called the Act), shall apply to this clause.

“Extraordinary nuclear occurrence” means an event that DOE has determined to be such an occurrence, as defined in the Act. A determination of whether or not there has been an extraordinary nuclear occurrence will be made in accordance with the procedures in 10 CFR part 840.

“Public liability,” referred to below, is public liability as defined in the Act, which (1) arises out of or in connection with the activities under this contract, including transportation; and (2) arises out of or results from a nuclear incident or precautionary evacuation.

(b) *Authority.* This clause is incorporated into this contract pursuant to the authority contained in subsection 170d. of the Act.

(c) *Financial protection.* Except as hereafter permitted or required in writing by DOE, the Contractor will not be required to provide or maintain, and will not provide or maintain at Government expense, any form of financial protection to cover public liability, as described in paragraph (a) of this clause. DOE may, however, at any time require in writing that the Contractor provide and maintain financial protection of such a type and in such amount as DOE shall determine to be appropriate to

cover such public liability, provided that the costs of such financial protection are reimbursed to the Contractor by DOE.

(d) *Indemnification.* To the extent that the Contractor and other persons indemnified are not compensated by any financial protection permitted or required by DOE, DOE will indemnify the Contractor and other persons indemnified against (i) claims for public liability as described in paragraph (a) of this clause; and (ii) such legal costs of the Contractor and other persons indemnified as are approved by DOE, provided that DOE's liability, including such legal costs, shall not exceed the amount set forth in section 170e(1)(B) of the Act in the aggregate for each nuclear incident or precautionary evacuation occurring within the United States or \$2,000,000,000 in the aggregate for each nuclear incident occurring outside the United States, irrespective of the number of persons indemnified in connection with this contract.

(e)

(1) *Waiver of defenses.* In the event of a nuclear incident (as defined in the Act) arising out of nuclear waste activities (as defined in the Act), the Contractor, on behalf of itself and other persons indemnified, agrees to waive any issue or defense as to charitable or governmental immunity.

(2) In the event of an extraordinary nuclear occurrence that—

- (i) Arises out of, results from, or occurs in the course of the construction, possession, or operation of a production or utilization facility; or
- (ii) Arises out of, results from, or occurs in the course of transportation of source material, by-product material, or special nuclear material to or from a production or utilization facility; or
- (iii) Arises out of or results from the possession, operation, or use by the Contractor or a subcontractor of a device utilizing special nuclear material or by-product material, during the course of the contract activity; or
- (iv) Arises out of, results from, or occurs in the course of nuclear waste activities, the Contractor, on behalf of itself and other persons indemnified, agrees to waive—
 - (A) Any issue or defense as to the conduct of the claimant (including the conduct of persons through whom the claimant derives its cause of action) or fault of persons indemnified, including, but not limited to negligence, contributory negligence, assumption of risk, or unforeseeable intervening causes, whether involving the conduct of a third person or an act of God;
 - (B) Any issue or defense as to charitable or governmental immunity; and any issue or defense based on any statute of limitations, if suit is instituted within three years of the date on which the claimant first knew, or reasonably could have known, of his injury or change and the cause thereof. The waiver of any such issue or defense shall be effective regardless of whether such issue or defense may otherwise

be deemed jurisdictional or relating to an element in the cause of action. The waiver shall be judicially enforceable in accordance with its terms by the claimant against the person indemnified.

- (v) For the purposes of making a determination of whether or not there has been an extraordinary nuclear occurrence, “offsite,” as used in 10 CFR part 840, means “away from the contract location,” a phrase that means any DOE facility, installation, or site at which contractual activity under this contract is being carried out, and any contractor-owned or controlled facility, installation, or site at which the Contractor is engaged in the performance of contractual activity under this contract.

(3) The waivers set forth in paragraph (e) of this clause—

- (i) Shall be effective regardless of whether such issue or defense may otherwise be deemed jurisdictional or relating to an element in the cause of action;
- (ii) Shall be judicially enforceable in accordance with its terms by the claimant against the person indemnified;
- (iii) Shall not preclude a defense based upon a failure to take reasonable steps to mitigate damages;
- (iv) Shall not apply to any injury or damage to a claimant (or claimant's property) that is intentionally sustained by the claimant, or that results from a nuclear incident intentionally and wrongfully caused by the claimant;
- (v) Shall not apply to injury to a claimant who is employed at the site of and in connection with the activity where the extraordinary nuclear occurrence takes place, if benefits therefor are either payable or required to be provided under any workmen's compensation or occupational disease law;
- (vi) Shall not apply to any claim resulting from a nuclear incident occurring outside the United States;
- (vii) Shall be effective only with respect to those obligations set forth in this clause and in insurance policies, contracts or other proof of financial protection; and
- (viii) Shall not apply to, or prejudice the prosecution or defense of, any claim or portion of claim not within the protection afforded under
 - (A) the limit of liability provisions under subsection 170e. of the Act,
 - and (B) the terms of this agreement and the terms of insurance policies, contracts, or other proof of financial protection.

- (f) *Notification and litigation of claims.* The Contractor shall give immediate written notice to DOE of any known action or claim filed or made against the Contractor or other person indemnified for public liability as defined in paragraph (a) of this clause. Except as otherwise directed by DOE, the Contractor shall furnish promptly to DOE copies of all pertinent papers received by the Contractor or filed with respect to such actions or claims. DOE shall have the right to, and may collaborate with, the Contractor and any other person indemnified in the settlement or defense of any action or claim and shall have the right to:

- (1) Require the prior approval of DOE for the payment of any claim that DOE may be required to indemnify hereunder; and
 - (2) Appear through the Attorney General on behalf of the Contractor or other person indemnified in any action brought upon any claim that DOE may be required to indemnify hereunder, take charge of such action, and settle or defend any such action. If the settlement or defense of any such action or claim is undertaken by DOE, the Contractor or other person indemnified shall furnish all reasonable assistance in effecting a settlement or asserting a defense.
- (g) *Continuity of DOE obligations.* The obligations of DOE under this clause shall not be affected by any failure on the part of the Contractor to fulfill its obligation under this contract and shall be unaffected by the death, disability, or termination of existence of the Contractor, or by the completion, termination or expiration of this contract.
- (h) *Effect of other clauses.* The provisions of this clause shall not be limited in any way by, and shall be interpreted without reference to, any other clause of this contract, including the Disputes clause, provided, however, that this clause is subject to the clauses at 48 CFR 52.203-5, Covenant Against Contingent Fees, and 970.5232-3, Accounts, Records, and Inspection, and any provisions later added to this contract, as required by applicable Federal law, including statutes, Executive orders and regulations, to be included in Nuclear Hazards Indemnity Agreements.
- (i) *Civil penalties.* The Contractor and its subcontractors and suppliers who are indemnified under the provisions of this clause are subject to civil penalties, pursuant to section 234A of the Act, for violations of applicable DOE nuclear-safety related rules, regulations, or orders, and pursuant to section 234C of the Act, for violations of applicable DOE worker safety and health related rules, regulations, and orders. If the Contractor is a not-for-profit contractor, as defined by section 234Ad.(2), the total amount of civil penalties paid shall not exceed the total amount of fees paid within any one-year period (as determined by the Secretary) under this contract.
- (j) *Criminal penalties.* Any individual director, officer, or employee of the Contractor or of its subcontractors and suppliers indemnified under the provisions of this clause are subject to criminal penalties, pursuant to section 223(c) of the Act, for knowingly and willfully violating the Act, and applicable DOE nuclear safety-related rules, regulations or orders for which violation results in, or if undetected, would have resulted in a nuclear incident.
- (k) *Inclusion in Subcontracts.* The Contractor shall insert this clause in any subcontract that may involve the risk of public liability, as that term is defined in the Act and further described in paragraph (a) of this clause. However, this clause shall not be included in subcontracts in which the subcontractor is subject to Nuclear

Regulatory Commission (NRC) financial protection requirements under section 170b. of the Act or NRC agreements of indemnification under section 170c. or k. of the Act for the activities under the subcontract.

42.0 NFS 1852.228-78 CROSS-WAIVER OF LIABILITY FOR SCIENCE OR SPACE EXPLORATION ACTIVITIES UNRELATED TO THE INTERNATIONAL SPACE STATION (DEVIATION) (Sept 2025)

(a) The purpose of this clause is to extend a cross-waiver of liability to NASA contracts for work done in support of Agreements between Parties involving Science or Space Exploration activities that are not related to the International Space Station (ISS) but involve a launch. This cross-waiver of liability must be broadly construed to achieve the objective of furthering participation in space exploration, use, and investment.

(b) As used in this clause, the term:

(1) "Agreement" refers to any NASA Space Act agreement that contains the cross-waiver of liability provision authorized in 14 CFR Part 1266.104.

(2) "Damage" means:

- (i) Bodily injury to, or other impairment of health of, or death of, any person;
- (ii) Damage to, loss of, or loss of use of any property;
- (iii) Loss of revenue or profits; or
- (iv) Other direct, indirect, or consequential Damage;

(3) "Launch Vehicle" means an object, or any part thereof, intended for launch, launched from Earth, or returning to Earth which carries Payloads or persons, or both.

(4) "Party" means a party to a NASA Space Act agreement for Science or Space Exploration activities unrelated to the ISS that involve a launch and a party that is neither the prime contractor under this contract nor a subcontractor at any tier hereof.

(5) "Payload" means all property to be flown or used on or in a Launch Vehicle.

(6) "Protected Space Operations" means all Launch or Transfer Vehicle activities and Payload activities on Earth, in outer space, or in transit between Earth and outer space in implementation of an Agreement for Science or Space Exploration activities unrelated to the ISS that involve a launch. Protected Space Operations begins at the signature of the Agreement and ends when all activities done in implementation of the Agreement are completed. It includes, but is not limited to:

(i) Research, design, development, test, manufacture, assembly, integration, operation, or use of Launch or Transfer Vehicles, Payloads, or instruments, as well as related support equipment and facilities and services; and

(ii) All activities related to ground support, test, training, simulation, or guidance and control equipment, and related facilities or services. Protected Space Operations excludes activities on Earth which are conducted on return from space to develop further a payload's product or process other than for the activities within the scope of an Agreement.

(7) "Related entity" means:

- (i) A contractor or subcontractor of a Party at any tier;
- (ii) A user or customer of a Party at any tier; or
- (iii) A contractor or subcontractor of a user or customer of a Party at any tier.

The terms "contractors" and "subcontractors" include suppliers of any kind.

(8) "Transfer Vehicle" means any vehicle that operates in space and transfers Payloads or persons or both between two different space objects, between two different locations on the same space object, or between a space object and the surface of a celestial body. A Transfer Vehicle also includes a vehicle that departs from and returns to the same location on a space object.

(c) Cross-waiver of liability:

(1) The Contractor agrees to a waiver of liability pursuant to which it waives all claims against any of the entities or persons listed in paragraphs (c)(1)(i) through (c)(1)(iv) of this clause based on Damage arising out of Protected Space Operations. This cross-waiver must apply only if the person, entity, or property causing the Damage is involved in Protected Space Operations and the person, entity, or property damaged is damaged by virtue of its involvement in Protected Space Operations. The waiver must apply to any claims for Damage, whatever the legal basis for such claims, against—

- (i) A Party;
- (ii) A Party to another NASA Agreement or contract that includes flight on the same Launch Vehicle;
- (iii) A Related Entity of any entity identified in paragraphs (c)(1)(i) or (c)(1)(ii) of this clause; or
- (iv) The employees of any of the entities identified in (c)(1)(i) through (iii) of this clause.

(2) The Contractor agrees to extend the cross-waiver of liability as set forth in paragraph (c)(1) of this clause to its own subcontractors at all tiers by requiring them, by contract or otherwise, to:

- (i) Waive all claims against the entities or persons identified in paragraphs (c)(1)(i) through (c)(1)(iv) of this clause; and
- (ii) Require that their Related Entities waive all claims against the entities or persons identified in paragraphs (c)(1)(i) through (c)(1)(iv) of this clause.

(3) For avoidance of doubt, this cross-waiver of liability includes a cross-waiver of claims arising from the Convention on International Liability for Damage Caused by Space Objects, entered into force on 1 September 1972, in which the person, entity, or property causing the Damage is involved in Protected Space Operations and the person, entity, or property damaged is damaged by virtue of its involvement in Protected Space Operations.

(4) Notwithstanding the other provisions of this clause, this cross-waiver of liability must not be applicable to:

- (i) Claims between the Government and its own contractors or between its own contractors and subcontractors;
- (ii) Claims made by a natural person, his/her estate, survivors, or subrogees (except when a subrogee is a Party to an Agreement or is otherwise bound by the terms of this cross-waiver) for bodily injury to, or other impairment of health, or death of such person;
- (iii) Claims for Damage caused by willful misconduct;
- (iv) Intellectual property claims;
- (v) Claims for damages resulting from a failure of the contractor to extend the cross-waiver of liability to its subcontractors and related entities, pursuant to paragraph (c)(2) of this clause; or
- (vi) Claims by the Government arising out of or relating to a contractor's failure to perform its obligations under this contract.

(5) Nothing in this clause must be construed to create the basis for a claim or suit where none would otherwise exist.

(6) This cross-waiver must not be applicable when 49 U.S.C. Subtitle IX, Chapter 701 is applicable.

(End of clause)

43.0 RESERVED

44.0 RESERVED

**45.0 FAR 52.219-1 SMALL BUSINESS PROGRAM REPRESENTATIONS
(DEVIATION NOV 2025) ALTERNATE I (DEVIATION NOV 2025)**

(a) Definitions. As used in this provision-

Economically disadvantaged women-owned small business (EDWOSB) concern means a small business concern that is at least 51 percent directly and unconditionally owned by, and the management and daily business operations of which are controlled by, one or more women who are citizens of the United States and who are economically disadvantaged in accordance with 13 CFR part 127, and the concern is certified by SBA or an approved third-party certifier in accordance with 13 CFR 127.300. It automatically qualifies as a women-owned small business concern eligible under the WOSB Program.

HUBZone small business concern means a small business concern that meets the requirements described in 13 CFR 126.200, is certified by the Small Business Administration (SBA) and designated by SBA as a HUBZone small business concern in the Small Business Search (SBS) (13 CFR 126.103).

Service-disabled veteran-owned small business (SDVOSB) concern eligible under the SDVOSB Program means an SDVOSB concern that is designated in the System for Award Management (SAM) as certified by the Small Business Administration (SBA) in accordance with 13 CFR 128.300.

Small business concern—

- (1) Means a concern, including its affiliates, that is independently owned and operated, not dominant in its field of operation, and qualified as a small business under the criteria in 13 CFR part 121 and the size standard in paragraph (b) of this provision.
- (2) Affiliates, as used in this definition, means business concerns, one of whom directly or indirectly controls or has the power to control the others, or a third party or parties control or have the power to control the others. In determining whether affiliation exists, consideration is given to all appropriate factors including common ownership, common management, and contractual relationships. SBA determines affiliation based on the factors set forth at 13 CFR 121.103.

Small disadvantaged business concern means a small business concern that-

- (1) Is at least 51 percent unconditionally and directly owned (as defined at 13 CFR 124.105) by one or more socially disadvantaged (as defined at 13 CFR 124.103) and economically disadvantaged (as defined at 13 CFR 124.104) individuals who are citizens of the United States, and
- (2) The management and daily business operations of which are controlled (as defined at 13 CFR 124.106) by individuals who meet the criteria in paragraph (1) of this definition.

Women-owned small business (WOSB) concern eligible under the WOSB Program (in accordance with 13 CFR part 127) means a small business concern that is at least 51 percent directly and unconditionally owned by, and the management and daily business operations of which are controlled by, one or more women who are citizens of the United States, and the concern is certified by SBA or an approved third-party certifier in accordance with 13 CFR 127.300.

(b)(1) The North American Industry Classification System (NAICS) codes and corresponding size standards for this acquisition are as follows; the categories or portions these NAICS codes are assigned to are specified elsewhere in the solicitation:

NAICS Code	Size standard
_____	_____
_____	_____
_____	_____

[Contracting Officer to insert NAICS codes and size standards].

(2) The small business size standard for a concern that submits an offer, other than on a construction or service acquisition, but proposes to furnish an end item that it did not itself manufacture, process, or produce (i.e., nonmanufacturer), is 500 employees, or 150 employees for information technology value-added resellers under NAICS code 541519, if the acquisition—

- (i) Is set aside for small business and has a value above the simplified acquisition threshold;
- (ii) Uses the HUBZone price evaluation preference regardless of dollar value, unless the offeror waives the price evaluation preference; or
- (iii) Is an 8(a), HUBZone, service-disabled veteran-owned, economically disadvantaged women-owned, or women-owned small business set-aside or sole-source award regardless of dollar value.

(c) Representations. (1) The Offeror shall represent its small business size status for each one of the NAICS codes assigned to this acquisition under which it is submitting an offer.

NAICS Code	Small business concern (yes/no)

[Contracting Officer to insert NAICS codes.]

(d) Notice. Under 15 U.S.C. 645(d), any person who misrepresents a firm's status as a business concern that is small, HUBZone small, small disadvantaged, service-disabled veteran-owned small, economically disadvantaged women-owned small, or women-owned small eligible under the WOSB Program in order to obtain a contract to be awarded under the preference programs established pursuant to section 8, 9, 15, 31, and 36 of the Small Business Act or any other provision of Federal law that specifically references section 8(d) for a definition of program eligibility, will be—

- (1) Punished by imposition of fine, imprisonment, or both;
- (2) Subject to administrative remedies, including suspension and debarment; and
- (3) Ineligible for participation in programs conducted under the authority of the Act.

(End of provision)

46.0 RESERVED

47.0 FAR 52.240-90 Security Prohibitions and Exclusions Representations and Certifications (DEVIATION) NOV 2025

(a) Definitions. As used in this provision—

Backhaul, covered article, covered telecommunications equipment or services, critical technology, FASCSA order, Intelligence community, interconnection arrangements, national security system, roaming, sensitive compartmented information, sensitive compartmented information system, source, and substantial or essential component have the meanings provided in the clause 52.240-91, Security Prohibitions and Exclusions.

Business operations means engaging in commerce in any form, including by acquiring, developing, maintaining, owning, selling, possessing, leasing, or operating equipment, facilities, personnel, products, services, personal property, real property, or any other apparatus of business or commerce.

Marginalized populations of Sudan means—

(1) Adversely affected groups in regions authorized to receive assistance under section 8(c) of the Darfur Peace and Accountability Act (Pub. L. 109-344) (50 U.S.C. 1701 note); and

(2) Marginalized areas in Northern Sudan described in section 4(9) of such Act.

Restricted business operations means business operations in Sudan that include power production activities, mineral extraction activities, oil-related activities, or the production of military equipment, as those terms are defined in the Sudan Accountability and Divestment Act of 2007 (Pub. L. 110-174). Restricted business operations do not include business operations that the person (as that term is defined in Section 2 of the Sudan Accountability and Divestment Act of 2007) conducting the business can demonstrate—

(1) Are conducted under contract directly and exclusively with the regional government of southern Sudan;

(2) Are conducted under specific authorization from the Office of Foreign Assets Control in the Department of the Treasury, or are expressly exempted under Federal law from the requirement to be conducted under such authorization;

(3) Consist of providing goods or services to marginalized populations of Sudan;

(4) Consist of providing goods or services to an internationally recognized peacekeeping force or humanitarian organization;

(5) Consist of providing goods or services that are used only to promote health or education; or

(6) Have been voluntarily suspended.

Sensitive technology—

(1) Means hardware, software, telecommunications equipment, or any other technology that is to be used specifically—

(i) To restrict the free flow of unbiased information in Iran; or

(ii) To disrupt, monitor, or otherwise restrict speech of the people of Iran; and

(2) Does not include information or informational materials the export of which the President does not have the authority to regulate or prohibit pursuant to section 203(b)(3) of the International Emergency Economic Powers Act (50 U.S.C. 1702(b)(3)).

(b) Procedures.

(1) Covered telecommunications and video surveillance. The Offeror shall review the list of excluded parties in the System for Award Management (SAM) at <https://www.sam.gov> for entities excluded from receiving federal awards for “covered telecommunications equipment or services.”

(2) FASCSA Orders.

(i) The Offeror shall search in SAM for the phrase “FASCSA order” for any covered article, or any products or services produced or provided by a source, if there is an applicable FASCSA order described in paragraph (e) of FAR 52.240-91, Security Prohibitions and Exclusions.

(ii) The Offeror shall review the solicitation for any FASCSA orders that are not in SAM but are effective and apply to the solicitation and resultant contract (see FAR 40.204-1(c)(2)).

(iii) FASCSA orders issued after the date of solicitation do not apply unless added by an amendment to the solicitation.

(c) Covered telecommunications equipment or services representations. By submission of its offer, the Offeror represents that, after conducting a reasonable inquiry (that looks at any information in the Offeror’s possession but does not need to include an internal or third-party audit)—

(1) It will not provide covered telecommunications equipment or services to the Government in the performance of any contract, subcontract or other contractual instrument resulting from this solicitation, except as waived by the solicitation, or as disclosed in paragraph (g); and

(2) It does not use covered telecommunications equipment or services, or use any equipment, system, or service that uses covered telecommunications equipment or services, except as waived by the solicitation, or as disclosed in paragraph (g).

(d) FASCSA Representation. By submission of this offer, the offeror represents that it has conducted a reasonable inquiry, and that the offeror does not propose to provide or use in response to this solicitation any covered article, or any products or services produced or provided by a source, if the covered article or the source is prohibited by an applicable FASCSA order in effect on the date the solicitation was issued, except as waived by the solicitation, or as disclosed in paragraph (g). A reasonable inquiry will look at any information in the offeror’s possession but does not need to include an internal or third-party audit.

(e) Sudan certification. By submission of its offer, the offeror certifies, after conducting a reasonable inquiry (that looks at any information in the offeror’s possession but does not need to include an internal or third-party audit), that the offeror does not conduct any restricted business operations in Sudan.

(f) Iran Representation and Certifications.

(1) Except as provided in paragraph (f)(2) of this provision or if a waiver has been granted in accordance with FAR 40.203-3, the offeror, after conducting a reasonable inquiry (that looks at any information in the offeror's possession but does not need to include an internal or third-party audit), by submission of its offer—

(i) Represents, to the best of its knowledge and belief, that the offeror does not export any sensitive technology to the government of Iran or any entities or individuals owned or controlled by, or acting on behalf or at the direction of, the government of Iran;

(ii) Certifies that the offeror, or any person (as defined at section 15 of the Iran Sanctions Act of 1996, Pub. L. 104-172, 50 U.S.C. 1701 note) owned or controlled by the offeror, does not engage in any activities for which sanctions may be imposed under section 5 of the Act. These sanctioned activities are in the areas of development of the petroleum resources of Iran, production of refined petroleum products in Iran, sale and provision of refined petroleum products to Iran, and contributing to Iran's ability to acquire or develop certain weapons or technologies; and

(iii) Certifies that the offeror, and any person owned or controlled by the offeror, does not knowingly engage in any transaction that exceeds \$15,000 with Iran's Revolutionary Guard Corps or any of its officials, agents, or affiliates, the property and interests in property of which are blocked pursuant to the International Emergency Economic Powers Act (50 U.S.C. 1701 et seq.) (see OFAC's Specially Designated Nationals and Blocked Persons List at <https://www.treasury.gov/resource-center/sanctions/SDN-List/Pages/default.aspx>)

(2) Exception for trade agreements. The representation and certification requirements of paragraph (f)(1) of this provision do not apply if—

(i) This solicitation includes a trade agreements notice or certification (e.g., 52.225-6, Trade Agreements Certificate); and

(ii) The offeror has certified that all the offered products to be supplied are designated country end products or designated country construction material.

(iii) The offeror shall email questions concerning sensitive technology to the Department of State at CISADA106@state.gov.

(g) Disclosure.

(1) If the Offeror is not able to represent compliance with the prohibitions in paragraphs (c) or (d), then the Offeror shall disclose within 72 hours to the contracting office identified in paragraph (g)(2) the following information for each product or service not compliant:

(i) Contract number and order number, if applicable;

(ii) Identification of whether this disclosure relates to paragraph (c) on covered telecommunication equipment or services, or to paragraph (d) on FASCSA orders;

(iii) A description of the products or services that the Contractor identifies or has reason to suspect is prohibited (include brand; model number, such as the original equipment manufacturer (OEM) number, manufacturer part number, or wholesaler number; and item description, as applicable);

(iv) The entity that produced the product or service (include entity name, unique entity identifier, Contractor and Government Entity (CAGE) code, facilities responsible for design, fabrication, assembly, packaging, and test of the product, and whether the entity was the OEM or a distributor (provide manufacturer codes and distributor codes used for the product));

(v) Description of the functionality of the product or service and how that functionality impacts the risk to the product or service;

(vi) An explanation of any factors relevant to determining if the product or service should be permitted by an applicable exception, exemption, or waiver (if the offeror would like the Government to consider a waiver);

(vii) Whether alternative products or services are available that would be compliant with the prohibition;

(viii) If the product or service is related to item maintenance, include the following information on the item being maintained:

(A) Brand;

(B) Model number, OEM number, manufacturer part number, or wholesaler number; and

(C) Item description, as applicable.

(ix) Any readily available information about mitigation actions undertaken or recommended.

(2) If a disclosure is required to be submitted to a contracting office, the offeror shall submit the disclosure as follows:

(i) If a Department of Defense contracting office, the offeror shall submit the disclosure to the website at <https://dibnet.dod.mil>.

(ii) For all other contracting offices, the Offeror shall submit the disclosure to the Contracting Officer.

(3) If the disclosure provided does not contain any of the information required by paragraph (1), and the Offeror later discovers new information that is required by paragraph (1), then the Offeror shall submit a subsequent disclosure within 72 hours of discovering the new information.

(h) Executive agency review of disclosures. The Contracting Officer will review disclosures provided in paragraph (g) to determine if any applicable waiver may be sought. The Contracting Officer may choose not to pursue a waiver and may instead make an award to an Offeror that does not require a waiver.

(End of provision)

SECTION D

DOCUMENTS, EXHIBITS, AND ATTACHMENTS

The following eleven attachments are hereby incorporated into this contract in their entirety, made a part hereof, and shall apply in the performance of this contract.

Attachment D1	Statement of Work
Attachment D2	Contract Data Requirements List (CDRL)
Attachment D3	Supplemental Data Requirements List (SDRL)
Attachment D4	Small Business and Small Disadvantaged Business Subcontracting Plan
Attachment D5	Safety and Health Plan
Attachment D6	Reliability Plan
Attachment D7	Quality Management Plan
Attachment D8	Work Plan
Attachment D9	Most Favored Customer Certification Plan
Attachment D10	Acronyms and Abbreviations
Attachment D11	Definitions

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STATEMENT OF WORK

1.0 INTRODUCTION

1.1 Scope

This Statement of Work (SOW) and all Exhibits and documents attached or referenced herein define the Government's requirements for the Contractor to provide launch services in support of NASA's Launch Services Program (LSP). Unless expressly stated otherwise, Exhibits are supplemental to the SOW, and do not replace requirements in the SOW. The scope of this contract effort includes risk category 2 and 3 launch services capable of delivering, at a minimum, a 250kg payload to orbit at an altitude of 200 km and a launch inclination of 28.5°.

This SOW defines the overall launch service requirements for 'NASA or NASA-sponsored payloads' (hereinafter referred to as 'payloads'). The Contractor shall perform all tasks necessary to safely and reliably launch payloads in accordance with NASA-defined mission objectives. The Contractor shall support advance planning and perform analysis tasks as directed by the Contracting Officer.

It is the general contemplation of the parties to this contract that the Contractor shall have a broad mission in performing launch service related functions for the Government and designees. Therefore, the general scope of the contract covers any launch service and launch service related activities arising from the SOW in support of earth and space science exploration, and space station re-supply.

1.2 Objectives

The goal of the NASA Launch Services II (NLS II) contract is to provide the Agency with domestic launch services that are safe, successful, reliable, and affordable. The launch services will be provided at a fixed price. The contract will, to the maximum extent practical, incorporate best commercial practices.

The objectives of this contract are to:

- Ensure the safety of the public, as well as all personnel, hardware, and property associated with the launch services.
- Provide affordable, accurate, and on-time delivery of NASA and/or NASA-sponsored payloads to space on expendable launch vehicles.
- Provide a mechanism to incorporate new launch services, technology upgrades, improved systems engineering processes, and advances in manufacturing techniques.
- Provide risk mitigation while utilizing commercial practices.

- Provide flexible manifesting policy that recognizes the national priorities of NASA missions.
- Provide a capability to optimize cost, schedule, and performance to satisfy mission objectives.
- Provide for clear Government visibility into program schedule, technical performance, and risk.
- Foster competition and create opportunities for new, emerging launch service providers.
- Promote partnering among customers, launch service providers, and the LSP to maximize flexibility and responsiveness to customers' needs.

1.3 Compliance Documents

The Contractor shall comply with the requirements contained in the following documents.

Document No.	Revision	Document Title
Applicable Range Safety Requirements		(e.g., EWR-127-1, AFSPCMAN 91-710, RSM-2002)
AS9100	Rev D	Aerospace Quality Management System
KNPR 8715.3 ¹	Rev L	KSC Safety Practices Procedural Requirements
LSP-PD-120.05	D	Launch Telemetry Requirements
NPR 8715.6 ²	E	NASA Procedural Requirements for Limiting Orbital Debris
NASA-STD-8719.9 ¹	C	NASA Safety Standard for Lifting Devices and Equipment

¹ The Contractor shall comply with the latest revisions of these documents. These documents apply to NASA property, personnel, software, and processes performed in NASA-owned facilities.

² The Contractor shall support NASA's compliance with this document.

1.4 Reserved

2.0 STANDARD LAUNCH SERVICE

The Contractor shall perform all launch service tasks necessary to deliver payloads to defined orbital parameters in compliance with mission requirements. The launch service shall support missions to all orbital parameters, consistent with vehicle configuration capabilities and launch Range restrictions. The Contractor shall provide launch services, which are in compliance with all Range and responsible authority requirements. The Contractor shall make all arrangements with the responsible authorities for the required launch Range authorization and support for vehicle processing; integrated payload/vehicle processing, launch; and launch site maintenance and modifications. NASA reserves the right to approve the choice of launch site [e.g. Cape Canaveral Space Force Station (CCSFS), Vandenberg Space Force Base (VSFB)].

The Contractor shall furnish all services, maintain all equipment and infrastructure including, but not limited to: program management, mission integration, launch site support, ground and flight system safety, and performance assurance, necessary to accomplish the safe and successful launch of payloads to the required orbit conditions within required launch periods. The Contractor shall provide facilities and services at Contractor facilities for NASA personnel performing insight and approval functions during the performance of the contract. The Contractor shall provide access to launch vehicle documentation in support of NASA insight and approval functions.

The Contractor shall provide all necessary services, test hardware and software, and mission specific elements required to integrate the payload(s) to the launch vehicle systems. The Contractor shall meet all launch service performance requirements described in Exhibit 1, Capabilities, Specifications and Environments. All capabilities and conditions stated in Exhibit 1 must be consistent and compatible with all other capabilities and conditions stated in Exhibit 1 and in response to the requirements of the SOW. Any exceptions to the stated capabilities or conditions must be specifically noted.

The Contractor shall coordinate with NASA Public Affairs Office all press releases concerning launches under this contract. During vehicle build-up, payload integration, and launch countdown, the Contractor shall allow NASA Public Affairs access to facilities to photograph and videotape activities, including hazardous operations. The Contractor shall assist NASA Public Affairs in developing the launch commentary for NASA Television by furnishing launch countdown and operations background material. The Contractor may also be asked to provide information to support the development of the press kit document and the NASA pre-launch and post-launch news conferences. The Contractor shall coordinate with NASA Public Affairs Protocol and Guest Services a minimum of sixty (60) days in advance of each launch to determine any special requirements.

The Contractor shall provide standard launch services as delineated in Exhibit 2, Standard Launch Services List which shall comply with the requirements of the SOW. All hardware, software, analyses, and support required to provide each item listed in Exhibit 2, shall be included in the standard launch service.

All Contract Data Requirements List (CDRL) and Supplemental Data Requirements List (SDRL) items, identified in Attachment D2 and D3 respectively, shall be included in the standard launch service.

2.1 Launch Vehicle

The standard launch service shall include, as a minimum, the following:

- (A) Launch timing capabilities with
 - (i) Reserved
 - (ii) Multiple approximately twenty-four (24) hour re-launch attempts in the event of a launch scrub
 - (iii) Instantaneous launch window
 - (iv) Inertial targeting over a **TBP** (e.g. 30) minute launch window utilizing multiple discrete or continuous targeting
- (B) A launch vehicle and Payload Adapter (PA) with appropriate electrical and mechanical interfaces (as described in Exhibit 1) required for payload integration and testing.
- (C) A payload separation system with the following characteristics:
 - (i) The payload shall be protected from debris generated by the separation system.
 - (ii) The separation system shall function in a manner that prevents any re-contact with the payload, including Contractor-provided attach hardware on the payload, by the upper stage or any element of the separation system once separation has been initiated.
 - (iii) Redundant payload separation indications.
- (D) A Payload Fairing (PLF) with the following items, as a minimum:
 - (i) **TBP** payload access doors of sizes **TBP**, whose locations shall be mission unique within the stated structural limitations of the fairing. Additional or

different size doors shall be provided as a non-standard service. The Contractor shall provide payload personnel access through all standard and non-standard access doors from fairing mate through final vehicle closeout.

- (ii) Thermal environments specified not to exceed **TBP**°C, +/- **TBP**°C (**TBP**°F, +/- **TBP**°F) during ascent on any launch vehicle component inside the fairing with a view factor to the payload, with the exception of the upper stage motor which shall not exceed **TBP**°C (**TBP**°F). Maximum thermal environments shall be specified in the mission specific ICD (CDRL C2-1.1).
 - (iii) Thermal and/or acoustic blankets sealed or filtered such that venting of blanket material debris does not contaminate the payload.
- (E) Flight instrumentation, as specified in Exhibit 1, paragraph 9.0, Payload Environment Instrumentation, to provide data adequate to support Section C, paragraph 24.2, Mission Success Determination.
- (F) Capability to spin payloads, when applicable, prior to deployment over a minimum range of zero (0) to **TBP** degrees per second with an accuracy of **TBP** degrees per second.
- (G) Post mission maneuvers:
 - (i) A Collision/Contamination Avoidance Maneuver (CCAM), when applicable, following payload separation to minimize payload contamination and any chance of re-contact with the separated payload
 - (ii) End of mission maneuvers after CCAM shall support the limiting of orbital debris IAW mission requirements and vehicle capability.
- (H) Provisions for the prevention of static charging on umbilical connectors that may be subjected to static charging from plasma heating or atmospheric particulate matter (e.g., bleed resistors, dead facing, or connector covers).
- (I) A logo in accordance with NASA provided artwork. The size of the logo shall be a maximum of **TBP** dimensions. The logo installation shall be appropriately documented in the mission Interface Control Document (ICD) (CDRL C2-1.1).
- (J) Availability of a Test Payload Adapter (TPA), with appropriate mechanical interfaces and payload separation system, to support payload integration and testing at NASA/NASA Customer (NNC) facilities or such other locations as defined by the ICD. At this facility, the Contractor shall support, with necessary personnel and equipment, the following: a fit check of the payload with the flight PA, environmental testing with a TPA (personnel support is limited to one installation

and one removal of TPA), payload shock testing with a TPA, and sufficient pyrotechnics for two shock test firings.

- (K) Command, control, and power signal capability from the launch control facility to the payload umbilical Ground Support Equipment (GSE) IAW the mission specific ICD.
- (L) All mechanical GSE required to interface with the payload mechanical adapters and accomplish the payload to launch vehicle mating operation IAW the mission ICD.
- (M) Both halves of the electrical connector at each spacecraft-to-launch vehicle interface. The Contractor shall provide three (3) sets for each electrical connector interface: a flight set, a flight spare set, and a set for spacecraft testing.
- (N) All other hardware equipment, software, infrastructure, and logistics necessary to perform the contracted launch services.

2.2 Program Management

The Contractor shall provide all program management functions required to provide the launch services and to satisfy the mission requirements for each NASA mission. The program management function of this contract shall provide insight to NASA for all technical and programmatic activities performed under this contract.

The Contractor shall coordinate all program management functions and issues directly with the KSC LSP designated representative(s). The NASA Contracting Officer is the only NASA representative authorized to provide formal contract direction.

2.2.1 Formal Reviews

The Contractor shall conduct program reviews, design reviews, and readiness reviews, and shall provide for the participation of NNC. The Contractor shall provide minutes and action items resulting from each review to NASA within one week after the review. A copy of the presentation material shall be available at the review for all NNC attendees (CDRL C1-1)

2.2.1.1 Program Reviews

During the contract performance period, the Contractor shall conduct Program Reviews with NASA at least once per year to: report development and production status, ensure schedules support program objectives, review action items, review program schedules, and discuss any issues. The intent of the program review is to provide a forum for open dialog between NASA and the Contractor with respect to launch services. NASA will provide status of Agency direction at the reviews. The review location shall alternate between NASA and Contractor facilities unless mutually agreed upon to do otherwise.

2.2.1.2 Design Reviews – Mission Specific

The Contractor shall conduct and chair/co-chair design reviews, as described below, that apply to the system, subsystem, component, and software level for all mission specific items. Where there is not a direct match between a SOW specified mission specific design review(s) and the Contractor's standard review(s), the Contractor's review process will be acceptable provided it addresses equivalent content. All derived requirements from all system requirements must be identified and addressed in each of these reviews. NASA will approve the Mission Unique items in these reviews.

(A) Mission Specific Requirements Review (MSRR)

The Contractor shall conduct an MSRR prior to the Mission Specific Preliminary Design Review (MSPDR) with NNC to review the mission specific design requirements for the following items:

- (i) System requirements' identification and definition to a level adequate to verify launch vehicle performance capabilities.
- (ii) Design restrictions, limitations, and known violations.
- (iii) Physical and mechanical interfaces (e.g., payload to launch vehicle, payload envelope, and access provisions).
- (iv) Electrical interfaces (e.g., launch vehicle to payload, payload to umbilical, interfaces with electrical ground support equipment, pad electrical systems, ground batteries, telemetry, grounding, and power).
- (v) Functional interfaces (e.g., structures, structural loads, and vibration).
- (vi) Avionics systems and interfaces (e.g., payload avionics interfaces with launch vehicle, separation systems, telemetry interfaces, payload command and telemetry, and RF).
- (vii) Mass properties.
- (viii) Environmental requirements (e.g., thermal, contamination, vibration, pressure, Electromagnetic Compatibility (EMC), shock, launch complex RF, and lightning).
- (ix) Orbital requirements, launch vehicle performance, launch window injection, and deployment attitudes and rates.

- (x) Payload/Launch Vehicle separation requirements (e.g., separation conditions, launch vehicle post-separation maneuver requirements, and telemetry).

(B) Mission Specific Preliminary Design Review (MSPDR)

The Contractor shall conduct a preliminary detailed design review prior to major commitment to drawings and design. Mission unique trade studies shall be completed prior to the MSPDR. The Contractor shall discuss analyses performed and their results along with comparisons to any similar proven designs. The Contractor shall evaluate the safety of the design and its ability to meet safety requirements. The preliminary mission unique design shall be subject to NASA's approval. NASA reserves the right to withhold approval until all action items have been closed. As a minimum, the Contractor shall provide verification of the following items at the MSPDR:

- (i) All system requirements have been allocated to the subsystem and component level and the flow down is adequate to verify system performance.
- (ii) The design solutions being proposed are expected to meet the performance and functional requirements.
- (iii) The design does not pose major problems that may cause schedule delays.
- (iv) Overall system architecture has been established and all launch vehicle to payload interfaces have been identified and are verifiable.
- (v) The design solution can be produced based on existing processes and techniques; if not, risk areas, which require unique and unproved processes, are identified and risk mitigation plans are established.
- (vi) An acceptable operations concept has been developed.
- (vii) Preliminary launch vehicle interfaces have been defined.
- (viii) Preliminary plans are established for end-to-end testing methodologies.
- (ix) 30% mission unique drawings released.

(C) Mission Specific Critical Design Review (MSCDR)

The Contractor shall conduct an MSCDR prior to design freeze and before significant fabrication activity begins. The Contractor shall present a final detailed

design using drawings, analyses, and evaluation testing that shows the design meets final performance and interface specifications, safety requirements, and mission objectives. The Contractor shall provide selection criteria for the evaluation tests performed to prove validity. The mission unique critical design shall be subject to NASA's approval. NASA reserves the right to withhold approval until all action items have been closed. As a minimum, the Contractor shall provide verification of the following items at the MSCDR:

- (i) All technical problems and design anomalies have been resolved without compromising system performance, reliability and safety.
- (ii) The detailed design will meet performance, functional requirements, and schedule.
- (iii) Software simulations and prototyping results do not present any potential mission risks.
- (iv) All key subsystem and/or component engineering analyses are complete.
- (v) Integrated safety analysis identifying any remaining hazards and proposed resolution.
- (vi) Launch vehicle/payload compatibility test plans have been defined.
- (vii) 90% mission unique drawings released.

(D) Mission Specific System Acceptance Review (MSSAR)

The NLS II Contractor shall conduct an MSSAR after the design, fabrication, qualification testing and analysis is complete and shall incorporate any flight data available as part of the launch service. The Contractor shall address all mission specific items for the payload and mission requirements to verify qualification, compliance, and systems-level compatibility using completed analyses, test, inspection, and demonstrations results. The mission unique system acceptance shall be subject to NASA's approval. NASA reserves the right to withhold approval until all action items have been closed. As a minimum, the Contractor shall provide verification of the following items at the MSSAR:

- (i) Results of the mission specific acceptance reviews of the major suppliers or major subsystems;
- (ii) Design changes that occurred subsequent to MSCDR or changes as a result of new flight data;

- (iii) Summary of applicable component tests (test setups, test cases, results and significant anomalies), analyses, margins, or similarity assessments. Included in the summary is component qualification rationale (similarity, test, analysis) for all affected components that are new, changed or subject to new environments or functional requirements;
- (iv) Qualification rationale for the system as a whole;
- (v) Methodology and results of current analyses;

2.2.1.3 Readiness Reviews

The Launch Vehicle Systems Readiness Review, Pre-Mate Readiness Review, Launch Management Coordination Meeting (LMCM), Flight Readiness Review (FRR) and Launch Readiness Review (LRR) described herein will be conducted for each NASA mission.

(A) Launch Vehicle Systems Readiness Review

The Contractor shall conduct a Launch Vehicle Systems Readiness Review to demonstrate that the launch site and launch vehicle are ready to proceed with launch vehicle processing activities at the launch site. The Contractor shall present as a minimum:

- (i) Mission description including S/C and launch vehicle configuration; integration status; summary of waivers; post launch data review of previous flights (NASA and non-NASA); mission specific analyses results and status; vehicle hardware assignments, assessments and status; and launch site status
- (ii) A detailed schedule showing all activities (ground and flight) remaining to achieve a successful, on-time launch

(B) Pre-Mate Readiness Review

The Contractor shall conduct a Pre-Mate Readiness Review to demonstrate the launch site and launch vehicle are ready for payload mechanical and electrical integration. The Contractor shall conduct a launch vehicle/site walk-down [reference SOW Section 2.4.1.1(D)] with NNC participation prior to or in conjunction with the Pre-Mate Readiness Review. The Contractor shall present as a minimum:

- (i) Action item status, safety status, payload mating plan, closure plan, payload integration/launch site documentation, interface verifications, checkout and

launch software status, nonconformance reports, launch site status, S/C readiness, and review of flight profile.

- (ii) A detailed schedule showing all activities remaining to achieve an on-time launch.

(C) Launch Management Coordination Meeting (LMCM)

The Contractor shall participate in an LMCM conducted by NASA before each launch dress rehearsal or launch. The LMCM is used to ensure the readiness of the launch team to execute the procedures necessary to conduct the launch dress rehearsal (reference SOW Section 2.4.1.3) or launch. At this meeting, participant roles and responsibilities during countdown shall be identified. The Contractor shall develop and present a decision matrix for its launch team which defines who has authority to issue a GO, NO GO, and HOLD during launch countdown. The Contractor shall describe its launch day management activities, identify key team members, and define responsibilities and communications between the launch vehicle, NASA, and payload teams.

(D) Flight Readiness Review (FRR)

NASA will conduct/chair, and the Contractor shall participate in an FRR for each mission to ensure the specific launch vehicle is acceptable for flight and all Range and other mission requirements have been met, or will be satisfied prior to launch. The FRR is held approximately three (3) days before launch. As a minimum, the Contractor shall provide verification that:

- (i) All critical items required to proceed into final launch countdown are ready
- (ii) Vehicle configuration is defined and all vehicle systems have been verified IAW launch site test plans
- (iii) All previously recorded action items have been closed or are reflected on the schedule
- (iv) All previously held Contractor's readiness review actions have been closed or resolved
- (v) Launch site/Range support organizations have committed to launch
- (vi) Tracking and data support resources are committed to launch
- (vii) Any open work is identified and closeout plans and schedules are in place and supportable

- (viii) Any constraints to launch are identified and resolution plans developed
- (ix) Mission risks are known and documented
- (x) Launch commit criteria for payload and launch vehicle is approved and released

The Contractor shall also discuss:

- (i) Anomalies from previous missions, including non-NASA missions
- (ii) Hardware/software failures in the field either on the NASA assigned launch vehicle or in the fleet
- (iii) Open corrective actions/problems reports
- (iv) Mission Unique and First flight items

(E) Launch Readiness Review (LRR)

NASA will conduct/chair, and the Contractor shall participate in an LRR one day prior to launch to verify all actions from the FRR are complete and final processing has been successfully completed. NASA will appoint the chairperson for this review. At the conclusion of this review, an “approval to proceed with launch countdown” is given. Representatives from the Contractor, Range, and NNC agencies sign the Certificate of Flight Readiness. The Contractor shall prepare the Certificate of Flight Readiness. A sample Certificate of Flight Readiness is provided in Exhibit 4.

2.2.2 Risk Management

The Contractor shall implement risk management techniques that address the identification, analysis, mitigation, and tracking of potential impacts to mission success. The Contractor shall develop the criteria, methods, and procedures used for identifying critical items in the Risk Management Plan (CDRL C3-6).

2.2.3 Configuration Management

The Contractor shall perform configuration management of the launch vehicle design and production for all launch vehicle components/subsystems, hardware, and software.

2.2.4 Manifest Policy

The Contractor shall develop and provide a Manifest Policy as described in CDRL C1-4 that addresses the Contractor’s overall approach to ensuring timely launch of payloads.

The Contractor shall provide a Program Master Schedule and Launch Vehicle Planning Manifest (SDRL S1-3) to support Government mission scheduling.

2.2.5 New Launch Vehicle Orientation

The Contractor shall conduct an orientation briefing at KSC. This briefing shall introduce Contractor personnel and establish project interfaces with NASA personnel and describe the Contractor's organization and infrastructure. The briefing shall contain information summarizing the design, performance, fabrication, integration, testing, qualification and operational features of the launch vehicle systems and supporting facilities required to provide the launch service in the form of diagrams, schematics, pictures, drawings, videos, etc.

2.3 Mission Integration Services

2.3.1 Mission Integration Management

The Contractor shall be responsible for managing the mission integration of the payload flight and ground systems with the launch vehicle and its associated GSE. In accordance with this responsibility, the Contractor shall perform, as a minimum, the following services:

- (A) Provide a single point of contact with overall mission responsibility for each mission. This single point of contact shall be responsible for coordinating support from all technical disciplines and management during the integration process.
- (B) Conduct mission integration meetings (kick-off, working group, technical interchange) approximately quarterly from Mission Integration Start (L-30M) to launch.
- (C) Provide a co-chairperson along with NASA for all working groups and technical interchange meetings. The Contractor co-chairperson shall be responsible for preparing and distributing agendas, minutes and action item logs for each meeting (SDRL S1-2). The Contractor shall maintain the action item database and ensure closure of all actions.
- (D) Provide appropriate technical/engineering representation at payload preliminary and critical design reviews.
- (E) Plan, schedule, and manage mission analyses required to define and verify compatibility of the payload with the interface requirements and environments (reference Table D1-A, SOW paragraph 2.3.2).
- (F) Track development status of and resolve issues associated with mission specific hardware and software.

- (G) Coordinate interface and support requirements for the mission.
- (H) Plan/coordinate mission specific flight operations.
- (I) Manage integration activities at the launch site.
- (J) Manage and coordinate the launch vehicle safety approval process.
- (K) Prepare, maintain and implement a payload to launch vehicle ICD for each mission. The ICD shall include all mission requirements including launch vehicle and launch site interface definition and environments (including GSE). The ICD shall include the payload to launch vehicle/launch site electrical and mechanical interface drawings (CDRL C2-1.1). The baseline ICD (CDRL C2-1.1) will be incorporated by reference into Exhibit 8 of the contract via contract modification upon approval of the baseline ICD. The final revision to the ICD (CDRL C2-1.1), including all approved changes and waivers, will be incorporated by reference into Exhibit 8 of the contract via contract modification prior to the mission success determination. The Contractor shall comply with the latest approved revision of the ICD, to include all approved changes and waivers.
- (L) Create and maintain an ICD Verification Matrix (CDRL C2-1.2) which shall include a verification plan for each requirement in the ICD based on the appropriate method. The Contractor shall perform verification of each ICD requirement based upon the verification plan identified in the Verification Matrix.
- (M) Manage the design, development, qualification, testing and integration of mission unique requirements.
- (N) Evaluate the capability of the launch vehicle and define any performance and payload volume the Contractor may make available to NASA for secondary payloads.
- (O) Provide mission status to NNC throughout the launch campaign from Authority to Proceed (ATP) through mission success determination.
- (P) Provide appropriate representation and participation at NASA conducted Ground Operations Working Groups (GOWG's) approximately 2 times a year from L-27 months to spacecraft arrival at the launch site and at the NASA Ground Operations Review (GOR).

2.3.2 Mission Analyses

All vehicles provided under this contract shall include the following analysis for each mission as part of the standard launch service. The Contractor shall prepare and submit the standard mission analyses as listed in Table D1-A.

CDRL	ANALYSIS
C4-1	Performance and Guidance Accuracy Analysis
C4-2	Final Mission Analysis (FMA)
C4-3	Payload/Expended Stage Separation Analysis
C4-4	Payload Fairing Venting Analysis
C4-5	Payload Fairing Clearance Analysis
C4-6	Pre-Flight Controls and Stability Analysis
C4-7	Coupled Dynamic Loads Analysis
C4-8	Integrated Thermal Analysis
C4-9	RF Link Analysis
C4-10	Payload/Launch Vehicle EMC, RF, and EED Compatibility Analysis
C4-11	Mission Unique Software
C4-12	Quick Look Flight Report
C4-13	Final Flight Report

Table D1-A: Mission Analyses to Support Payload Integration

The standard launch service shall include all analyses required to evaluate compliance with NPR 8715.6A, NASA Procedural Requirements for Limiting Orbital Debris. The number of submittals for the Standard Mission Analyses are identified in Table D2-A.

2.3.3 Mission Success Determination

Mission success will be based on the criteria set forth in Section C, paragraph 24.1, Mission Success Criteria. Mission success determination requirements are detailed in Section C, paragraph 24.2, Mission Success Determination.

The Contractor shall prepare and submit a Quick Look Flight Report (CDRL C4-12) and a Final Flight Report (CDRL C4-13). Within thirty (30) days of receipt of the Final Flight Report, the Contracting Officer will either determine the launch a Mission Success or inform the Contractor of the Government's intent to withhold determination for payload evaluation.

2.4 Launch Site Support

2.4.1 Pre-Launch Checkout and Launch Support

2.4.1.1 Launch Vehicle Preparation and Launch

For launch services provided under this contract, the Contractor shall, as a minimum;

- (A) Perform all launch vehicle preparations and launch site operations necessary to safely and successfully deliver the payload to the desired orbit.
- (B) Generate the required documents (CDRL C3-7) and obtain all required safety approvals for the launch vehicle system and integrated payload/launch vehicle system operations including payload/launch vehicle system integration and launch operations.
- (C) Provide NASA access to all meetings including, but not limited to, scheduling meetings, test briefings, and technical meetings. Upon request, the Contractor shall provide copies of schedules, test briefings, and other material presented at technical meetings.
- (D) Conduct launch vehicle/launch site walk-downs with NNC participation.
- (E) Comply with the submittal and operational requirements for all NASA provided facilities occupied during launch vehicle processing (CDRL C1-5). Failure to comply may result in denied access to the NASA provided facility.

2.4.1.2 Telemetry Data and Launch Countdown Support

For all NASA and non-NASA missions, the Contractor shall provide all real-time, full-rate launch vehicle telemetry data, GSE data, operational voice, video, and countdown timing, as defined below. Full-rate is defined as the rate at which the sensor is transmitted to the ground. Services shall be operational for all integrated launch vehicle powered-on operations at the launch site.

2.4.1.2.1 Vehicle and Ground Support Equipment (GSE) Telemetry

The Contractor shall provide sufficient information for LSP to independently process all full-rate, real-time and recorded launch vehicle and GSE telemetry (RF and hardwire). (CDRL C6-1) (SDRL S6-1) NASA's preference is raw telemetry defined as directly off the GSE or launch vehicle. Any pre-processing of telemetry (including encryption) prior to delivery to LSP must be coordinated in accordance with CDRL C6-1.

For NASA missions, the Contractor shall provide all real-time, full-rate, integrated launch vehicle telemetry and GSE data to LSP's Communication Center in accordance with IAW NASA Policy LSP-PD-120.05. For non-NASA missions the Contractor shall provide all available ground and vehicle telemetry throughout pre-launch and flight in real-time, including through disposal burn to LOS. The Contractor shall also provide

all vehicle telemetry that was received and stored for later retransmission when requested by NASA (SDRL S6-1).

2.4.1.2.2 Network Requirements

The Contractor shall provide each of the defined signals in 2.4.1.2.1 Vehicle & GSE Telemetry requirement and 2.4.1.2.3 Operational Voice requirement, to a mutually agreed upon interface. The transport between the Contractor's facility to LSP's Communication Center shall be redundant and the implementation shall be negotiated between NASA and the Contractor. For NASA Spacecraft Command and Telemetry, the Contractor shall provide a mutually agreed upon transport interface between the launch complex and LSP's Communication Center.

2.4.1.2.3 Operational Voice Requirements

For all missions, the Contractor shall provide LSP's Communication Center all Contractor operational voice nets. Talk/listen or monitor-only net configuration will be determined on a net-by-net, operation-by-operation basis. A max number of 24 unique Spacecraft or other NASA nets shall be supported as requested by NASA.

2.4.1.2.4 Video Requirements

For all missions, the Contractor shall provide LSP's Communication Center full-time video feeds of Launch Vehicle and GSE views. For NASA missions, If not all views are available, a minimum of 8 video feeds shall be provided, with the ability to coordinate switching during dynamic operations. For NASA missions, the Contractor shall provide on-board Launch Vehicle videos in real-time and/or recordings as available. For non-NASA missions the Contractor shall provide on-board Launch Vehicle videos in real-time and/or recordings if not in conflict with non-NASA S/C requirements.

2.4.1.2.5 Timing Requirements

For NASA missions, the Contractor shall provide LSP's Communications Center a countdown clock signal for each countdown operation. NASA's preference is a CS-5 timing signal that represents the values of the L-Time, T-Time and Expected Liftoff values. For non-NASA missions, any countdown operations for which the countdown clock is generated shall be provided to LSP's Communications Center.

2.4.1.3 NASA Launch Flight Operations

The Contractor shall be responsible for coordinating and ensuring all tracking and data recovery support meets mission requirements (CDRL C2-3).

The Contractor shall coordinate all downrange assets. The contractor shall provide NASA the latest mission specific support requirements provided to all the downrange assets at L-3 months. Updates after L-3 months shall be provided, as required. The Contractor shall provide the tracking assets with the technical coordination needed to execute the end-to-end telemetry link tests (e.g. telemetry playback tapes, mission trajectory disks, Bit Error Rate Test, etc.). The Contractor shall coordinate with NASA on telemetry coverage.

The Contractor shall perform end to end telemetry link tests to verify end to end links from all ground, space-based and/or other tracking assets used to LSP's Communication Center to ensure readiness prior to launch to meet LSP's telemetry coverage requirements IAW LSP-PD-120.05, Rev. D (CDRL C6-2) The tests shall be completed NLT L-2 days.

The Contractor shall receive and record the raw launch vehicle telemetry data for all phases of powered flight, IAW NASA Policy LSP-PD-120.05, Rev. D. The Contractor shall receive and record launch vehicle telemetry data encompassing payload separation IAW NASA Policy LSP-PD-120.05, Rev. D.

The Contractor shall ship required electronic hardware and satisfy export control and frequency authorizations as required by local and international governments.

On day of launch, the Contractor shall maintain in contact with the downrange assets and provide LSP's communication center access to the down range voice communications and the systems while also providing status/readiness to the launch team as requested.

2.4.1.4 NASA Launch Countdown Support

The Contractor shall implement all pre-launch and launch support requirements in a "see and hear what the Contractors sees and hears" philosophy for the applicable NASA team member(s).

The Contractor shall conduct a minimum of one Dress Rehearsal prior to launch. The rehearsal shall exercise the launch countdown procedure, and will have the participation of the NNC and launch vehicle Contractor launch teams. The rehearsal may use an abbreviated countdown script with simulated payload/vehicle conditions, including anomalies.

The Contractor shall provide launch countdown procedures, Mission Constraints Documents (CDRL C1-2) and a Mission Console Notebook (CDRL C1-3) for the launch dress rehearsal and launch in hard copy and a mutually agreed upon electronic format.

The Contractor shall provide **TBD** NASA personnel access to consoles, with vehicle monitoring capabilities, co-located within the Contractor's launch control center. The Contractor shall provide access to real-time telemetry (RF and hardwire), voice communication channels with talk/listen capabilities, video and telephones for the co-located NASA personnel. The Contractor shall implement the NASA provided console communications configuration for those consoles occupied by the NASA team within the Contractor's launch control center. The Contractor shall support validation of this configuration with the designated NASA LSP Data Center prior to launch dress rehearsal and launch.

During the Launch Countdown, the Contractor shall poll NASA at key milestone events for GO/NO-GO status. The final poll shall include the NASA Launch Manager's GO/NO-GO status for launch. The Contractor shall provide NASA access to anomalies and concurrence with anomaly resolution prior to proceeding with Launch Countdown. The Contractor shall poll NASA for any recycle of the Launch Countdown.

2.4.2 Launch Site Payload Support

2.4.2.1 Payload Processing Facility

For launches conducted from CCSFS, VSFB, and WFF, NASA will provide the Payload Processing Facility (PPF) unless processing services are purchased as a non-standard service. In the event the Contractor proposes and NASA approves an alternate launch site, the Contractor shall be responsible for providing a PPF, subject to NASA approval, meeting the payload requirements.

2.4.2.2 Launch Site Payload Integration Support

The Contractor shall provide all services, equipment, and support required for the integration and launch of each payload. Services shall include, as a minimum, the following:

- (A) Encapsulation of the payload complement at the PPF, for either flight or transport to the launch complex, transport, mate with the launch vehicle and performance of integrated checkout activities.
- (B) Verification of the end-to-end functionality of the umbilical lines provided for payload use.
- (C) Support for the installation and checkout of the payload GSE (provided by the payload customer) at the launch site, and payload communication accommodations from the pad to the PPF. The Contractor shall also provide for a NASA console at the Contractor's launch control center.

- (D) Provision of the mountings for, and the installation of, the payload GSE (provided by the payload customer) at the launch complex and/or remote sites.
- (E) Provision for contingency off-loading of payload propellants in accordance with the appropriate Range Safety requirements and appropriate payload procedures.
- (F) Provision for launch site services to meet payload requirements such as: power, air conditioning, GN₂/GHe purges, and contamination control.
- (G) Support of all activities required to de-mate and to return the payload to the processing facility if necessary.
- (H) Preparation of the procedures for integrated launch vehicle/payload operations for NASA approval and incorporation of the payload procedures as appropriate (CDRL C5-5).
- (I) Coordination of payload and launch vehicle operations into an integrated operational flow, preparation of the schedules, and provision of the schedules with updates as needed to NNC.
- (J) Provision of contamination control for payloads while they are in the possession of the Contractor, or in a Contractor-provided PPF IAW the SOW, paragraph 2.4.2.3.
- (K) Provision of a payload protective cover for use after payload mate to the launch vehicle, if the fairing is installed at the launch pad. The cover shall be purged with conditioned air meeting the minimum requirements stated in the SOW, paragraph 2.4.2.3(C). Provisions shall be made for payload personnel ingress and egress.
- (L) Certification of payload contamination control requirements for: fairing surface cleanliness, cleanroom environments, and purge system cleanliness.
- (M) Support for the installation of customer provided RF re-radiating equipment.
- (N) Provision for the planning and execution of activities associated with the integrated testing of the customer's payload with the launch vehicle on the launch pad.

2.4.2.3 Contamination and Environmental Control

The Contractor shall provide and implement a generic payload contamination control plan and, if required, provide and implement a contamination control implementation plan for each payload. Using IEST-STD-CC1246E, MIL-PRF-27401, ISO 14644-1, ISO 14644-2, and NASA RP-1124 (Rev-4) as guidelines, the launch service shall meet the following minimum requirements:

- (A) Payload/Vehicle Integration Environment. For all Contractor provided facilities where the payload resides and is exposed (including payload processing facilities,

integration facilities, or facilities at the launch pad), a Class 8 clean room environment IAW ISO 14644-1 shall be provided. In addition, the temperature and humidity environment shall have the ability to be controlled and maintained within a range of **TBP** to **TBP**°C (**TBP** to **TBP**°F) and 30% to **TBP**%, respectively. This environment shall be maintained at all times unless the payload is encapsulated within a transportation container or payload fairing and purge air has been established IAW 2.4.2.3(B) or (C), herein. If the Contractor provides a payload protective cover as identified in paragraph 2.4.2.2(K) of the SOW, and purge air has been established, the facility environment may be exceeded. Prior to removal of the payload protective cover, the facility shall be returned to a Class 8 IAW ISO 14644-1 clean room environment. The Contractor shall provide contamination and environmental monitoring when the payload is exposed.

- (B) Transportation Environment. The following shall apply to all payload transportation containers (including the payload fairing, if used for transportation) provided by the Contractor. Following payload encapsulation, the transportation container shall be purged with conditioned filtered air or GN₂ and the payload environment shall be maintained within a temperature and humidity range of **TBP** to **TBP**°C (**TBP** to **TBP**°F) and **TBP**% to **TBP**%, respectively. The Contractor shall identify periods or configurations where the temperature and humidity requirements cannot be met. Conditioned air filtration shall be capable of removing 99.97% of all particles greater than 0.3 microns and removing 95% of all hydrocarbons with a molecular weight greater than 70. GN₂ shall be Grade B, Type 1, (or better) IAW MIL-PRF-27401F. Transportation GN₂ purging or conditioned air shall not impinge directly on the payload at any point.
- (C) Fairing Environment. Following fairing installation and payload mate to the launch vehicle, the PLF environment shall be purged with conditioned filtered air and shall maintain a temperature set point range, selectable by the payload, of **TBP** to **TBP**°C (**TBP** to **TBP**°F), controllable to +/- **TBP**°C (**TBP**°F), through T-0. Relative humidity shall be maintained within 30% to 50%, controllable to **TBP**%. The Contractor shall identify periods or configurations where the temperature and humidity requirements cannot be met. Conditioned air filtration shall be capable of removing 99.97% of all particles greater than 0.3 microns and removing 95% of all hydrocarbons with a molecular weight greater than 70. After the fairing has been closed out for flight, Grade B, Type 1, GN₂ (or better) IAW MIL-PRF-27401F, may be used for purging the fairing environment. Fairing GN₂ purging or conditioning air shall not impinge directly on the payload at any point.
- (D) Fairing Internal and Payload Adapter Surface Cleaning. The internal surfaces of the payload fairing compartment (including Payload Adapter) shall be cleaned, certified, and maintained to IEST-STD-CC1246E Level 750A or better, with the number and locations of samples to be determined by the integration team. Cleaning and certification shall be accomplished in a clean room rated at or above Class 8 IAW ISO14644-1. In lieu of IEST-STD-CC1246E Level 750A, the payload fairing compartment (including Payload Adapter) may be cleaned and certified such that

contribution from launch vehicle sources measured from S/C turnover (Launch Vehicle Contractor control of S/C for integrated operations) through CCAM shall not exceed 1% total obscuration and 150 Angstroms molecular contamination. The sampling methodology for particulate shall be per ASTM E1216 and for molecular per ASTM E1235. The number and locations of samples to be determined by the integration team.

(E) Reserved

(F) Clean Room Garments. Personnel garments used, at a contractor facility, in the integration of the payload shall be provided and cleaned by the Contractor. Personnel garments used in the integration of the payload shall comply with accepted clean room and personnel safety operating standards as specified in the mission specific contamination control plan.

(G) Materials. All materials used in areas in close proximity to the payload shall be selected based on NASA RP-1124 (Rev-4). Materials shall have a Total Mass Loss (TML) of less than 1.0% and Collected Volatile Condensable Materials (CVCM) of less than 0.10%, or be expressly identified and submitted to NASA for approval.

2.4.2.4 Operational Support Services

The Contractor shall provide safety training, instruction, and certification for all Contractor-operated or provided integration facilities and launch sites to ensure users are aware of facility, launch site, launch vehicle and payload hazards and have adequate knowledge to carry out their tasks unescorted in a safe manner. The Contractor shall provide access for payload personnel to the payload/launch vehicle or storage facilities to accommodate payload customer requirements. The Contractor shall provide security to meet the requirements for payload or personnel protection. The Contractor shall provide to NASA, upon request, launch service user's guide and facility information (SDRL S1-1).

2.4.3 Range Support and Services

Launch vehicles provided under this contract shall include all hardware, software, analysis and support necessary to meet the applicable Range Safety Requirements (e.g., EWR-127-1, AFSPCMAN 91-710, or RSM-2002).

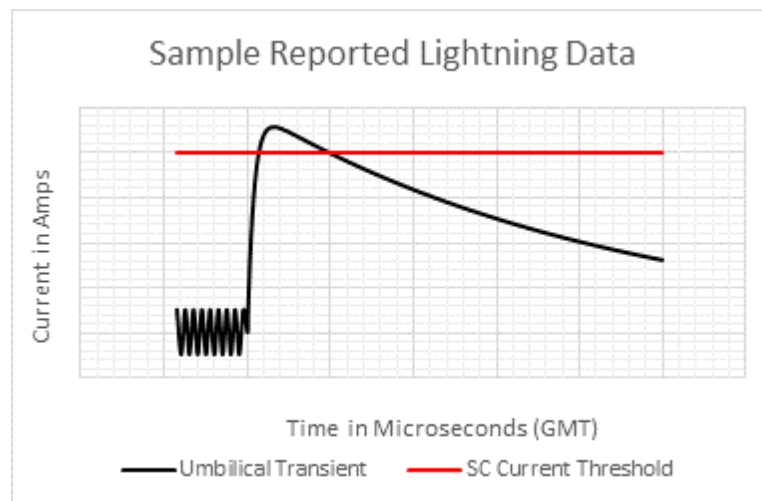
As part of the launch service, the Contractor shall make all launch Range support arrangements for: scheduling Range for launch and integrated testing, Range Safety functions, communications and timing, metric C-band beacon (radar) coverage, telemetry coverage, camera coverage of launch, and tracking and telemetry station acquisition predictions. The Contractor shall be responsible for coordinating and ensuring all tracking and data recovery support meets mission requirements (CDRL C2-3).

The Contractor shall make arrangements for Range provided services necessary to support the launch service. As a minimum, the following services are to be provided: fluids, gases, propellants, ordnance storage, facility usage, equipment support, shop and laboratory services, meteorology, base security, fire protection and environmental health.

The mission specific Program Requirements Document/Operational Requirements (PRD/OR)(CDRL C2-2), or equivalent mission specific Range support documentation, shall be submitted to obtain Range support. The Contractor, with support from NASA, shall complete all forms pertinent to the mission and submit them to the appropriate Range for formal acceptance.

2.4.5 Lightning Monitoring

1. The Contractor shall use a calibrated system to monitor the Spacecraft umbilical currents for lightning induced transients at the launch site after encapsulated assembly mate to the launch vehicle while contractor-provided umbilical connectivity is maintained between the spacecraft and spacecraft's electrical ground support equipment.
2. The Contractor shall provide payload umbilical electrical characteristics (transfer impedance plot) to estimate the induced pin voltages based on monitored current.
3. The Contractor shall establish a reporting threshold based on Spacecraft provided voltage inputs and launch vehicle umbilical electrical characteristics.
4. The Contractor shall record and deliver any induced transients over the threshold no later than 2 hours after the transient event. The Contractor shall report data according to parameters below.



Lightning Data Example

2.5 Safety, Reliability, and Quality Assurance

During the period of performance the Contractor shall establish, implement, and maintain comprehensive safety and health, reliability and quality assurance programs covering program management, mission integration management, and the design, development, production, test, integration and launch of the LVS.

2.5.1 Safety and Health Program

The Contractor shall provide a Safety and Health Plan that will implement safety and health requirements consistent with federal, state, and local government regulations and applicable launch processing site Safety and Health requirements. This Plan shall describe the Safety organization including structure of management interfaces and also system safety methods employed to ensure compliance with applicable launch processing site safety requirements.

The Contractor shall implement a system safety program to identify hazards of systems/subsystems and impose applicable design requirements and management controls to prevent mishaps. The Contractor shall develop system safety documentation, plans, procedures, technical analyses, etc., IAW safety requirements of the applicable launch processing site. Launch processing site safety required documentation shall be made available to NASA (SDRL S3-4) upon request. The Contractor shall participate in all safety working groups (i.e. payload safety working group (PSWG), etc.) as the launch vehicle safety representative for each NASA mission under this contract.

When the contractor is performing work in any NASA owned facility (e.g., buildings 836 and SLC-2W at VSFB and PHSF at KSC), all NASA requirements and safety documentation as contained in the SOW, paragraph 1.3, Compliance Documents, shall be adhered to. The Contractor shall submit system safety documentation (baseline and any changes) to NASA for review/approval IAW CDRL C3-7 for processing at all NASA provided facilities. NASA will review the safety documentation to ensure designs, processing and operations do not pose an unacceptable safety risk to NASA personnel/contractors, resources or risk to Mission Success.

The Contractor shall report Close Calls/Mishaps to NASA IAW CDRL C3-2. The Contractor shall make provisions for NASA safety representative insight into integrated Payload/launch vehicle processing.

2.5.2 Reliability Program

The Contractor shall implement and maintain a Reliability Program with an overall vehicle design reliability of no less than 95% at an 80% confidence factor. The Contractor's program shall facilitate evaluation of the Contractor and subcontractor's programs to determine if the product meets the overall design reliability requirements. Overall vehicle reliability predictions shall be incrementally revised to reflect design modifications.

2.5.3 Quality Assurance Program

The Contractor shall maintain a quality management system that is AS9100:2016, third party certified. The Registrar shall be accredited by a body recognized by either the ANSI-ASQ National Accreditation Board (ANAB) or the International Aerospace Quality Group (IAQG). In the event the Contractor certification is revoked, NASA shall be notified within five (5) business days (CDRL C3-3).

The Contractor shall maintain a Software Assurance Program using AS9115 as a guideline.

The Contractor shall accommodate NASA participation in Contractor and subcontractor audits and AS9100 audits. NASA insight will consist of monitoring audits with the Contractor's auditors and inspectors in order to provide understanding of the Contractor's quality system and insight of their processes.

The Contractor shall support NASA performance of AS9100 2nd party audits, as required. The audits will be performed IAW AS9101 requirements.

The Contractor shall provide a current Audit Plan and schedule for in-house and subcontractor audits upon request from NASA (CDRL C3-4). The Contractor shall provide a copy of both the Contractor performed internal Quality Audit Report and the subcontractor/vendor Quality Audit Reports (CDRL C3-5).

The Contractor shall provide for NASA attendance at any flight hardware reviews the Contractor performs at Contractor or subcontractor facilities. The Contractor shall make available to NASA any build paper, test results, nonconformance reports, discrepancy history, statistical process control, and failure analyses that are relevant to the reviews.

The Contractor shall provide read-only quality information to NASA (via remote terminal) from such Quality Assurance on-line database systems as exist and to which the Contractor has regular and timely input. An example is an on-line Problem/Failure Reporting (P/FR) system.

The contractor shall provide NASA all anomaly resolutions that affect the integrated payload/launch vehicle assembly, including both hardware and software. NASA will approve all integrated payload/launch vehicle anomaly resolutions. The contractor shall provide insight to Material Review Board (MRB) and failure reporting for all launch vehicle nonconformances.

The Contractor shall participate in the Government/Industry Data Exchange Program (GIDEP) and provide Alert System Documentation (CDRL C3-1).

2.6 NASA Insight and Approval

The Contractor shall comply with the Government's implementation of NPD 8610.23 and NPD 8610.24 as defined under Section C, Clause 25.0 and the SOW paragraph 2.2.1, Formal Reviews. As part of the standard launch service, the Contractor shall provide the data, documentation, drawings, analytical models, and support services as necessary to accommodate the requirements specified under contract Section C, Clause 25.0, Government Insight and Approval.

The Contractor shall notify NASA of qualification or test anomalies involving similar launch vehicles, systems, subassemblies and components. The Contractor shall make available to NASA all problem reports or discrepancy reports on launch vehicle systems' failures and anomalies. This shall include insight into fleet-wide problems, anomalies, MRB actions, deviations or waivers to systems, subsystems, materials, processes, and test equipment including those used on non-NASA missions.

In the event of an anomaly or launch failure, the Contractor shall support NASA's Failure Review Board, if activated, or shall allow NASA to fully participate in the Contractor's Failure Investigation Board including those for non-NASA missions.

NASA may elect to have representation as a resident office at the Contractor's major manufacturing and engineering facilities for the life of the contract. The Contractor shall provide accommodations and services, such as badging, furniture, telephones, and use of easily accessible fax and copy machines from one to three residents and up to four visitors at each location. A minimum of two voice and two data telephone lines shall be provided. Electronic data transfer compatibility between the resident office and off-site NASA institutions is required. A resident office will not be required at CCSFS or VSFB, but NASA will require operational support accommodations (i.e. office space, power, telephone, communication end instruments for monitoring vehicle testing, etc.) at the launch complex during NASA launch operations.

2.7 Small Business and Small Disadvantaged Business Subcontracting Plan

The Contractor shall update the Small Business and Small Disadvantaged Business Subcontracting Plan (Attachment D4) on an annual basis in accordance with CDRL C7-1.

3.0 NON-STANDARD SERVICES

The Contractor shall provide the non-standard services identified in Exhibit 3 as directed by the Contracting Officer. Implementation of all non-standard services shall be fully compliant with this SOW.

4.0 MISSION UNIQUE REQUIREMENTS

The Contractor shall provide the mission unique services identified in Exhibit 9 as directed by the Contracting Officer. Implementation of all mission unique services shall be fully

compliant with this SOW.

4.1 Mission Unique Hardware

The Contractor shall design, manufacture, test, and qualify for flight the mission unique hardware that is required to support the payload and mission. The Contractor shall prepare and submit drawings (CDRL C5-4), test plans (CDRL C5-2), and test reports (CDRL C5-3) to support NASA insight and approval of mission unique hardware.

For vehicle changes initiated by the Contractor that are not fleet-wide changes, the Contractor shall prepare and submit a vehicle data package (CDRL C5-1).

The Contractor shall use MIL-STD-1540 B or C as a guideline when developing environmental qualification and acceptance criteria and related test and analysis. Factors of safety for mission unique hardware shall meet the requirements of Table D1-B.

Application	Test		Analysis (No Test)	
	Yield	Ultimate	Yield	Ultimate
Non-Man Rated	1.10	1.25	1.60	2.0

Table D1-B: Mission Unique Hardware Factors of Safety

4.2 Mission Unique Software

The Contractor shall provide source code and mission constants' listings (CDRL C4-11) with appropriate requirement specifications to support NASA approval of mission unique software. The Contractor shall prepare and submit a pre-flight control system and stability analysis report (CDRL C4-6) for the vehicle and mission unique software used for each mission.

4.3 Unanticipated Mission Unique Services

At the time of contract award, the Government may not have identified all mission unique services required for each mission. As unanticipated mission unique services are identified, the Contracting Officer will authorize the Contractor, in accordance with FAR 52.243-1, Changes – Fixed Price Alt I, to perform these services. In performing unanticipated mission unique services, the Contractor shall design the necessary ground and flight hardware/software; conduct appropriate design reviews; and manufacture, test and qualify for flight launch vehicle mission unique hardware/software, i.e., other than that provided as a standard service (Exhibit 2) or described in the non-standard services list (Exhibit 3).

In addition, NASA may request the co-manifesting of secondary payloads on NASA primary payloads utilizing the excess performance available to NASA, in which case, the Contractor shall be required to perform all services necessary for the integration and flight of the secondary payloads on the primary payloads.

5.0 SPECIAL TASK ASSIGNMENTS

The Contractor shall be required to perform special studies and analyses, provide materials, or fabricate hardware in support of this contract. Each task will be initiated by written direction from the NASA Contracting Officer. These tasks generally include: advance planning and feasibility studies in support of future contemplated missions; analyses in support of change requirements to authorized missions; development, fabrication, and test of hardware/software to support planning studies or special tests; mission unique studies; material provision; and hardware fabrication in support of potential missions prior to mission authorization.

6.0 INFORMATION TECHNOLOGY SECURITY

The contractor shall comply with NPR 2810.1A, NASA's Policy on the Security of Information Technology for LSC IT hardware and/or software provided to NASA. New IT hardware and/or software systems delivered to NASA shall be compliant prior to authorization to process. The contractor shall develop, update and implement an IT Security Plan compliant with NPR 2810.1A. NASA IT Security personnel will evaluate the LSC IT hardware and/or software security and determine compliance prior to NASA acceptance.

7.0 GUIDELINE DOCUMENTS

The following documents are to be used as guidelines to the extent specified in this SOW. If a revision number or date is not specified, the latest revision of the documents shall be used as guidelines to the extent specified in the SOW.

Document No	Rev.	Document Title
ISO 14644-1	2015	Clean rooms and associated controlled environments – Part 1: classification of air cleanliness
ISO 14644-2	2015	Clean rooms and associated controlled environments – Part 2: specifications for testing and monitoring to prove continued compliance with ISO 14644-1-First Edition
IRIG-106-19	2019	Telemetry Standards
AS9115	A	Software Engineering - Guidelines for the Application of AS9115 to Computer Software
AS9101	G	Quality Management Systems - Audit Requirements for Aviation, Space, and Defense Organizations

Document No	Rev.	Document Title
MIL-STD-1553 ¹	C	Digital Time Division Command/Response Multiplex Data Bus
MIL-STD-1540	B or C	Test Requirements for Launch, Upper-Stage, and Space Vehicles
MIL-STD-176	A, Notice 1	Weight and Balance Data Reporting Forms for Guided Missiles and Space Launch Vehicles
MIL-STD-1773	Notice 1	Fiber Optics Mechanization of an Aircraft Internal Time Division Command/Response Multiplex Data Bus
IEST-STD-CC1246	E 2013	Product Cleanliness Levels and Contamination Control Program
MIL-PRF-27401	H 2023	Propellant Pressurizing Agent, Nitrogen
ASTM E1216	2021	Standard Practice for Sampling for Particulate Contamination by Tapelift
ASTM E1235	2020	Standard Test Method for Gravimetric Determination of Nonvolatile Residue (NVR) in Environmentally Controlled Areas for Spacecraft
NASA RP-1124	4 1997	Outgassing Data for Selecting Spacecraft Materials
NPR 2810.1A	2011	Security of Information Technology
NASA-STD-8729.1	A, 2017	Planning, Developing and Managing an Effective Reliability and Maintainability (R&M) Program
NPD 8610.23	C	Launch Vehicle Technical Oversight Policy
NPD 8610.24	C	Launch Services Program (LSP) Pre-Launch Readiness Reviews
NPD 8610.7	D	Launch Services Risk Mitigation Policy for NASA-Owned and/or NASA-Sponsored Payloads/Missions
LSP-PLN-324.01	C	Expendable Launch Vehicle Certification
NPR 8621.1	D	NASA Procedural Requirements for Mishap and Close Call Reporting, Investigating, and Recordkeeping

Document No	Rev.	Document Title
LSP-PLN-353.01		Launch Services Program Risk Management Plan
NPD 1280.1	A	NASA Management System Policy
NPR 7120.5	F	NASA Space Flight Program and Project Management Requirements
NPR 8715.3	E	NASA General Safety Program Requirements

¹ MIL-STD-1773 may be substituted

8.0 RESERVED

EXHIBIT 1

CAPABILITIES, SPECIFICATIONS AND ENVIRONMENTS

Exhibit 1 represents the capabilities, specifications, and environments of the proposed launch services, including all standard launch service and non-standard services.

1.0 VEHICLE CONFIGURATION

The Contractor shall define (in Table D1-C) the vehicle, Solid Rocket Motor (SRM), upper stage, Payload Adapter (PA), and Payload Fairing (PLF) configuration options available for each proposed vehicle class. These configurations shall include all standard electrical systems, mechanical systems, propulsion systems, ordnance devices, and flight instrumentation.

Vehicle Configuration	SRM Configuration	Upper Stage	PA	Fairings	

Table D1-C: Vehicle Configurations
(Example Only – Contractor to Specify Format and Data)

2.0 PAYLOAD DELIVERY CAPABILITY

The Contractor shall provide where applicable, in Tables D1-D through D1-G3, the maximum payload capability from appropriate launch site with associated ground rules for each vehicle configuration defined in paragraph 1.0, Vehicle Configuration. Performance shall be expressed in terms of separated payload mass (assuming standard PAs as outlined in paragraph 1.0, Vehicle Configuration) and shall assume a 3-sigma performance reserve for the following orbits:

Low Earth Orbits (LEO): Data shall be provided for circular orbit altitudes from 200 to 2000 km (in increments of 100 km) for inclinations of 0, 5, 10, 20, 28.5, 38, 51.6, 60, 70 and 90 degrees.

Sun-Synchronous Orbits: Data shall be provided for circular orbit altitudes from 200 to 2000 km.

High Energy Missions: Data shall be provided as a function of C_3 from $-10 \text{ km}^2/\text{sec}^2$ to $200 \text{ km}^2/\text{sec}^2$.

Elliptical Orbits: Data shall be provided as a function of apogee altitude and inclination using the launch vehicle's standard ground rules. In addition, provide performance with a perigee altitude of 2000 km.

Geo-Synchronous Orbits (GSO): Data shall be provided for insertion directly into Geo-Synchronous Orbits (GSO).

Orbit Altitude (km)	Separated Payload Mass (kg)									
	Inclinations (Degrees)									
	0	5	10	20	28.5	38	51.6	60	70	90
200										
300										
400										
500										
600										
700										
800										
900										
1000										
1100										
1200										
1300										
1400										
1500										
1600										
1700										
1800										
1900										
2000										

Table D1-D: Maximum LEO Payload Capability for **TBP** Vehicle Configuration,
Launch Site, and Ground Rules
(Data **TBP**)

Orbit Altitude (km)	Separated Payload Mass (kg)				
	Vehicle Configuration				
	A	B	C	D	E
200					
300					
400					
500					
600					
700					
800					
900					
1000					
1100					
1200					
1300					
1400					
1500					
1600					
1700					
1800					
1900					
2000					

Table D1-E: Maximum Sun-Synchronous Payload Capability for **TBP** Vehicle Configuration, Launch Site, and Ground Rules
(Data **TBP**)

C ₃ Value (km ² /sec ²)	Separated Payload Mass (kg)				
	Vehicle Configuration				
	A	B	C	D	E
-10					
-5					
0					
5					
10					
15					
20					
25					
30					
35					
40					
45					
50					
60					
70					
80					
90					
100					
110					
120					
130					
140					
150					
160					
170					
180					
190					
200					

Table D1-F: High Energy Mission Payload Capability for **TBP** Vehicle Configuration, Launch Site, and Ground Rules
(Data **TBP**)

Apogee Altitude (km)	Separated Payload Mass (kg)							
	Inclination (Degrees)							
	28.5	27.0	24.0	20.0	15.0	10.0	5.0	0.0
100,000								
90,000								
80,000								
70,000								
60,000								
50,000								
40,000								
35,788								
30,000								
20,000								
10,000								
5,000								

Table D1-G1: Maximum Elliptical Orbit Payload Capability with Optimized Perigee for **TBP**
Vehicle Configuration, Launch Site, and Ground Rules
(Data **TBP**)

Apogee Altitude (km)	Separated Payload Mass (kg) (Perigee Altitude of 2000 km)							
	Inclination (Degrees)							
	28.5	27.0	24.0	20.0	15.0	10.0	5.0	0.0
100,000								
90,000								
80,000								
70,000								
60,000								
50,000								
40,000								
35,788								
30,000								
20,000								
10,000								
5,000								

Table D1-G2: Elliptical Orbit Payload Capability with Perigee Altitude of 2000 km for **TBP**
Vehicle Configuration, Launch Site, and Ground Rules
(Data **TBP**)

Geo-Synchronous Orbit Altitude (km)	Separated Payload Mass (kg)	
	Inclination (Degrees)	
	28.5	0.0
35,788		

Table D1-G3: Maximum Payload Capability for Direct Insertion into GSO for **TBP** Vehicle Configuration, Launch Site, and Ground Rules
(Data **TBP**)

3.0 INSERTION ACCURACIES

The Contractor shall provide standard insertion accuracies for each vehicle configuration defined in paragraph 1.0 for each of the maximum and minimum mission profiles defined in paragraph 2.0. Table D1-H lists example orbital parameters that should be considered.

Required 3-Sigma Insertion Error Limit Magnitude	
Variable	Accuracy
Inclination	
Argument of Periapsis	
Right Ascension of Ascending Node	
Apogee	
Perigee	
Right Ascension of Launch Asymptote	
Declination of Launch Asymptote	
C ₃	

Table D1-H: Standard Insertion Accuracy Requirements
(Example Only – Contractor to Specify Format and Data)

4.0 PAYLOAD SEPARATION ATTITUDE ACCURACY AND RATES

Standard payload deployment attitudes/rates shall be provided in Table D1-I. Errors are defined for each vehicle configuration and consist of all known dispersion sources (guidance, navigation, ACS limitations, etc.) Values provided reflect the generic capabilities of the launch vehicle, and all assumptions made with respect to payload mass properties shall be stated.

Vehicle Configuration	Errors	
	Attitude	Rate

Table D1- I: Standard Payload Deployment Attitudes/Rates (TBP)

5.0 PAYLOAD FAIRING (PLF)

The Contractor shall provide a diagram of the maximum payload static envelope and the associated dynamic envelope for a rigid payload hard mounted to the PA. *A separate diagram for each PLF/PA combination specified in paragraph 1.0 shall be provided.*

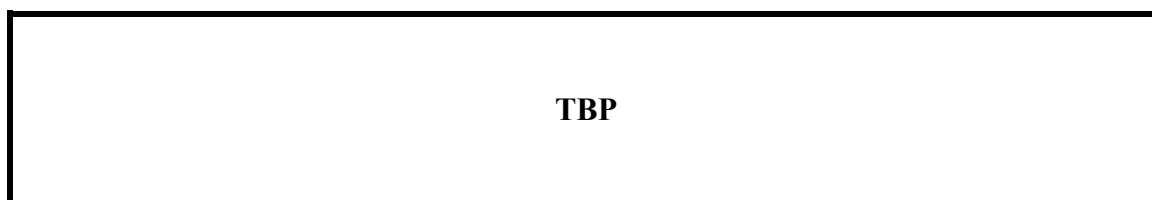


Figure D1-1: Maximum Payload Static Envelopes

6.0 MECHANICAL INTERFACE

The Contractor shall provide a diagram that illustrates the various payload adapters and associated interfaces available through the launch service. *A separate diagram for each adapter shall be provided.*

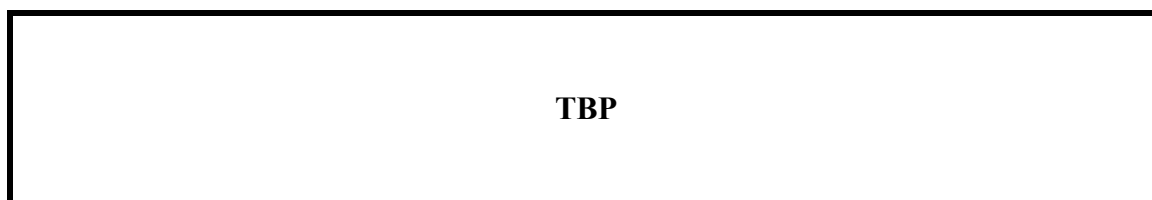


Figure D1-2: Payload Adapters and Interfaces

7.0 ELECTRICAL INTERFACE

The Contractor's payload electrical interface shall provide for payload separation sensing by the payload, and shall provide **TBP** interface connections between the payload and payload GSE for the purpose of providing power, command and control, and payload status monitoring during launch processing and liftoff. The characteristics of these wires shall be as specified in Table D1-J.

Function	Number of Wires	Peak Voltage (VDC)	Continuous Current (Amps)	Resistance from GSE to Payload (Ohms)	Shielding Coverage
Power	TBP	TBP	TBP	TBP	TBP
Data ¹	TBP	TBP	TBP	TBP	TBP
Continuity Sense	TBP	TBP	TBP	N/A	N/A

¹ Data lines shall use twisted, shielded pairs unless otherwise required by the payload.

Table D1-J: Electrical Interface Minimum Wire Requirements

For missions with low tip-off requirements, switches may be required on either the payload or launch vehicle. Mounting details will be specified in the ICD.

8.0 PAYLOAD ENVIRONMENT

The Contractor shall provide data for the maximum payload acoustic, vibration, shock, and acceleration, and thermal environment levels during launch and ascent. The maximum electromagnetic radiation levels encountered by the payload shall be depicted in Figure D1-3. The Contractor shall illustrate the pressure profiles and depressurization rates for each PLF in Figure D1-4. The maximum expected payload flight environments due to acoustics (50% fill), vibration, and shock shall be shown in Tables D1-K, D1-L, and D1-M, respectively. The maximum expected total acceleration (limit load factors), including uncertainty factors, shall be shown in Table D1-N (key dynamic events may be broken out as appropriate). The maximum expected sinusoidal vibrations at the base of the PA shall be shown in Table D1-O. The maximum quasi-steady acceleration as a function of payload separated mass shall be shown in Figure D1-5. The Contractor shall provide the maximum thermal environment during ascent for every launch component (i.e., upper stage, unblanketed skin, thermal/acoustic blankets, etc.) inside the fairing with a view factor to the payload in Figure D1-5A.

TBP

Figure D1-3: Maximum Electromagnetic Radiation Levels Encountered on Payload

TBP

Figure D1-4: PLF Pressure Profiles and Depressurization Rates

TBP

Table D1-K: Maximum Flight-Level Payload Acoustic Environment

TBP

Table D1-L: Maximum Flight-Level Payload Interface Random Vibration Environment
(If Applicable)

TBP

Table D1-M: Maximum Flight-Level Payload Interface Shock Response Spectrum

Axis	Maximum Acceleration (g)
Axial	TBP
Lateral	TBP

Notes:(1) Sign convention: positive axial acceleration produces compression
(2) Axial and lateral accelerations are simultaneous

Table D1-N: Payload Design CG Limit Load Factors

TBP

Table D1-O: Maximum Expected Sinusoidal Vibrations at the Base of the PA
(If Applicable)

TBP

Figure D1-5: Maximum Quasi-Steady Acceleration as a Function of Payload Separated Mass

TBP

Figure D1-5A: Maximum Thermal Environment During Ascent

9.0 PAYLOAD ENVIRONMENT INSTRUMENTATION

To support mission success determination (Section C, paragraph 24.2), the following flight instrumentation shall be provided to monitor the payload environment. Table D1-P establishes the baseline requirements for payload environment instrumentation.

Environment	Type	Number and Locations	Range, Sensitivity, and Bandwidth
Acoustics	TBP	TBP	TBP
Loads & Dynamics	TBP	TBP	TBP
Static Pressure	TBP	TBP	TBP
Temperature	TBP	TBP	TBP

Table D1-P: Contractor-Provided Payload Environment Instrumentation

10.0 GENERIC SECONDARY PAYLOAD CHARACTERISTICS/INTERFACE

The allowable characteristics of generic secondary payloads shall be specified in Table D1-Q. The mechanical interface, allowable mass properties and envelope shall be standardized for both separating and non-separating secondary payloads. The Contractor shall illustrate both the separating and non-separating secondary payload interfaces in Figures D1-6 and D1-7 respectively. Sufficient clearance between the secondary payload envelope and the launch vehicle shall be allowed so as to not require measurement of the secondary payload. A fit check of the secondary payload shall be performed. A generic separation system shall be provided for separating secondary payloads. The separation event shall be sequenced and controlled by the launch vehicle. The launch vehicle must be able to verify command sequencing.

The Contractor shall provide the capability for secondary payload battery trickle charge through an existing fairing access door. Charging may be required until fairing close-out. Charging equipment and cabling will be provided by the secondary payload agency.

Characteristics	Separating	Non-Separating
Weight/CG Distance from Separation Plane (not to exceed)	TBP	TBP
Volume (not to exceed)	TBP	TBP
Electrical Interfaces	TBP	TBP
Attachment	TBP (See Figure D1-6)	TBP (See Figure D1-7)
Coupled Frequency	TBP	TBP

Table D1-Q: Generic Secondary Payload Characteristics

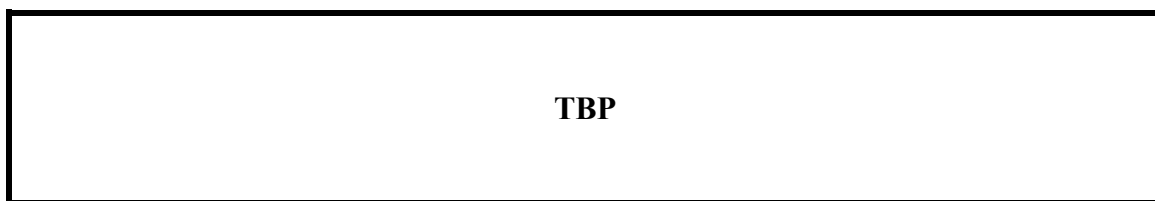


Figure D1-6: Separating Secondary Payload Interface

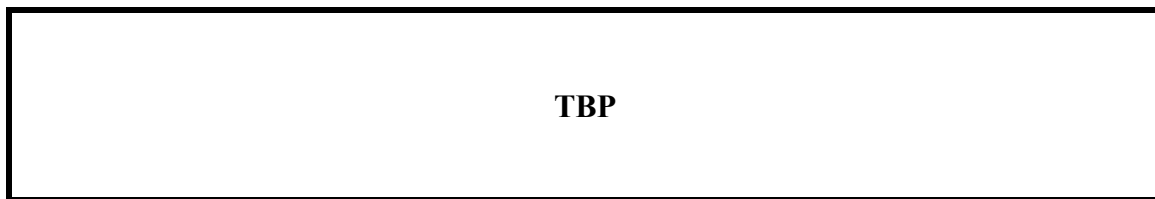


Figure D1-7: Non-Separating Secondary Payload Interface

EXHIBIT 2

STANDARD LAUNCH SERVICES LIST

The Contractor shall provide the following as part of their standard launch service. These services shall be performed for each mission in accordance with the contract terms and conditions. Each item shall include all hardware, software, analyses, and support required to perform the service. All capabilities and conditions in Exhibit 1 that pertain to the services listed herein shall apply.

TBP

EXHIBIT 3

NON-STANDARD SERVICES LIST

The Contractor shall provide the capability to perform the following **TBP** non-standard services. These services shall be performed for each mission, as requested and authorized by the NASA Contracting Officer, in accordance with the contract terms and conditions. Each non-standard item listed shall include all hardware, software, analyses, and support required to provide the service. Any impacts to the capabilities or conditions defined in Exhibit 1 introduced by the non-standard service shall be explained in detail. The Contractor shall not be required to design and develop a non-standard service in support of this contract.

The Contractor shall list all non-standard services provided for each proposed launch vehicle configuration, which will be negotiated and incorporated into the contract. All non-standard services provided shall be fully compliant with the SOW. Any non-standard services provided that are not fully compliant shall be designated as such in the proposal with details supplied as to the impact of making the service compliant. All proposed non-standard services shall have an associated, no-later-than ordering date with respect to the launch date. In addition, the Contractor shall provide any non-standard services deemed useful for future NASA missions based on past experience.

If authorized, the Contractor shall implement the following services:

- 1.0 Alternate Vehicle Configurations or Performance Enhancements** (i.e., Extended Nozzles, Additional Motors/Thrust Augmentation)
- 2.0 Alternate Fairing Configurations and/or Modifications**
 - 2.1 Alternate configurations
 - 2.2 Additional access doors in excess of two included in standard service
 - 2.3 Access doors of non-standard size
 - 2.4 Mounting provisions, cabling and antenna systems for re-radiating signals from the payload to remote sites
 - 2.5 S, C, X, and Ku-band re-radiation equipment
 - 2.6 RF-transparent doors
- 3.0 Alternate Payload Adapters (PAs)**
 - 3.1 Different size or different payload interface PAs
 - 3.2 Low tip-off rate PAs
- 4.0 Upper Stage Hardware**
 - 4.1 Motor
 - 4.2 PA
 - 4.3 Spin system with enhanced capability
 - 4.4 Despin system with enhanced capability
 - 4.5 Improved insertion accuracy options

5.0 Multiple PA, including Related Mission Integration Support

6.0 Secondary PA for Non-Separating Secondary Payload

7.0 Secondary PA/Separation System for Separating Secondary Payload

7.1 Secondary Payload Adapter (with S/C-provided separation system)

7.2 Secondary Payload Adapter (with contractor-provided separation system)

8.0 RF Through-the-Fairing Transmission Capability

9.0 Enhanced Electrical Interface

9.1 Increased capacity payload-to-GSE interface.

9.2 Launch vehicle command and control of payload

9.3 Launch vehicle supplied payload power capability

9.4 Enhanced telemetry capabilities including interleaving/deinterleaving of payload telemetry data

9.5 Additional launch vehicle environmental instrumentation

10.0 Launch Vehicle Mounted Cameras

Launch Vehicle Mounted Cameras including necessary lighting and the real time transmission of the video signal to the designated NASA LSP Data Center. Identification of potential ground coverage sites and coordination of video transmission between all ground sites and NASA LSP Data Center is to be included.

11.0 Special Contamination Control Options

11.1 Enhanced fairing environment

11.2 Enhanced fairing internal surface cleaning (IEST-STD-CC1246E level 500A)

11.3 Optional payload/vehicle integration environments

11.4 Additional instrument purge system(s) up to T-0 (IEST-STD-CC1246E level 100A)

11.5 Grade B Nitrogen (GN₂) IAW MIL-PRF-27401F or conditioned air cooling supply for spot cooling of payload components from payload mate or encapsulation up to T-0.

11.6 S/C Grade C Nitrogen (GN₂) IAW MIL-PRF-27401F Purge System

11.7 Enhanced PLF and PLA Cleaning IAW IEST-STD-CC1246E level 500A. The internal surfaces of the payload fairing compartment (including Payload Adapter) shall be cleaned, certified and maintained to IEST-STD-CC1246E Level 500A or better, with the number and locations of samples to be determined by the integration team. Cleaning and certification shall be accomplished in a clean room rated at or above Class 8 IAW ISO14644-1. In lieu of IEST-STD-CC1246E Level 500A, the payload fairing compartment (including Payload Adapter) may be cleaned and certified such that contribution from launch vehicle sources measured from S/C turnover (Launch Vehicle Contractor control of S/C for integrated operations) through CCAM shall not exceed 0.5% total obscuration and <1mg/0.1m² Nonvolatile Residue (NVR). The sampling methodology for particulate shall be per ASTM 1216 and for NVR per ASTM 1235 with the number and locations of samples to be determined by the integration team.

11.8 Certified Portable Class 7 Clean Enclosure(s) for Open PLF Access Doors

12.0 Reserved

13.0 Co-Manifested Payload Mission Service

14.0 NASA Secondary Payload Mission Service

15.0 Payload Compatibility Assessment

The Contractor shall provide a document compiling all compatibility analyses performed on the co-manifested or secondary payload to verify the co-manifested or secondary payload does not unacceptably impact the primary. The specifics of the different analyses performed will vary from mission to mission and depend greatly on the nature of both primary and secondary payloads. Some items which are common elements of the document are: combined coupled loads analysis, critical clearance analysis, interface failure modes and effects analysis, safety assessment, combined thermal analysis, and mass properties/GN&C analysis.

16.0 Secondary Payload Mission Feasibility Study

A secondary payload mission feasibility study shall assess the performance capability and payload volume available to accommodate secondary payload missions on a non-interference basis with a given primary mission. Separating and non-separating secondary payloads shall be accommodated as a non-standard service.

17.0 Secondary Payload Accommodation Study

A secondary payload accommodation study shall address the suitability of and the approach for accommodating secondary payloads on a given vehicle configuration. The study shall develop a design, general payload requirements (secondary payload interface, volume, mass, environments) and implementation scheme for implementing both separating and/or non-separating secondary payloads. Dynamic loading analysis for generic secondary payloads shall be performed so that mission unique analyses are minimized.

18.0 Secondary Payload Mass Simulator

A mass simulator shall be designed and fabricated to match the requirements (e.g., moment of inertia, rigidity, mass, and CG) of the secondary payload that will be replaced. The mass simulator shall also adhere to the requirements (e.g., contamination, cleanliness, and interface) identified in the ICD. The Contractor shall complete any other analyses or documentation required, for replacement of the secondary payload with the mass simulator.

19.0 Payload Processing Facilities

- 19.1 The Contractor shall provide the capability to provide ground processing facilities for payloads at the Eastern Range.
- 19.2 The Contractor shall provide the capability to provide ground processing facilities for payloads at the Western Range.
- 19.3 The Contractor shall provide the capability to provide ground processing facilities for payloads at the Wallops Flight Facility.

20.0 Additional Mission Analysis and Support

- 20.1 Reserved
- 20.2 Support for additional mission integration working group meetings in excess of four per calendar year.
- 20.3 Reserved
- 20.4 Performance and Guidance Accuracy Analysis
- 20.5 Coupled Dynamic Loads Analysis
- 20.6 Pre-Flight Controls and Stability Analysis
- 20.7 Integrated Thermal Analysis
- 20.8 RF Link Analysis
- 20.9 Payload Fairing Clearance Analysis
- 20.10 Payload/Expended Stage Separation Analysis
- 20.11 Analysis of ECS flow impingement on payload surfaces
- 20.12 RF Resonance Analysis to determine maximum RF field strengths that payload and launch vehicle avionics would be exposed to if spacecraft RF transmission systems were activated while encapsulated within the fairing.
- 20.13 Advanced mission performance analysis support (limited feasibility assessments in response to new mission concepts)
- 20.14 Launch vehicle Collision Avoidance (COLA) trajectory and covariance data.
- 20.15 Launch vehicle flight software code and associated documentation.

21.0 Reserved

- 21.1 Reserved
- 21.2 Reserved
- 21.3 **Reserved**

22.0 NASA Public Affairs Office Support

Accommodations for NASA guests, in units of 50, to view launch, including local transportation, launch viewing facilities, lavatory accommodations, food and beverage service, first aid and medical support for each proposed launch site.

23.0 Launch Service Contractor (LSC) provided Telemetry

The LSC shall provide all telemetry data (main vehicle telemetry and special instrumentation/wideband telemetry system) IAW *LSP-PD-120.05, "Launch Telemetry Requirements."*

24.0 Launch Service Contractor (LSC) provided Telemetry Acquisition Assistance Message (AAM)

The Contractor shall provide Acquisition Assistance Messages (AAMs) to the down-range telemetry sites utilized by the mission for antenna pointing to acquire launch vehicle telemetry and to payload-designated control centers/telemetry sites for acquisition of space vehicle telemetry. The Government utilizes down-range telemetry sites to receive record and relay telemetry data from the Contractor's launch vehicle in accordance with *LSP-PD-120.05, "Launch Telemetry Requirements."* Mission-specific telemetry sites are selected to acquire data from the launch vehicle during the launch, powered-flight, and spacecraft separation phases of the mission. These telemetry sites may be owned/operated by NASA, the DoD, a U.S. commercial entity, or a foreign entity.

Generation of the AAMs will be treated as no greater than REQUIRED capability for launch commit criteria purposes. Should any greater capability be imposed, then the requirements to meet that greater capability shall be considered outside the scope of this NSS and subject to a separate contract action.

Format for AAMs will be in accordance with industry standard.

25.0 TDRSS Compatible Telemetry System

Design, build, qualify, integrate and test a TDRSS compatible telemetry system in the Launch Vehicle. After first flight, the TDRSS compatible telemetry system will be a standard launch service capability provided for NASA missions.

26.0 Extended Integration Periods

- 26.1 Three Month Extended Integration Period
- 26.2 Six Month Extended Integration Period
- 26.3 Nine Month Extended Integration Period
- 26.4 Twelve Month Extended Integration Period

27.0 Contractor Defined Non-Standard Services

The Contractor may propose other services that are not listed.

EXHIBIT 4

SAMPLE CERTIFICATE OF FLIGHT READINESS

**FLIGHT CERTIFICATION DOCUMENT
FOR THE (TBD) MISSION**

The following agencies certify that the necessary launch test operations to date have been completed for the launch vehicle and payload. Pending satisfactory closure of in-line work and any open items identified in the Launch Readiness Review, the launch vehicle, payload, and all supporting systems are ready for the launch operation.

(Launch Service Contractor)

Launch Services Program Chief Engineer
NASA / KSC

Chief Safety Officer
NASA

Launch Services Program Manager
NASA / KSC

S/C Project Manager
NASA / **TBP** Center

Launch Services Program Launch Director
NASA / KSC

The following Government agencies, involved in supporting this **(TBD)** mission, certify that all applicable requirements to date have been met and that all Government support elements are making final preparations to support this mission. Pending satisfactory closure of in-line work to prepare for support of this mission and any open items identified in the Launch Readiness Review, all Government support elements are ready for the launch operation. Furthermore, pending satisfactory closure of in-line launch vehicle and observatory processing and any open items identified in the Launch Readiness Review, the **(TBD)** Space Wing foresees no national resource protection or Range Safety issues which would preclude launch.

(TBD) Space Wing

Date

EXHIBIT 5

RESERVED

Table D1-R: Reserved

Table D1-S: Reserved

CLINS 1 THROUGH 10 DATA SHEETS

RESERVED

EXHIBIT 6

RESERVED

Table D1-Y: Reserved

IDIQ 1 THROUGH 10 DATA SHEETS

RESERVED

EXHIBIT 7

LAUNCH VEHICLE CERTIFICATION DATA

The following data provides the necessary information to establish the common launch vehicle configuration certification level, either Category 2 or 3, for every common launch vehicle configuration offered.

Refer to Attachment E2, Section 14, Launch Vehicle Certification Criteria, for data submission requirements.

TBP

EXHIBIT 8

MISSION INTERFACE CONTROL DOCUMENTS (ICD)

The baseline ICD (CDRL C2-1.1) will be incorporated by reference into Exhibit 8 of the contract via contract modification upon approval of the baseline ICD. The final revision to the ICD (CDRL C2-1.1), including revisions and waivers approved after the ICD is baselined, will be incorporated by reference into Exhibit 8 of the contract via contract modification prior to the mission success determination. The Contractor shall comply with the latest approved revision of the ICD, to include all approved changes and waivers.

EXHIBIT 9

LAUNCH SERVICE TASK ORDER MISSION SOLUTIONS

Launch service task order mission solutions will be attached to the contract upon award of task order(s). In accordance with Section C, Clause 1.0(s), Order of Precedence, any inconsistencies between the ICD (Exhibit 8) and the Mission Solution (Exhibit 9) shall be resolved by giving precedence to the ICD over the Mission Solution once the ICD is incorporated by reference into the contract.

TBD

ATTACHMENT D2

CONTRACT DATA REQUIREMENTS LIST (CDRL)

The Contract Data Requirements List (CDRL) identifies critical elements of the contracted effort where NASA requires aspects of mission integration insight and approval. The following CDRL defines the scope of documentation required; however, NASA will utilize the Contractor's existing documentation to the extent practicable. Where there is not a direct match between a CDRL item and the Contractor's standard documentation, the Contractor's documentation will be acceptable provided it contains equivalent data requirements. CDRL approval may be assumed unless the Contractor is notified by NASA of disapproval within thirty (30) days. Under certain circumstances, NASA may elect to eliminate certain submittal cycles.

The number of copies listed in Table D2-A represents the number of hard copies to be delivered to NASA only if the CDRL data is not available electronically, except CDRLs C1-1, C1-2, C1-3, C5-4, and C5-5 which must also be delivered in hard copy. All electronic formats shall be mutually acceptable. All data requirements shall be delivered to the KSC ELV Library. The Contractor shall notify the Contracting Officer in writing of CDRL delivery.

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C1: Documentation				
C1-1*	Formal Meeting & Review Documentation	Approval (B & E(vi); Review all other	(A)(i) at Review minus 1W; (A)(ii) at Review; (B,C,D,E) Review plus 1W	A/R	A/R
C1-2*	Countdown Documentation	Approval (A,B); Review (C)	L-1W	As Changed	5
C1-3*	Mission Console Notebook	Review	24 hours prior to Mission Dress Rehearsal	As Changed	100
C1-4	Manifest Policy	Review	Contract Award	As Changed	5
C1-5*	Operations Documentation for Processing in NASA Provided Payload Processing Facilities	Review	A/R in CDRL	As Changed	2

* Required per mission

TABLE D2-A: CDRL INDEX

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C2: Mission Integration				
C2-1.1*	Interface Control Document (ICD)	Approval	Preliminary ICD at ATP + 6M	Baseline ICD at L-15M and A/R	10
C2-1.2*	Interface Control Document (ICD) Requirements Verification Matrix	Approval	NLT 3 months after Preliminary ICD CDRL approval	A/R or with each ICD revision	3
C2-2*	Mission Specific Program Requirements Document/ Operational Requirements (PRD/OR)	Approval	A/R	As Changed	3
C2-3*	Launch vehicle Detailed Mission Requirements (DMR)	Approval	A/R	As Changed	3
C2-4*	Reserved				
C2-5*	Mission Success Criteria & Determination Methodology	Approval	NLT L-4M	As Changed	5

* Required per mission

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C3: Safety & Mission Assurance				
C3-1	Alert System Documentation	Review	Dispo + 1M	A/R	2
C3-2	Mishap Notification, Investigation and Mishap Preparedness, and Contingency Plan	Review	A/R in CDRL	A/R	2
C3-3	AS9100 Certification Certificate	Review	Upon Recertification	Upon Recertification	1
C3-4	Quality Management System (QMS) and Supplier/Vendor Audit/Assessment Schedules	Review	Contract Award +45 Days	When Revised	2
C3-5	QMS and Supplier/Vendor Audit/Assessment Report	Review	As Generated	Per Internal or External Audit	2
C3-6	Risk Management Plan	Approval	At Contract Award Plus 90 Days	Annual Review And Updates A/R	2

				1 Electronic and 1 Paper Copy	
C3-7*	Safety Documentation for Processing In NASA Provided Facilities	Approval for NASA owned facilities; Review for NASA procured facilities	A/R in CDRL	A/R	2

* Required per mission

TABLE D2-A: CDRL INDEX (Cont.)

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C4: Analysis				
C4-1*	Performance and Guidance Accuracy Analysis (PGAA)	Initial Submittal = Review, Subsequent Submittals = Approval	ATP+ TBP	L-12M; L-6M	3
C4-2*	Final Mission Analysis (FMA)	Approval	Receipt of P/L Data + TBP	A/R	3
C4-3*	Payload/Expended Stage Separation Analysis	Approval	L-TBP	L-TBP	3
C4-4*	Payload Fairing Venting Analysis	Approval	Receipt of P/L Data + TBP	A/R	3
C4-5*	Payload Fairing Clearance Analysis	Approval	Receipt of P/L Model + TBP	L-TBP ; L-TBP	3
C4-6*	Pre-Flight Controls and Stability Analysis	Approval	L-TBP	A/R	3
C4-7*	Coupled Dynamic Loads Analysis	Approval	Receipt of P/L Model + 3M	L-15M L-26W	3
C4-8*	Integrated Thermal Analysis	Approval	L-TBP	A/R	3
C4-9*	RF Link Analysis	Approval	L-12M	A/R	3
C4-10*	Payload/Launch Vehicle EMC, RF, and EED Compatibility Analysis	Approval	(B,C,D) L-12M (A & E) A/R in CDRL	(B,C,D) L-3M (A & E) A/R in CDRL	3
C4-11*	Mission Unique Software	Approval	Preliminary = L-TBP ; Final = L-TBP	A/R	3

C4-12*	Quick Look Flight Report	Review	L+ TBP	N/A	3
C4-13*	Final Flight Report	Review	L+ TBP	A/R	3

* Required per mission

TABLE D2-A: CDRL INDEX (Cont.)

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C5: Engineering				
C5-1*	Vehicle Data Package	Review	TBP	As Changed	4
C5-2*	Mission Unique Hardware Test Plan	Approval	ATP + TBP	As Changed	4
C5-3*	Mission Unique Hardware Test Report	Review	Test Completion + TBP	As Changed	4
C5-4*	Mission Specific Drawings	Review	As Generated per design review criteria, but no later than 30D prior to fabrication, installation, assembly	As Changed	4
C5-5*	Integrated Procedures	Approval	Review and comment, 45D Prior to Use; Approval, 1W Prior to Use	A/R	5

* Required per mission

Item	Document	Approval/ Review	Initial Submittal Date	Subsequent Submittal Date	No. of Copies
	C6: Telemetry and Communications				
C6-1*	Vehicle and GSE Telemetry Formats	Review	L-3M	As Changed with Configuration	2
C6-2*	Vehicle and GSE Telemetry (Test, Pre-Launch, and Launch)	Review	Test + 1D; Pre-Launch + 1D; Launch + 1W	N/A	1
	C7: Miscellaneous				
C7-1	Small Business and Small Disadvantaged Business Subcontracting Plan	Approval	A/R in CDRL	A/R in CDRL	1

* Required per mission

TABLE D2-A: CDRL INDEX (Cont.)

CDRL DESCRIPTION

Number: C1-1

FORMAL MEETING & REVIEW DOCUMENTATION

DESCRIPTION/PURPOSE:

To provide presentation and handout material, minutes, and accompanying action item lists from each formal review.

DATA REQUIREMENTS:

Formal review documentation shall include:

- (A) Presentation packages and hand-out materials provided at the review for NASA attendees
 - (i) Mission Specific Design Reviews
 - (ii) All other formal reviews
- (B) Minutes from each formal review
- (C) List of attendees
- (D) Agreements from each formal review
- (E) An action item list maintained and updated by the Contractor. The list shall include:
 - (i) Due Date
 - (ii) Subject
 - (iii) Assigned tracking number for each action item
 - (iv) Person or organization responsible for completing the action
 - (v) Status of action (i.e., open, closed, or deleted)
 - (vi) Closure rationale

CDRL DESCRIPTION

Number: C1-2

COUNTDOWN DOCUMENTATION

DESCRIPTION/PURPOSE:

To provide the documents necessary to launch the vehicle and also assist NASA management in determination of a GO or NO-GO.

DATA REQUIREMENTS:

Appendices for the documents listed below shall be provided when necessary to present supplemental or incidental information, detailed tabulations or deviations, or graphic representations.

(A) Countdown

The Countdown document shall provide:

- (i) All necessary steps and built-in holds required to successfully launch the vehicle and its payload during the launch window
- (ii) Procedures required to safe the vehicle after a launch attempt
- (iii) Procedures required to recycle the countdown for another launch attempt

The necessary steps shall be listed as a sequence of events with each entry including a T-time. T-time is defined as the time before launch, not including planned holds, and with T-0 being the time of launch vehicle lift-off from the launch pad. T+ time is any event after T-0. Each member of the launch team shall be identified by call sign and communication channel. All mission constraints to launch shall also be identified.

(B) Mission Constraints Document

This document is a summary of all the launch vehicle, payload and range mission constraints to launch. The launch vehicle constraints shall include, but not be limited to:

- (i) Weather
- (ii) Communication
- (iii) Range requirements
- (iv) Collision avoidance (COLA)
- (v) Tracking and telemetry

The document shall also identify critical measurements during the launch countdown. Not to exceed values shall be identified for all critical measurements.

(C) Redline Document

A document that identifies monitored measurements, applicable in test and countdown, used to verify that the Launch Vehicle and ground systems are operating within normal limits. This

document shall include Operational and Redline limits which, if specified parameters minimum or maximum values are exceeded, indicate: a condition that may be hazardous to personnel, vehicle, or site; a condition which may compromise the successful completion of the vehicle mission or; a condition that is outside of the expected range based on previous experience or analysis.

CDRL DESCRIPTION

Number: C1-3

MISSION CONSOLE NOTEBOOK

DESCRIPTION/PURPOSE:

A mission console notebook to assist on-console users during pre-count, terminal count, and plus count operations.

DATA REQUIREMENTS:

As a minimum, the notebook shall include:

- (A) Terminal or abbreviated countdown procedure
- (B) Flight sequence of events
- (C) Range, payload, launch vehicle mandatory items
- (D) Communication channel matrix
- (E) Down-range telemetry coverage plan
- (F) Launch window for all possible attempts
- (G) Launch decision flowchart
- (H) Control center seating
- (I) Call sign matrix
- (J) Protocol for calling holds
- (K) Day of launch winds plan

CDRL DESCRIPTION

Number: C1-4

MANIFEST POLICY

DESCRIPTION/PURPOSE:

The contractor shall provide at a minimum: the definition of standard setback guidelines (including setback guidelines against other product lines that are offered to other non-NASA customers), manifest separation ground rules, priorities, and assumptions, or equivalent.

DATA REQUIREMENTS:

Documentation shall be delivered in a mutually acceptable format.

CDRL DESCRIPTION

Number: C1-5

OPERATIONS DOCUMENTATION FOR PROCESSING IN NASA PROVIDED PAYLOAD PROCESSING FACILITIES

DESCRIPTION/PURPOSE:

This item covers the submission of documentation to NASA when performing operations in a NASA provided facility in support of this contract.

DATA REQUIREMENTS:

The following items shall be submitted for the launch vehicle hardware and operations to be brought in a NASA provided PPF.

- Launch Vehicle facility, processing, security, contamination requirements and processing contingency plans.
 - Initial submittal date 18 months prior to launch.
- Documentation of any hazardous waste that the launch vehicle will produce in the facility.
 - Initial submittal date sixty (60) days prior to arrival.
- Non-hazardous Launch vehicle standalone procedures.
 - Initial submittal date forty-five (45) days prior to use.
- Launch vehicle personnel badging and training requirements to include any medical certifications.
 - Initial submittal date ninety (90) days prior to arrival.

CDRL DESCRIPTION

Number: C2-1.1

INTERFACE CONTROL DOCUMENT (ICD)

DESCRIPTION/PURPOSE:

The ICD identifies and defines the functional and environmental interfaces, and performance requirements of the payload, and payload GSE with the launch vehicle and associated payload/launch processing facilities. The baseline ICD (CDRL C2-1.1) will be incorporated by reference into Exhibit 8 of the contract via contract modification upon approval of the baseline ICD. The final revision to the ICD (CDRL C2-1.1), including all approved changes and waivers, will be incorporated by reference into Exhibit 8 of the contract via contract modification prior to the mission success determination. The Contractor shall comply with the latest approved revision of the ICD, to include all approved changes and waivers.

Preliminary ICD is the first CDRL release of the ICD and is expected to have several TBD's. Baseline ICD is the next release of the CDRL and is expected to have very few TBD's.

DATA REQUIREMENTS:

The number of interface requirement changes incorporated in each new revision and subsequent release of the ICD shall not exceed ten (10) approved requirements changes.

Where there is not a direct match between the specified ICD data requirements identified below and the Contractor's standard ICD, the Contractor's standard ICD will be acceptable provided it addresses equivalent content:

1.0 INTRODUCTION

- 1.1 Purpose
- 1.2 Scope
- 1.3 Definitions
- 1.4 Interface Change Control
 - 1.4.1 ICD Change Procedure
 - 1.4.2 Interface Requirement Discrepancy
- 1.5 Requirement Traceability
- 1.6 Requirements Status

2.0 APPLICABLE DOCUMENTS

- 2.1 Government Documents
- 2.2 Contractor Documents
- 2.3 Other Documents

3.0 INTERFACE REQUIREMENTS

- 3.1 Structural /Mechanical Interfaces
 - 3.1.1 Structural Interfaces
 - 3.1.1.1 Static Payload Envelope
 - 3.1.1.2 PLF Clearance Requirements
 - 3.1.1.3 SC Clearance Requirements
 - 3.1.1.4 Separation Clearance Requirements
 - 3.1.1.5 SC Access Requirements
 - 3.1.2 Structural Loads
 - 3.1.2.1 SC Stiffness
 - 3.1.2.2 SC Interface Loads
 - 3.1.2.3 Strength
 - 3.1.2.4 SC Load Factors
 - 3.1.3 Mass Properties
 - 3.1.3.1 SC Mass Properties
 - 3.1.3.2 SC Propellant Data
- 3.2 Electrical Interfaces
 - 3.2.1 Airborne Interfaces
 - 3.2.1.1 Standard Electrical Interfaces
 - 3.2.1.2 Separation Indication
 - 3.2.1.3 Interface Electrical Constraints
 - 3.2.1.4 SC Separation System Electrical Interface
 - 3.2.1.5 SC to LV Interleaved Telemetry
 - 3.2.2 SC Umbilical Interface
 - 3.2.2.1 T-0 Umbilical Interface
 - 3.2.2.2 EGSE Interface Electrical Constraints
 - 3.2.3 EGSE Telemetry/Command/Data Link
 - 3.2.3.1 EGSE Hardline Links
 - 3.2.3.2 SC Interleaved Data
 - 3.2.4 Electrical Grounding
 - 3.2.4.1 Spacecraft Grounding
 - 3.2.4.2 Support Equipment Grounding
 - 3.2.4.3 Personnel Grounding
 - 3.2.4.4 Ground Continuity
- 3.3 Environmental Interfaces

- 3.3.1 Thermal
 - 3.3.1.1 SC Thermal Environments
- 3.3.2 Contamination
 - 3.3.2.1 Payload Fairing Purge
 - 3.3.2.1.1 Spacecraft Transport to the Launch Complex
 - 3.3.2.1.2 Spacecraft at Launch Complex
 - 3.3.2.2 GHe Concentration
 - 3.3.2.3 Payload Compartment Cleanliness
 - 3.3.2.3.1 Non-Volatile Residue
 - 3.3.2.3.2 PLF Inspection
 - 3.3.2.3.3 PLA Inspection
 - 3.3.2.3.4 Witness Plate Access
 - 3.3.2.3.5 Spacecraft Cleanliness
 - 3.3.2.4 Spacecraft Contaminant Deposition Limits
 - 3.3.2.4.1 Particulate
 - 3.3.2.4.2 Molecular
 - 3.3.2.5 Spacecraft Instrument GN₂ Purge
- 3.3.3 Ascent Pressure
- 3.3.4 Dynamics
 - 3.3.4.1 Sine Vibration
 - 3.3.4.2 Acoustics
 - 3.3.4.3 Shock
- 3.3.5 Electromagnetic Compatibility
 - 3.3.5.1 EMI Safety Margin
 - 3.3.5.2 Radiated Emissions
 - 3.3.5.2.1 Spacecraft Maximum Allowable Radiated Emissions
 - 3.3.5.2.2 Launch Vehicle Maximum Allowable Radiated Emissions
 - 3.3.5.2.3 SC RF Susceptibility
 - 3.3.5.2.4 Launch Site RF Environment
 - 3.3.5.2.5 Portable Sources Radiated Emissions
 - 3.3.5.3 PLF Electrostatic Discharge
 - 3.3.5.4 Thermal Blanket Electrostatic Discharge
 - 3.3.5.5 RF Transmitter/Receiver Systems EMC
 - 3.3.5.6 Lightning
 - 3.3.5.7 SC/LV Interface Electrical Bonding

- 3.3.6 Mission Satisfaction Kit
- 3.4 Flight Design
 - 3.4.1 Contract Orbit Requirements
 - 3.4.1.1 Mission Orbit Specification
 - 3.4.1.2 Reference Mission
 - 3.4.2 Launch Window
 - 3.4.3 SC Separation
 - 3.4.4 Sun Angle Constraint
 - 3.4.5 Thermal Roll
 - 3.4.6 LV Telemetry Data
 - 3.4.6.1 SC Separation State Vector
 - 3.4.6.2 SC Provided Post Separation Data
- 3.5 Ground Operations
 - 3.5.1 Spacecraft Contractor Supplied Ground Support Equipment
 - 3.5.2 Payload Processing Facility
 - 3.5.3 Launch Complex Ground Operations
 - 3.5.3.1 GSE
 - 3.5.3.2 Ground Operations Communications
 - 3.5.3.2.1 Countdown Clock
 - 3.5.3.2.2 Closed Circuit TV
 - 3.5.3.2.3 Public Address
 - 3.5.3.2.4 Telephone
 - 3.5.3.2.5 Operating Voice System
 - 3.5.3.2.6 Contingency Operations
 - 3.5.4 SC Transport
 - 3.5.4.1 Processing Facility to Launch Complex Transport
 - 3.5.4.2 SC Hoist and Mate Loads
 - 3.5.5 Environmental Control System
 - 3.5.5.1 PLF Environmental Control System
 - 3.5.6 Fluids And Gases Requirements
- 3.6 Safety Requirements

CDRL DESCRIPTION

Number: C2-1.2

INTERFACE CONTROL DOCUMENT (ICD) REQUIREMENTS VERIFICATION MATRIX

DESCRIPTION/PURPOSE:

A verification matrix will document and track all ICD requirements and methods by which they will be verified and responsible organization. The verification matrix shall list the final documentation of the verification approved by the cognizant LSC engineer.

DATA REQUIREMENTS:

The verification matrix shall be released no later than three (3) months after the Preliminary ICD CDRL approval and shall be updated upon release of each ICD revision.

Where there is not a direct match between the specified ICD Verification Matrix data requirements identified below and the Contractor's standard ICD Verification Matrix, the Contractor's standard ICD Verification Matrix will be acceptable provided it addresses equivalent content:

- ICD requirement language
- ICD requirement paragraph number
- Verification Method (Test, Inspection, Analysis, Demonstration)
- Responsible Organization
- Verification Plan
- Verification Source
- Notes/Comments
- Status

CDRL DESCRIPTION

Number: C2-2

MISSION SPECIFIC PROGRAM REQUIREMENTS DOCUMENT/OPERATIONAL REQUIREMENTS (PRD/OR)

DESCRIPTION/PURPOSE:

The Mission Specific Program Requirements Document (PRD) and Operational Requirements (OR) identifies specific support services provided by the Range. Range support services required by NASA and the Contractor are submitted to the Range through the PRD and OR.

DATA REQUIREMENTS:

The Mission Specific PRD/OR, or equivalent mission specific Range support request documentation for the proposed launch site(s), shall be submitted to obtain Range support. The Contractor, with support from NASA/KSC, completes all forms appropriate to the mission in the format required by the Range and submits them to the appropriate Range for formal acceptance.

CDRL DESCRIPTION

Number: C2-3

LAUNCH VEHICLE DETAILED MISSION REQUIREMENTS (DMR)

DESCRIPTION/PURPOSE:

The Launch Vehicle DMR documents formal requests for tracking, communications, and data system services that are either managed by or negotiated through the Exploration, Operations, Communication and Navigation Systems Division (Code 450) at Goddard Space Flight Center (GSFC).

DATA REQUIREMENTS:

The DMR, consisting of sets of Universal Documentation System (UDS) standard forms (with associated instructions), shall be completed to obtain network support from NASA for the baseline launch vehicle. The Contractor, with the support of NASA/KSC, completes all forms appropriate to the vehicle and submits them to NASA/KSC for review and GSFC Code 450 for formal acceptance. The DMR shall follow the format provided in the NASA/GSFC Instruction Manual for the NASA DMR document.

CDRL DESCRIPTION

Number: C2-4

RESERVED

CDRL DESCRIPTION

Number: C2-5

MISSION SUCCESS CRITERIA & DETERMINATION METHODOLOGY

DESCRIPTION/PURPOSE:

This data item proposes the criteria and methodology to support the Contracting Officer's determination of mission success.

DATA REQUIREMENTS:

- (A) A list of proposed criteria to be used to determine the success of the mission. These criteria shall be based on applicable flight related ICD requirements.
- (B) The Contractor's proposed methods, including required vehicle telemetry measurements, ground observation, and analysis/reconstruction techniques, for verifying that each proposed mission success criterion is satisfied. Allowances for sensor measurement accuracy shall be explicitly identified and justified.

CDRL DESCRIPTION

Number: C3-1

ALERT SYSTEM DOCUMENTATION

DESCRIPTION/PURPOSE:

A Government Industries Data Exchange Program (GIDEP) Alert Report will be generated for matches of all hardware covered by this contract.

DATA REQUIREMENTS:

- (A) This report will identify all matches of hardware provided under this contract.
- (B) The report shall identify the GIDEP Alert number, flight hardware impacted, analysis of the problem, and any corrective action required.
- (C) The report shall include results from any subcontractor GIDEP Alert reviews.

CDRL DESCRIPTION

Number: C3-2

MISHAP NOTIFICATION, INVESTIGATION AND MISHAP PREPAREDNESS, AND CONTINGENCY PLAN

DESCRIPTION/PURPOSE:

To provide NASA with an understanding of the Contractor's processes notifying NASA of mishaps, as well as how the Contractor plans to investigate and establish corrective action plans for mishaps that occur while performing the work required in support of this SOW. Contractor employees on NASA-owned property, custodian of NASA assets elsewhere to the extent those assets are involved, or contractors involved in payload to launch vehicle integrating operations, shall report mishaps or close calls. Mishap Preparedness and Contingency Plan (MPCP) describes the immediate actions required in response to a mishap.

DATA REQUIREMENT:

a. GENERAL:

Mishap reporting, investigation and corrective action requirements differ according to whether the specific mishap occurs on NASA property or on contractor property. Mishaps occurring on third party property will be handled in the same manner as those occurring on contractor property.

- a.1. MPCPs shall describe how the Contractor responds to mishaps described below. This plan shall identify, by name, and responsibility all contractor personnel with responsibilities for mishap response, data impoundment, and hardware impoundment (includes retrieval of mishap generated debris). It shall describe: impoundment process, notification of NASA and other authorities, public release of information, and debris collection process. Additionally it shall describe special procedures for emergency response and debris retrieval personnel and identify potential post mishap hazards along with PPE and other hazard mitigation procedures. MSDSs shall be made available to first responders if requested. The MPCP shall also contain a description how the contractor interfaces with the NASA IRT and the process for transferring impounded data and material to the investigation authority. A template MPCP plan shall be delivered no later than six months prior to the first launch and will be updated after each launch based on lessons learned. The mission unique MPCP shall be delivered no later than one week prior to launch.

CDRL DESCRIPTION

Number: C3-2 (cont)

2. NASA MISHAPS:

Classification Level & Investigation Type	<u>Property Damage</u>	<u>Injury</u>
Type A Mishap	Total direct cost of mission failure and property damage is \$1,000,000 or more, or Crewed aircraft hull loss has occurred, or Occurrence of an unexpected aircraft departure from controlled flight (except high performance jet/test aircraft such as F-15, F-16, F/A-18, T-38, OV-10, and T-34, when engaged in flight test activities).	Occupational injury and/or illness that resulted in: A fatality, or A permanent total disability, or The hospitalization for inpatient care of 3 or more people within 30 workdays of the mishap.
Type B Mishap	Total direct cost of mission failure and property damage of at least \$250,000 but less than \$1,000,000.	Occupational injury and/or illness has resulted in permanent partial disability. or The hospitalization for inpatient care of 1-2 people within 30 workdays of the mishap.
Type C Mishap	Total direct cost of mission failure and property damage of at least \$25,000 but less than \$250,000.	Nonfatal occupational injury or illness that caused any workdays away from work, restricted duty, or transfer to another job beyond the workday or shift on which it occurred.
Type D Mishap	Total direct cost of mission failure and property damage of at least \$1,000 but less than \$25,000.	Any nonfatal OSHA recordable occupational injury and/or illness that does not meet the definition of a Type C mishap.
Close Call	An event in which there is no equipment/property damage or minor equipment/property damage (less than \$1000), but which possesses a potential to cause a mishap.	An event in which there is no injury or only minor injury requiring first aid, but which possesses a potential to cause a mishap.

CDRL DESCRIPTION

Number: C3-2 (cont)

3. ONSITE MISHAPS OCCURRING ON NASA PROPERTY:

Any unplanned occurrence, event, or anomaly involving NASA personnel, equipment, or facilities that meet one of the mishap classifications as defined above shall follow the Mishap and Close Call Reporting, Investigating, and Recordkeeping procedural requirements as documented in NPR 8621.1. For occurrences that meet one of the mishap classifications as defined above and occurring on NASA KSC owned facilities, launch sites or equipment, the Mishap and Close Call reporting, investigation and recordkeeping procedural requirements as documented in KNPR 8715.3 shall be followed.

4. MISHAPS OCCURRING ON CONTRACTOR PROPERTY, THIRD PARTY PROPERTY OR OTHER NON-NASA GOVERNMENT PROPERTY:

4.1 Immediate Notification and Reporting of Mishaps Occurring on Contractor Property, Third Party Property or Other Non-NASA Government Property.

- a. **Telephonic Report:** The Contractor shall notify the NASA Launch Services Division Safety Office within one hour of the occurrence of a Type A mishap, Type B mishap, high-visibility mishap, or high-visibility close call that involves NASA personnel or property while performing work in support of this SOW. For Type C or D Mishaps and Close Calls involving NASA personnel or property, the contractor shall notify the Launch Services Division Safety Office within four (4) hours of the occurrence.
- i. During this notification, the Contractor shall provide the following information: the Facility name, location of incident, time of incident, number of fatalities (if known), number of hospitalized employees (if known), type of injury (if known), type of damage (if known), contact person, contact person's phone number, and a brief description of the mishap.
- b. **24-Hour Report:** The Contractor shall provide an initial incident report to the Launch Services Division Safety Office within twenty-four (24) hours of the incident for all Mishaps, or Close Calls involving NASA personnel or property while performing work in support of this SOW. The following information, as a minimum, shall be included: author of report; author's phone number and mail code; date report submitted; time report submitted; incident date; incident time; incident_general location; exact location (if known); responsible organization; organization's point of contact; point of contact's phone number and mail code; mission affected; program impact (if known); number and type of injuries or fatalities (if known); type of damage to equipment, flight hardware, flight software, or facilities; estimate of direct cost of damage; and a brief description of the mishap or close call.

CDRL DESCRIPTION

Number: C3-2 (cont)

- c. Notify OSHA: Within eight (8) hours of a work-related mishap involving death of a Federal employee, or the hospitalization for inpatient care of three or more employees (provided at least one is a Federal employee) the contractor shall notify OSHA by calling the area office nearest the site of the mishap or OSHA's toll-free number, 1.800.321.6742. OSHA notification is required for any fatality or three or more hospitalizations that occur up to thirty (30) workdays after the respective mishap. In notifying OSHA, the Contractor shall provide OSHA with the following information: the establishment name, location of incident, time of incident, number of fatalities (if known), number of hospitalized employees (if known), contact person, contact person's phone number, and a brief description of the mishap.
- 4.2 Mishap Investigation and Corrective Action for Mishaps on Contractor Property, Third Party Property, or Other Non-NASA Government Property.
- a. An initial investigation by the Contractor is required for all mishaps which have been reported to NASA. NASA reserves discretionary authority to investigate mishaps which involve NASA personnel or resources regardless of location. The Contractor has the discretion to perform any collateral investigations. However, investigations implemented by NASA will take priority with regard to access to evidence, data, and witnesses. The proceedings of NASA investigations will remain confidential. The Contractor will have an opportunity to comment on the investigation report in accordance with NASA protocols.
 - b. Contractor Final Report. After the Contractor completes its investigation of the mishap and, when appropriate, has developed a plan of corrective action, the Contractor shall deliver this completed report to NASA within fifteen (15) days of completion. This plan will include any verification activities identified by the Contractor to ensure that corrective actions have been implemented or, if applicable, accomplished.

CDRL DESCRIPTION

Number: C3-3

AS9100 CERTIFICATION CERTIFICATE

DESCRIPTION/PURPOSE:

The certificate that is provided by the Registrar upon successful completion of AS9100 certification.

DATA REQUIREMENTS:

After completion of AS9100 3rd party certification, a copy of the certificate shall be provided to NASA. Upon loss of certification, NASA shall be notified within five (5) business days.

CDRL DESCRIPTION

Number: C3-4

**QUALITY MANAGEMENT SYSTEM (QMS) AND SUPPLIER/VENDOR
AUDIT/ASSESSMENT SCHEDULES**

DESCRIPTION/PURPOSE:

A QMS and Supplier/Vendor schedule that lists all planned audits/assessments and visits to subcontractors, suppliers, vendors, and internal sites.

DATA REQUIREMENTS:

The list shall include the subject of the audit/assessment, a tracking number or identifier, assigned assessor, and a schedule by month indicating the planned event.

CDRL DESCRIPTION

Number: C3-5

QMS AND SUPPLIER/VENDOR AUDIT/ASSESSMENT REPORT

DESCRIPTION/PURPOSE:

Audit/Assessment Report documenting the findings and corrective actions of Audit/Assessments conducted. Reports will be used to monitor Quality Management System and the Supplier Management/Procurement System.

DATA REQUIREMENTS:

The report shall include the date and location of the audit/assessment, participating individuals, purpose of the audit/assessment, assessment duration, method of assessment (i.e., AS9101, internal policies/procedures), objective, findings, corrective action, discussions/ comments, and conclusion.

CDRL DESCRIPTION

Number: C3-6

RISK MANAGEMENT PLAN

DESCRIPTION/PURPOSE:

In accordance with NPD 8610.23A, the contractor shall submit a Contractor's Risk Management Plan to be provided to the COTR for subsequent distribution.

DATA REQUIREMENTS:

- A. This Risk Management Plan shall describe how the contractor proposes to manage and accept risk throughout the program and on all activities required to perform this contract. It shall describe how the contractor will identify, assess, and mitigate/control risks as an integral part of the Contractor's management and decision process.
- Information that illustrates the functional relationships and lines of communication, including the involvement of sub-contractors and major vendors.
 - Descriptions of the risk management function within the organization including the process through which management decisions will be made. This will include a description of interfaces with NASA organizations and the NASA decision process for areas such as critical and catastrophic hazards, corrective actions, mishaps, anomaly resolution, safety and technical variances/waivers/exceptions, cost, schedule and mission success.
 - Description of the management controls that will be used to ensure compliance with the risk mitigation process.
- B. Risk reports shall be delivered quarterly (at a minimum) to the COTR. The reports shall include all identified risks, their likelihood, consequence and status of all activities associated with the mitigation, avoidance or acceptance of each risk. The report shall indicate the relative prioritization of all risks.
- C. Access to the contractor's risk database shall be provided to NASA personnel to be identified by the COTR or Contracting Officer.

CDRL DESCRIPTION

Number: C3-7

SAFETY DOCUMENTATION FOR PROCESSING IN NASA PROVIDED FACILITIES

DESCRIPTION/PURPOSE:

This item covers the submission of documentation to NASA when performing operations in a NASA provided facility in support of this contract. NASA will approve this documentation when processing in NASA owned facilities and NASA will review this documentation when processing in NASA procured facilities. This includes system safety documentation associated with launch vehicle systems, flight hardware and any associated equipment and/or ground support equipment.

DATA REQUIREMENTS:

Safety Documentation:

- Safety Data Package (Missile System Prelaunch Safety Package – MSPSP)
 - System Descriptions
 - Hazards Analyses/Hazard Reports
 - Single Failure Point (SFP) Analyses (where applicable)
 - Test Plans and Nondestructive Examination Plans
 - Material Compatibility
 - Material Safety Data Sheets (MSDS's)
- Tailored Range Safety Requirements
- Any Waivers or Equivalent Level of Safety Requests
- Mishap Preparedness and Contingency Plans
 - Initial submittal date forty-five (45) days prior to shipment to the facility
- Radiation control and use authorization for all ionizing and non-ionizing (radio frequency and high intensity light) sources
 - Initial submittal date 4 months prior to use
- Hazardous launch vehicle standalone procedures
 - Initial submittal date forty-five (45) days prior to use

If the LV, hardware, equipment, and/or system is of a new design or is being upgraded/modified, the Contractor shall provide draft and preliminary revision of the safety documentation. The Safety Data Package, tailored Range Safety requirements, and any waivers or Equivalent Level of Safety requests shall be delivered to NASA thirty (30) days prior to each design review milestone (i.e. MSPDR, MSCDR). Final revisions of safety documentation shall be delivered to NASA for approval at least forty-five (45) calendar days prior to first use for hazardous operations.

The Contractor shall update the safety data documentation and any technical analyses as necessary to keep the data current and resubmit to NASA for review.

CDRL DESCRIPTION

Number: C4-1

PERFORMANCE AND GUIDANCE ACCURACY ANALYSIS (PGAA)

DESCRIPTION/PURPOSE:

This analysis is prepared using the best available mission requirements (payload mass properties, insertion state vector requirements, tracking requirements, etc.) to identify any potential problems inherent in accomplishing the mission objectives. This analysis shall include parametric trade studies.

DATA REQUIREMENTS:

The analysis shall include, as a minimum:

- (A) The nominal and 3-sigma limits for orbit elements and associated covariance matrix
- (B) Performance impacts as a function of launch date and time
- (C) Performance capability, margins and reserves, including description of how performance reserve is calculated, for contract and reference mission case(s) as applicable
- (D) Sequence of events and tracking coverage
- (E) Time history of trajectory from launch to targeted condition and end of mission maneuvers
- (F) Launch vehicle summary weight statement
- (G) Sources and magnitude for all dispersions used. If statistical dispersions are used, then the distribution method (e.g., normal, gaussian) shall be specified for each parameter
- (H) Definition of all coordinate systems used
- (I) Reference to the source(s) for all payload and launch vehicle inputs to this analysis (e.g., requirements, mass properties)
- (J) Evaluation of compliance with applicable ICD requirements and constraints.
- (K) Launch windows determined by analysis
- (L) Input file
- (M) An itemized list of all modeling changes made since the previous CDRL C4-1 release (including launch vehicle propulsion, mass properties, aerodynamics, and mission design ground rules), along with an assessment of the performance effect of each change.

CDRL DESCRIPTION

Number: C4-2

FINAL MISSION ANALYSIS (FMA)

DESCRIPTION/PURPOSE:

This analysis is prepared based on payload requirements resulting from NASA review of the PGAA and contains information similar to that in the PGAA. This analysis supports pre-flight verification of compliance with mission requirements.

DATA REQUIREMENTS:

The FMA shall include, as a minimum:

- (A) The nominal 5 or 6 Degree of Freedom (DOF) trajectory simulation for each targeted insertion condition utilizing actual weights and propulsion models (electronic format is preferred).
- (B) All Data Requirements identified in PGAA Data Requirement Description.

CDRL DESCRIPTION

Number: C4-3

PAYLOAD/EXPENDED STAGE SEPARATION ANALYSIS

DESCRIPTION/PURPOSE:

The Payload/Expended Stage Separation Analysis is required to demonstrate that adequate separation distance exists between the payload and the upper stage during the relative motion predicted following payload deployment. The analysis is also required to demonstrate that payload tip-off rates satisfy requirements. This analysis supports pre-flight verification of compliance with mission requirements.

DATA REQUIREMENTS:

- (A) Provide nominal, 3-sigma, or worst-case payload tip-off rates and cone angles (cone angles if deployed while spinning) at deployment.
- (B) The analysis shall include the effects of residual thrust, vehicle and payload mass properties uncertainties, and separation mechanism uncertainties. Additionally, include any methods utilized to ensure contamination upon separation is prevented.
- (C) The report documenting this analysis shall list and describe all dispersion sources used for the analysis, mass properties data sources, and all coordinate systems used.
- (D) Expended stage/payload relative motion analysis following payload deployment

CDRL DESCRIPTION

Number: C4-4

PAYLOAD FAIRING VENTING ANALYSIS

DESCRIPTION/PURPOSE:

This analysis shall evaluate and verify that the payload depressurization rate requirements are satisfied.

DATA REQUIREMENTS:

The Payload Fairing Venting Analysis shall be based on the mission trajectory and payload geometry and volume.

CDRL DESCRIPTION

Number: C4-5

PAYLOAD FAIRING CLEARANCE ANALYSIS

DESCRIPTION/PURPOSE:

This analysis documents/demonstrates that sufficient clearance exists under dynamic conditions is available to prevent payload damage from lift-off through payload fairing jettison.

DATA REQUIREMENTS:

The Payload Fairing Clearance Analysis shall include, as a minimum, the effects of:

- (A) Thermal pre-load
- (B) Disconnect forces
- (C) Actuator forces
- (D) Dynamic response

CDRL DESCRIPTION

Number: C4-6

PRE-FLIGHT CONTROLS AND STABILITY ANALYSIS

DESCRIPTION/PURPOSE:

This report describes analyses conducted to ensure an acceptable vehicle mission unique autopilot design.

DATA REQUIREMENTS:

- (A) List all flight conditions analyzed. Reference source for flight condition data (e.g., nominal mission trajectory).
- (B) Provide tabular-format gain and filter histories for all times during the course of the mission and provide detailed descriptions of all filters used (by reference, if previously supplied). Mission unique software electronic-format files are acceptable.
- (C) List all vehicle models used including, but not limited to: aerodynamics, mass properties, structural frequencies, thrust profiles, and liquid slosh representations. Reference the source for each vehicle model.
- (D) List all dispersions used and describe (or reference) how they were defined.
- (E) List all software tools used to develop the analyses documented in this report.
- (F) Provide detailed descriptions of all linear analysis results, to include rigid body and flexible mode gain and phase margins at all flight conditions analyzed.
- (G) Provide vehicle structural mode frequency histories in graphical or tabular format.
- (H) Describe the results of any and all time-domain controllability analyses either generally applicable to the design or unique to design of the autopilot in the mission unique software.
- (I) Provide detailed flight control system consumables budget, including, but not limited to: any reaction/attitude control system propellant usage, thrust vector control system hydraulic fluid expelled, and vernier engine propellant consumed.

CDRL DESCRIPTION

Number: C4-7

COUPLED DYNAMIC LOADS ANALYSIS

DESCRIPTION/PURPOSE:

This analysis defines flight loads to major vehicle and payload structure. This analysis also supports pre-flight verification of compliance with mission requirements.

DATA REQUIREMENTS:

- (A) Describe the models, methodology and forcing functions used.
- (B) The flight events and conditions that cause the greatest loads on the payload shall be identified and included in the analysis. The flight events and conditions that cause the greatest deflections in the fairing and payload (including instances of minimum clearance) shall be identified and included in the analysis.
- (C) Output from each flight event shall include maximum/minimum tables of payload selected Acceleration Transformation Matrices (ATM), interface forces, and internal Load Transformation Matrices (LTM).
- (D) Worst-case payload fairing dynamic relative deflections shall be identified and included.
- (E) Three load cycles will be required per mission: A preliminary loads cycle, intermediate loads cycle, and a verification loads cycle.

CDRL DESCRIPTION

Number: C4-8

INTEGRATED THERMAL ANALYSIS

DESCRIPTION/PURPOSE:

The purpose of this CDRL is to quantify payload thermal environments in various operational configurations (pre-launch and in-flight). The scope of this analysis may vary depending on the complexity of the payload and upper stage thermal interfaces.

DATA REQUIREMENTS:

The integrated thermal analysis that is performed using the PGAA shall utilize current standard methodologies (e.g. Monte Carlo ray tracing techniques), not restrict the payload model size to less than 1000 thermal nodes or 2000 geometric surfaces and shall include, to the fidelity required by the thermal interfaces:

- (A) Temperature time histories at selected locations (identified by the payload) in the payload thermal model.
- (B) Description of all inputs, assumptions, boundary conditions, and analytical methods used to perform the analysis, including the following components:
 - (i) Pre-launch payload fairing ECS gas conditioning analyses
 - (ii) Payload fairing aeroheating
 - (iii) Trajectory simulation assumptions relative to the actual PGAA.
 - (iv) Ascent and on orbit (up to payload separation) analyses.
- (C) Summary of results identifying maxima and minima (for the selected locations of interest, identified by the payload) and the times and flight events at which they occur.
- (D) Input and output thermal analysis files used for the documented analyses.

CDRL DESCRIPTION

Number: C4-9

RF LINK ANALYSIS

DESCRIPTION/PURPOSE:

This analysis is performed for each nominal mission trajectory and is required to ensure sufficient RF Link margins exist for both the telemetry and flight termination systems during all phases of powered flight through payload separation. This analysis also supports pre-flight verification of compliance with mission requirements.

DATA REQUIREMENTS:

The document shall include the following data as a minimum:

- (A) Definition and description of all coordinate systems used.
- (B) Identification of the mission trajectory used.
- (C) Nominal, maximum and minimum loss / gain factors for all elements in each TX/RX signal propagation path.
- (D) A listing of all ground and mobile telemetry station locations and space based telemetry assets.
- (E) An electronic copy of respective antenna patterns with Theta, θ , having values from zero (0) to one hundred eighty (180) degrees and Phi, ϕ , having values from zero (0) to three hundred sixty (360) degrees for both magnitude and phase in a maximum of five (5) degree increments.
- (F) Analysis outputs, which must capture all information presented in the sample figure below (see Figure D2-1).

CDRL DESCRIPTION

Number: C4-9 (cont)

(G) Descriptive contractor recommendations to mitigate any risks identified by this analysis.

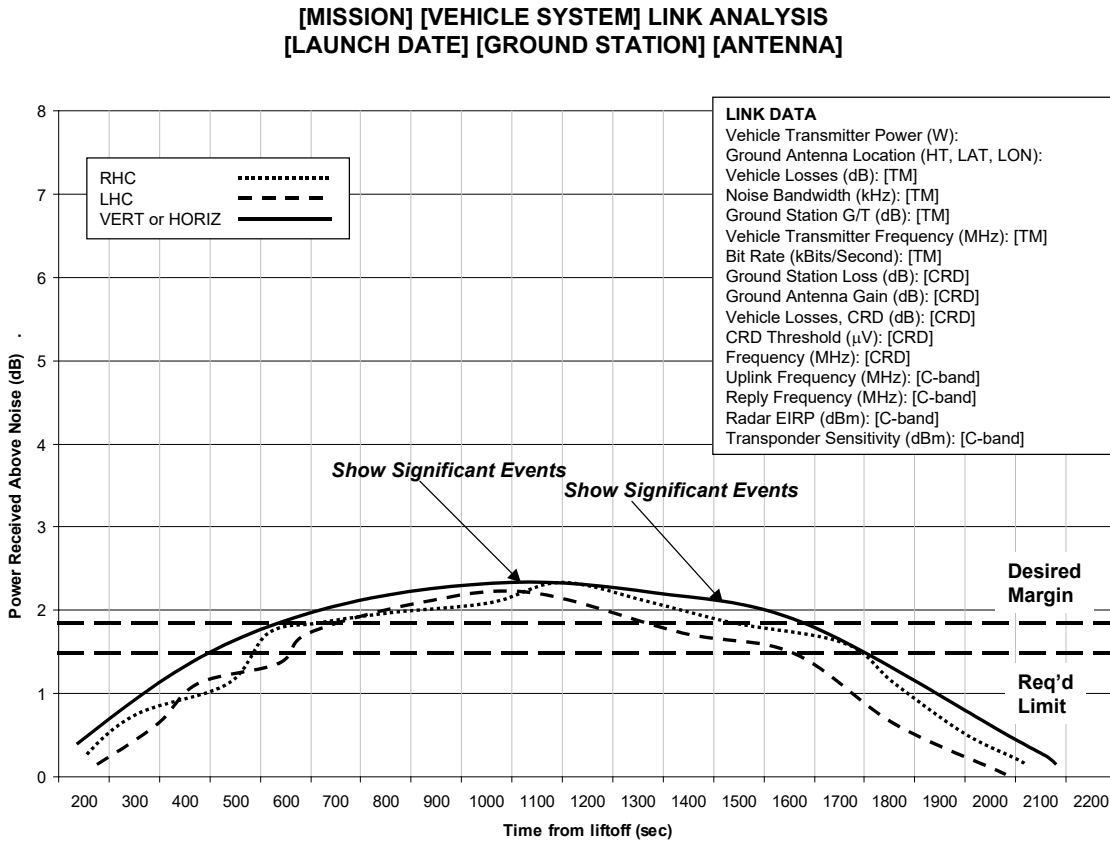


Figure D2-1: Power -Vs- Plus Count Time

CDRL DESCRIPTION

Number: C4-10

PAYLOAD/LAUNCH VEHICLE EMC, RF, AND EED COMPATIBILITY ANALYSIS

DESCRIPTION/PURPOSE:

This Electromagnetic Compatibility (EMC), Radio Frequency (RF), and EED compatibility analysis documents the results of EMC/RF acceptance tests, demonstrates that design requirements in the ICD are satisfied, and ensures that S/C and launch vehicle ordnance sub-systems will not be unintentionally triggered by electromagnetic disturbances generated by the S/C, launch vehicle, Range or other external sources. This is to ensure payload, launch vehicle, and Range are compatible and that payload electromagnetic (EM) energy exposure requirements will not be exceeded.

DATA REQUIREMENTS:

At a minimum, the following data items shall be delivered in approved electronic formats:

- (A) An EMC Control Plan including EMC management processes, design guidelines, test requirements and standards used by the launch vehicle program per fleet and as changed.
- (B) A Spacecraft/Launch Vehicle EM Compatibility Analysis which should include:
 - (i) Identification of test levels for S/C and launch vehicle susceptibility, intentional and unintentional emissions.
 - (ii) Field strengths expected from known ground and external emitters.
 - (iii) Calculation of EMI safety margins between S/C and launch vehicle hardware susceptibility levels and S/C, launch vehicle and external emissions sources.
- (C) A Lightning Effects Analysis detailing:
 - (i) The methods by which S/C and launch vehicle hardware will be protected from direct and indirect lightning effects.
 - (ii) The operational procedures and retest criteria to be used during lightning events.
- (D) An RF Compatibility Analysis to ensure all communication systems will operate as intended without fundamental or harmonic interference from other transmission systems onboard or external to the vehicle stack.
- (E) An EED Compatibility Analysis per fleet and as assessed/changed per mission detailing:
 - (i) Identification of EED Sub-System components and wiring.

- (ii) Identification of discrete EED trigger/firing circuits inclusive of circuit topologies, component identification and circuit parameters.
- (iii) Identification of potential failure modes on EED devices and command/control circuits from electromagnetic sources.
- (iv) Verification Analysis using CAE tools, Franklin analysis or other appropriate method, demonstrate EEDs and all ordnance circuitry meet NASA and Range requirements for EMI safety margins.
- (v) Descriptive contractor recommendations to mitigate any risks identified by this analysis.

CDRL DESCRIPTION

Number: C4-11

MISSION UNIQUE SOFTWARE

DESCRIPTION/PURPOSE:

Ensure the mission unique software is consistent with the analyses and data products delivered under CDRLs C4-2, C4-3, C4-6, C4-7, and C4-9.

DATA REQUIREMENTS:

- (A) Provide a report describing the methods used to check validity of the mission unique software (by reference if previously supplied).
- (B) Provide a memorandum or report documenting the results of mission unique software validation tests. At a minimum, this report shall include comparison of hardware-in-the-loop flight simulation test results with all required event sequences and the target insertion state.
- (C) Copy, in electronic format (ASCII text), of all mission constants, data loads, or I-loads for the mission. This data shall match the flight values used in the vehicle flight program.

CDRL DESCRIPTION

Number: C4-12

QUICK LOOK FLIGHT REPORT

DESCRIPTION/PURPOSE:

After each launch, a quick look post-flight analysis shall be performed using preliminary evaluations of all available vehicle telemetry and insertion performance. A briefing may satisfy the intent of this report.

DATA REQUIREMENTS:

The data report shall include:

- (A) Preliminary trajectory and performance data
- (B) Orbital accuracy estimates
- (C) Preliminary vehicle subsystem system performance
- (D) Preliminary evaluations

CDRL DESCRIPTION

Number: C4-13

FINAL FLIGHT REPORT

DESCRIPTION/PURPOSE:

After each launch, a final post-flight report shall be provided. This report will support the NASA Contracting Officer's mission success determination.

DATA REQUIREMENTS:

- (A) This report shall include, in its entirety, the Contractor's internal post-flight report produced for each mission.

This report shall also include:

- (i) Orbit elements determined from vehicle guidance data
 - (ii) Vehicle data indicating payload separation
 - (iii) Pre-flight prediction of expected flight environments (i.e., acoustic/vibration, quasi-static acceleration, thermal, and pressure). Generic vehicle environmental data may be submitted unless mission unique environmental requirements are identified in the ICD
 - (iv) Post flight determination of actual flight environments
 - (v) Explanation of significant differences between the predicted and actual flight environments
 - (vi) Vehicle sequence and attitudes data
 - (vii) Payload tracking and telemetry data that is interleaved in the launch vehicle data stream
- (B) This report shall also specifically address each ICD orbital parameter and environmental condition using flight data, ground observations, or other data sources in accordance with the determination methods proposed by the Contractor under CDRL C2-5.

CDRL DESCRIPTION

Number: C5-1

VEHICLE DATA PACKAGE

DESCRIPTION/PURPOSE:

This item is a data package of vehicle hardware and software changes that are neither dictated by payload requirements nor represent fleet-wide changes.

DATA REQUIREMENTS:

Package shall include, as a minimum:

- (A) A description of the item, the modification, and the purpose of the modification
- (B) For all hardware changes, an indented part list of the item, which shall define the difference between the assigned configuration and the as-built configuration and supporting rationale for differences. For purposes of this requirement, the as-built configuration excludes standard usage hardware (e.g., nuts, bolts, washers, shims, and pins). The as-built configuration list shall contain:
 - (i) Deliverable equipment part name, number, and serial number
 - (ii) Quantity
 - (iii) Drawing number or traceability code
 - (iv) Drawing dash number, change letter and unincorporated Engineering Orders (EOs)
 - (v) For EEE parts: circuit reference designators, part numbers, part manufacturers, and part lot data codes/serial numbers
- (C) For software changes, a copy of all change documentation shall be provided, along with a laboratory and field testing plan
- (D) Test Report/Data/Conclusions - A copy of the test procedures, a description of all changes since item qualification, a description of major fabrication anomalies and their disposition, a copy of the test data, a copy of any test failure reports and their disposition, and final approval signatures. For ordnance, inspection data records and a copy of the lot test data shall also be submitted.
- (E) Operating Time/Cycle - Status at the time of delivery of accumulated operating time and/or critical cycles history of parts designated as time/cycle critical
- (F) Allowable (specification requirement) and remaining operating time and/or cycles from point of delivery

CDRL DESCRIPTION

Number: C5-2

MISSION UNIQUE HARDWARE TEST PLAN

DESCRIPTION/PURPOSE:

This plan describes the qualification, test and analysis effort for mission unique hardware.

DATA REQUIREMENTS:

The mission unique hardware test plan shall include, as a minimum:

- (A) Description, sequence, configuration, factors of safety, and success criteria for all tests and analyses that support qualification of mission unique hardware
- (B) A verification matrix of requirements versus tests/analyses
- (C) Reference to or description of any applicable industry standards used to support testing

CDRL DESCRIPTION

Number: C5-3

MISSION UNIQUE HARDWARE TEST REPORT

DESCRIPTION/PURPOSE:

This plan describes the qualification and acceptance test results for mission unique hardware.

DATA REQUIREMENTS:

The mission unique hardware test report shall include, as a minimum:

- (A) Sequence, configuration, setup, and results for all tests described in the mission unique hardware test plan (CDRL C5-2)
- (B) A compliance matrix of test requirements versus tests successfully completed
- (C) Reference to or description of any applicable industry standards used to support testing
- (D) Detailed description of any test failures, anomalies, open items, and closure plan

CDRL DESCRIPTION

Number: C5-4

MISSION UNIQUE DRAWINGS

DESCRIPTION/PURPOSE:

This data item identifies the mission unique configuration.

DATA REQUIREMENTS:

All Launch Vehicle and Ground Support equipment mission unique drawings showing all sub-tier mission specific installations, including all mechanical and electrical interfaces. This shall include, but is not limited to:

- (A) Upper stage and payload adapter mission specific drawings showing all sub-tier mission unique installations, including all payload mechanical interfaces.
- (B) All payload adapter and fairing installation drawings showing all sub-tier mission unique installations.
- (C) Top-level vehicle system specification drawings showing all sub-tier mission unique installations.
- (D) All schematics harness drawings, and assembly drawings for mission unique electrical hardware.
- (E) A drawing tree showing all mission specific drawings.

CDRL DESCRIPTION

Number: C5-5

INTEGRATED PROCEDURES

DESCRIPTION/PURPOSE:

Site procedures for various operations that involve the payload and/or the launch vehicle/payload interfaces.

DATA REQUIREMENTS:

Provide as a minimum, the following integrated procedures (If applicable):

- (A) Payload/ launch vehicle electrical interface verification test
- (B) Payload encapsulation and transport
- (C) Payload/launch vehicle mate (includes payload adapter and vehicle mates, if separate)
- (D) Launch vehicle electrical readiness/flight simulation
- (E) Fairing installation
- (F) Combined system test/final readiness test
- (G) A list of all procedures used for vehicle and integrated processing with brief description of their purpose.

Procedures shall have incorporated all mission specific requirements and must be provided to NASA forty-five (45) days prior to use for coordination, review and comments. Final, coordinated procedures, ready for NASA approval, must be provided to NASA 1 week prior to use.

CDRL DESCRIPTION

Number: C6-1

VEHICLE AND GSE TELEMETRY FORMATS

DESCRIPTION/PURPOSE:

To provide all telemetry (RF and hardwire) formats for vehicle systems, GSE and sub-systems testing, including detailed description of formats that will allow NASA to process, display and verify all launch vehicle and GSE telemetry (RF and hardwire).

DATA REQUIREMENTS:

The raw vehicle and GSE telemetry (RF and hardwire) formats shall include as a minimum:

- (A) PCM Data Stream Description: this file will describe the format of the Pulse Code Modulation (PCM) telemetry data stream. A new file must be delivered if the formatting of the PCM data stream is modified. It must include:
 - (i) Bit rate in bits per second
 - (ii) Location and description of sync word
 - (iii) Number of minor frames in major frame
 - (iv) Size of minor frame
 - (v) Number of bits per word or syllable
- (B) Measurement Description File: this file must be standard ASCII text or another format easily read with a personal computer using commercially available software. A new file must be delivered if any measurement or calibration changes are made to the launch vehicle and/or GSE telemetry stream. It must include:
 - (i) Measurement ID
 - (ii) Measurement description
 - (iii) Engineering units of measurement
 - (iv) Location of measurement in minor frame
 - (v) Word step rate of measurement if supercommutated
 - (vi) Size of measurement in bits or bytes

CDRL DESCRIPTION

Number: C6-1(cont)

- (vii) Type of measurement
 - (viii) Initial minor frame number where measurement is found
 - (ix) Frame step rate of measurement if found in more than 1 minor frame
 - (x) Special conversion algorithm from raw to engineering units if not standard polynomial
 - (xi) Calibration coefficients for conversion from raw to engineering units
 - (xii) Bit number in word if measurement is a discrete
 - (xiii) List of guidance modes that measurement is valid
 - (xiv) Method of determining guidance modes
 - (xv) Measurement mask if required
- (C) Description of rules used to format measurement description file including:
- (i) Description of each field in file
 - (ii) Description of all codes or abbreviations
 - (iii) Method of determining revision level or revision date
- (D) Non-PCM Data Stream Description: This file will describe the format of the data stream location and description of all measurements, and the calibration data required to convert from raw to engineering units.
- (E) A list of flight events which will occur for each mission and the nominal time for event occurrence. If event can occur at variable times, this must include a description for determining event time during flight. This list must be delivered for every mission.
- (F) A list of predicted position and velocity state vectors for significant events during flight. This list must include, at a minimum, the end of all powered flight phases and payload separation. A description of the coordinate system used to provide the state vectors must also be included. This list must be delivered for every mission.
- (G) A list of vehicle attitude measurements and description of format used.

All files must be delivered using an electronic medium.

CDRL DESCRIPTION

Number: C6-2

VEHICLE AND GSE TELEMETRY (TEST, PRE-LAUNCH, AND LAUNCH)

DESCRIPTION/PURPOSE:

To provide real time launch vehicle and GSE telemetry (RF and Hardwire) to the NASA designated NASA LSP Data Center which will allow NASA to perform independent data processing and analysis.

DATA REQUIREMENTS:

Provide as a minimum:

- (A) All real time launch vehicle and GSE telemetry (RF and hardwire) test data to the NASA designated NASA LSP Data Center for all system and sub-system testing upon request. This data must be able to be time correlated in a manner that enables time synchronization and must conform to the standards defined in IRIG Standard 106-07.
- (B) All real time launch vehicle and GSE telemetry (RF and Hardwire) to the NASA designated NASA LSP Data Center from the start of Launch Countdown through all phases of powered flight up to LOS. This data must be able to be time correlated in a manner that enables time synchronization and must conform to the standards defined in IRIG Standard 106-07.

CDRL DESCRIPTION

Number: C7-1

SMALL BUSINESS AND SMALL DISADVANTAGED BUSINESS SUBCONTRACTING PLAN

DESCRIPTION/PURPOSE:

In accordance with FAR 19.702(a)(1), each solicitation of offers to perform a contract or contract modification, that individually is expected to exceed \$700,000 and that has subcontracting possibilities, shall require the apparently successful offeror to submit an acceptable subcontracting plan. NLS II Contractors will not be required to submit an updated plan on a mission by mission basis, but will be required to submit an annual update to the plan if they have met or exceeded the \$700,000 threshold since the last plan update.

The purpose of this CDRL is to update the Small Business and Small Disadvantaged Business Subcontracting Plan on an annual basis. The updated plan shall be submitted annually on October 1st unless the Contractor has not reached the \$700,000 threshold since the previous plan update. If the Contractor has not reached the threshold since the previous plan update, the Contractor shall notify the Contracting Officer in writing that it has not reached the threshold. An electronic copy of the plan shall be submitted to the Contracting Officer.

DATA REQUIREMENTS:

At a minimum, the plan should be updated to include revised goals (if applicable) and contract value.

ATTACHMENT D3

SUPPLEMENTAL DATA REQUIREMENTS LIST (SDRL)

The SDRL identifies the non-CDRL documents generated in the course of production, integration, test, pre-launch, and launch activities that shall be provided to NASA. These items shall be delivered to NASA upon request within ten (10) days. Where there is not a direct match between an SDRL item and the Contractor's standard documentation, the Contractor's documentation will be acceptable provided it contains equivalent data requirements.

ITEM	DESCRIPTION
	S1: Documentation
S1-1	Launch Service User's Guide and Facility Information
S1-2	Mission Integration Meeting Documentation
S1-3	Program Master Schedule and Launch Vehicle Planning Manifest

	S3: Safety & Mission Assurance
S3-1	Reserved
S3-2	Reserved
S3-3	As-Built Configuration List
S3-4	Safety Data Package

	S4: Analysis
S4-1	Mission Battery Budget Analysis
S4-2	Reserved

	S5: Engineering
S5-1	Vehicle Test Procedures

	S6: Telemetry and Communications
S6-1	Telemetry Test Data
S6-2	End to End Link Test Plan
S6-3	Recorded Launch Vehicle and GSE Telemetry (Test, Pre-Launch, Launch)
S6-4	Data Flow Diagrams, Schedules and Down Range Asset Status

Table D3-A: SDRL Index

SDRL DESCRIPTION

Number: S1-1

LAUNCH SERVICE USER'S GUIDE AND FACILITY INFORMATION

DESCRIPTION/PURPOSE:

The Launch Service User's Guide and facility information to provide customers with information and capabilities regarding launch services and facilities.

DATA REQUIREMENTS:

Provide the commercially available Launch Service User's Guide and information on the LSC provided facilities, including:

- (A) Payload GSE Space and Capabilities
- (B) Crane Size and Hook Height
- (C) Electrical Service
- (D) Environment, i.e., cleanliness level, relative humidity
- (E) Environment Monitoring Equipment
- (F) Temperature
- (G) Hazardous Capability
- (H) Power
- (I) Processing Space Floor Capability
- (J) Processing Space Size
- (K) Support Space
- (L) Grounding Provisions
- (M) Communications Capabilities
- (N) Security Systems
- (O) Mechanical Support Equipment (Lifts, etc)

SDRL DESCRIPTION

Number: S1-2

MISSION INTEGRATION MEETING DOCUMENTATION

DESCRIPTION/PURPOSE:

To provide presentation and handout material, minutes, and accompanying action item lists from each mission integration meeting (e.g., MIWG kick-off, telecons, working groups, and technical interchange).

DATA REQUIREMENTS:

Mission integration meeting documentation shall include:

- (A) Presentation packages and hand-out materials provided at the meeting for NASA attendees
- (B) Agendas (delivered prior to meeting)
- (C) Minutes from each Mission integration meeting
- (D) List of attendees
- (E) Agreements from each Mission integration meeting
- (F) An action item list maintained and updated by the Contractor. The list shall include:
 - (i) Due Date
 - (ii) Subject
 - (iii) Assigned tracking number for each action item
 - (iv) Person or organization responsible for completing the action
 - (v) Status of action (i.e., open, closed, or deleted)
 - (vi) Decisions/resolutions of action items as determined by joint NASA/Contractor mission integration team

SDRL DESCRIPTION

Number: S1-3

PROGRAM MASTER SCHEDULE AND LAUNCH VEHICLE PLANNING MANIFEST

DESCRIPTION/PURPOSE:

The contractor shall develop and provide a Program Master Schedule to include sub-tier schedules, production schedules, and launch operations schedules for each launch service. In addition, a Launch Vehicle Planning Manifest shall be provided which forecasts launch services for a period of no less than five (5) years. To the extent possible, each release should be consistent with the overall Manifest Policy. The contractor shall address mitigation steps for when these submittals are not consistent with the Manifest Policy.

DATA REQUIREMENTS:

Documentation shall be delivered in a mutually acceptable format.

SDRL DESCRIPTION

Number: S3-1

RESERVED

SDRL DESCRIPTION

Number: S3-2

RESERVED

SDRL DESCRIPTION

Number: S3-3

AS BUILT CONFIGURATION LIST

DESCRIPTION/PURPOSE:

A document describing the as-built configuration of each launch vehicle.

DATA REQUIREMENTS:

TBP

SDRL DESCRIPTION

Number: S3-4

SAFETY DATA PACKAGE

DESCRIPTION/PURPOSE:

This item covers the submission of system safety documentation required to be delivered to the applicable Federal or Commercial Range Safety Organizations for launch vehicle operations for NASA Missions under this contract. This includes, but is not limited to, United States Space Force (USSF) Eastern/Western Ranges, Reagan Test Site, Kodiak Island Launch Complex, and Wallops Flight Facility.

DATA REQUIREMENTS:

Contactors performing launch vehicle operations for NASA under this contract shall make available to NASA the launch processing site required system safety documentation for launch vehicle systems and associated ground support equipment (e.g. Tailored Requirements, MSPSP, Hazard Reports, Ground Operations Plan, Test Plans, Nondestructive Examination (NDE) Test Plans). All Range required documentation shall also be available to NASA upon request.

The contractor shall update the system safety documentation and any technical analyses as necessary to keep the data current.

This shall also include all agreements, determinations, interpretations, waivers, deviations, Equivalent Level of Safety (ELS), mishaps and close calls documented for the launch processing site safety organization.

SDRL DESCRIPTION

Number: S4-1

MISSION BATTERY BUDGET ANALYSIS

DESCRIPTION/PURPOSE:

This analysis verifies sufficient energy and peak load margin exists to satisfy both vehicle and payload requirements.

DATA REQUIREMENTS:

Contractor formats acceptable.

SDRL DESCRIPTION

Number: S4-2

RESERVED

SDRL DESCRIPTION

Number: S5-1

VEHICLE TEST PROCEDURES

DESCRIPTION/PURPOSE:

Procedures for various operations involving launch vehicle processing, testing, and checkout.

DATA REQUIREMENTS:

Launch Service Contractor (LSC) shall provide a list of all launch vehicle test or processing procedures.

Provide upon request the following vehicle processing procedures:

List: **TBP**

SDRL DESCRIPTION

Number: S6-1

TELEMETRY TEST DATA

DESCRIPTION/PURPOSE:

To provide raw telemetry signals, on electronic medium, that are representative of the launch vehicle and GSE telemetry allowing NASA to process, display and verify launch vehicle and GSE telemetry prior to launch vehicle testing.

DATA REQUIREMENTS:

Provide as a minimum:

- (A) An electronic copy representative of the telemetry signals of the launch vehicle and GSE.
- (B) Data on the electronic copy shall be time correlated in a manner that enables time synchronization of all telemetry data and must conform to IRIG Standard 106-07.

SDRL DESCRIPTION

Number: S6-2

END TO END LINK TEST PLAN

DESCRIPTION/PURPOSE:

To provide a test plan that verifies the telemetry (RF and hardwire), voice communication channels and video links from the Contractor's site to NASA's designated NASA LSP Data Center. The plan will describe the telemetry links from the launch vehicle through the Range and down range deployable telemetry receiving assets specific to that mission. This plan will verify NASA's ability to receive, process, display and verify telemetry, voice communication channels and video from the Contractor's site prior to launch vehicle testing.

DATA REQUIREMENTS:

The end-to-end link test plan shall address how the Contractor will test all links from the Contractor's site to NASA's designated NASA LSP Data Center. The plan shall address scope; organizations involved and points of contact for each; facilities used and/or involved; detailed information on telemetry signals, voice communication channels and video sources and transmission rates. The plan shall identify and/or suggest orbiting or ground locations and/or mobile assets to meet LSP telemetry coverage requirements prior to launch. The plan shall describe the coordination/scheduling/conducting of pre-launch data flow tests to ensure real-time connectivity of voice and data paths between the down range assets and the NASA designated NASA LSP Data Center.

SDRL DESCRIPTION

Number: S6-3

RECORDED LAUNCH VEHICLE AND GSE TELEMETRY (TEST, PRE-LAUNCH, LAUNCH)

DESCRIPTION/PURPOSE:

To provide a copy of all launch vehicle and GSE telemetry (RF and Hardwire) allowing NASA to perform independent data processing and analysis.

DATA REQUIREMENTS:

To provide:

- (A) A copy of all system and sub-system test telemetry (RF and hardwire) data on a compatible media or in an electronic format and corresponding processing software, if an electronic copy is provided. Telemetry data must be time correlated in a manner enabling time synchronization of all telemetry and conforming to the standards defined in IRIG Standard 106-07.
- (B) Electronic copy of the raw telemetry (RF and hardwire) data recorded from all phases of powered flight from two (2) minutes prior to stage ignition through **TBP** seconds after shutdown. The payload separation event from **TBP** seconds prior to and **TBP** seconds after separation shall also be included.
- (C) All RF signal strengths as a function of time from launch until completion of vehicle safing.

SDRL DESCRIPTION

Number: S6-4

DATA FLOW DIAGRAMS, SCHEDULES AND DOWN RANGE ASSET STATUS

DESCRIPTION/PURPOSE:

To provide data flow path diagrams, schedules and down range asset status allowing NASA to verify status/readiness of critical assets and data flow planning/coordination.

DATA REQUIREMENTS:

Provide data flow path diagrams, schedules and down range asset status between the NASA designated NASA LSP Data Center and down range assets to NLT fifteen (15) calendar days prior to launch and at the Flight Readiness Review.

ATTACHMENT D4

**SMALL BUSINESS AND SMALL DISADVANTAGED BUSINESS
SUBCONTRACTING PLAN**

TBP

ATTACHMENT D5

SAFETY AND HEALTH PLAN

TBP

ATTACHMENT D6

RELIABILITY PLAN

TBP

ATTACHMENT D7

QUALITY MANAGEMENT PLAN

TBP

ATTACHMENT D8

WORK PLAN¹

Payment Number	Program Event (Launch +/- months)		Accomplishment Criteria
1	L-“X” ² / L-30	ATP/Award/Mission Integration Start	Authority to Proceed received from the Contracting Officer. Long Lead Hardware Procurement Initiated as required. Launch slot reserved. Mission Integration Kickoff Meeting date proposed.
2	L-27	Sharable Mission Solution Mission Integration Kickoff Meeting Mission Specific Requirement Review (MSRR)¹	Provide a shareable Mission Solution document or mutually agreed to equivalent material that can be shared with the Spacecraft project team. Submitted at ATP+1 month. Mission Integration Kickoff Meeting complete MSRR completed and CDRL C1-1 submitted to NASA.
3	L-24	Preliminary Interface Control Document (ICD)¹ MSRR	Preliminary ICD submitted and CDRL C2-1.1 approved. Formal review documentation or MSRR CDRL C1-1 approved by NASA.
4	L-21	Preliminary Coupled Loads Analysis (CLA) Verification Matrix	Preliminary CLA complete and CDRL C4-7 has been submitted and approved by NASA. Verification Matrix CDRL C2-1.2 submitted
5	L-18	Mission Specific PDR (MSPDR)	Mission Specific Preliminary Design Review Complete and CDRL C1-1 has been submitted to NASA.
6	L-15	Baseline Interface Control Document (ICD) Launch Vehicle Certification	Baseline ICD and Verification Matrix complete and under configuration control and CDRL C2-1.1 and C2-1.2 have been approved by NASA. Launch vehicle certification activities on schedule (NASA will provide specific schedule milestones).

Payment Number	Program Event (Launch +/- months)		Accomplishment Criteria
7	L-12	Mission Specific Critical Design Review (MSCDR) Performance and Guidance Accuracy Analysis (PGAA)	Mission Specific Critical Design Review complete, CDRL C1-1 submitted to NASA. PGAA completed and C4-1 (L-12M) has been submitted to NASA.
8	L-09	Matchmate Subassemblies Fabrication MSCDR PGAA	Contractor verification adapter and separation system are available to support structural matchmate. Detailed fabrication of subassemblies approximately 50% complete. Formal review documentation on CDRL C1-1 approved for MSCDR. Approval of PGAA (L-12M) CDRL C4-1
9	L-05	Launch Vehicle Final Assembly and Test MSSAR Verification Coupled Dynamic Loads Cycle PGAA (L-6M)	Production schedules show NASA mission Launch Vehicle hardware is assigned, final assembly has started and is on schedule for shipment to launch site. Mission Specific System Acceptance Review (MSSAR) or equivalent complete and CDRL C1-1 has been submitted to NASA. Verification loads cycle complete and CDRL C4-7 has been approved by NASA. Approval of PGAA (L-6M) CDRL C4-1
10	L-03	PLF Hardware Acceptance Review (HAR) Launch Vehicle Hardware Acceptance	Final assembly and checkout of the PLF are complete and ready for PLF HAR or equivalent. Final assembly and checkout of the Launch Vehicle is substantially completed to support HAR or equivalent

Payment Number	Program Event (Launch +/- months)		Accomplishment Criteria
11	L-02	Launch Vehicle Systems Readiness Review Mission Success Criteria & Determination Methodology	Obtain concurrence to erect L/V on pad, or to continue with L/V processing at the field site. Mission Success Criteria & Determination and Methodology, CDRL C2-5 has been submitted and approved by NASA.
12	L	Launch	Launch event occurred and CDRL 4-12 has been provided to NASA.
13	L+03	Mission Success Determination	Final Flight Report CDRL C4-13 has been submitted to NASA and the NASA Contracting Officer has determined mission success.

¹ The “Program Event” and “Accomplishment Criteria” descriptions may be tailored to align with the contractor’s standard launch service process as long as the “Program Event” and “Accomplishment Criteria” descriptions remain substantially the same as is stated above. If proposing a launch service that has not achieved the certification category required, the work plan shall include critical certification “Program Events” and “Accomplishment Criteria.”

² If ATP is greater than L-30M the milestone date will be defined in the RLSP.

ATTACHMENT D9

MOST FAVORED CUSTOMER CERTIFICATION PLAN

TBP

ATTACHMENT D10

ACRONYMS AND ABBREVIATIONS

A/R	As Required
AAM	Acquisition Assistance Message
ACS	Attitude Control System
AFS	Agency Filing Scheme
AFSPCI	Air Force Space Command Instruction
AFSPCMAN	Air Force Space Command Manual
Amps	Amperes
ASCII	American Standard Code for Information Interchange
ASTM	American Society for Testing and Materials
ATM	Acceleration Transformation Matrix
ATP	Authority to Proceed
Attn.	Attention
BVS	Best Value Solicitation
C	Celsius
C ₃	Earth escapes energy expressed in km ² /sec ²
CAE	Computer Aided Engineering
CCAM	Collision/Contamination Avoidance Maneuver
CCSFS	Cape Canaveral Space Force Station
CDRL	Contract Data Requirements List
CFR	Code of Federal Regulations
CG	Center of Gravity
CLIN	Contract Line Item Number
COLA	Collision Avoidance
COTR	Contracting Officer's Technical Representative
COTS	Commercially available off-the-shelf
CVCM	Collected Volatile Condensable Materials
CY	Calendar Year
D	Day
dB	Decibel
dBm	Decibel Millivolts
DD	Day

DIG	Designated Industry Group
DLA	Declination of Launch Asymptote
DMR	Detailed Mission Requirements
DoD	Department Of Defense
DoDSSP	DOD Single Stock Point
DOF	Degrees of Freedom
DOT	Department of Transportation
DPAS	Defense Priority Allocation System
DUNS	Data Universal Numbering System
E.O.	Executive Order
EAR	Export Administration Regulations
ECS	Environmental Control System
EED	Electro Explosive Device
EEE	Electrical, Electronic and Electromechanical
EFT	Electronic Funds Transfer
ELS	Equivalent Level of Safety
ELV	Expendable Launch Vehicle
EMC	Electromagnetic Compatibility
EM	Electromagnetic
EO	Engineering Order
ETR	Eastern Test Range
EWB	Eastern/Western Range
F	Fahrenheit
FAR	Federal Acquisition Regulation
FED-STD	Federal Standard
FFP	Firm-Fixed-Price
FMA	Final Mission Analysis
FOIA	Freedom of Information Act
FPMR	Federal Procurement Material Record
FRR	Flight Readiness Review
FSC	Federal Supply Class
FSG	Federal Supply Group
g	Acceleration of Gravity
G&A	General and Administrative
GAO	General Accounting Office

GFP	Government-Furnished Property
GHe	Gaseous Helium
GIDEP	Government/ Industries Data Exchange Program
GN&C	Guidance Navigation and Control
GN ₂	Gaseous Nitrogen
GOR	Ground Operations Review
Govt.	Government
GOWG	Ground Operations Working Group
GPA	Agreement on Government Procurement from World Trade Organization
GSA	General Services Administration
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
GSO	Geo-Synchronous Orbit
GTO	Geosynchronous Transfer Orbit
HBCU	Historically Black Colleges and Universities
Horiz	Horizontal
HUBZone	Historically Underutilized Business Zones
Hz	Hertz
I-loads	Inertial Loads
IAW	In Accordance With
ICD	Interface Control Document
ID	Identification
IDIQ	Indefinite Delivery/Indefinite Quantity
IEEE	Institute of Electrical and Electronics Engineers
IEST-STD	Institute of Environmental Sciences and Technology Standard
IFB	Invitation For Bid
IRCA	International Registrar of Certified Auditors
IRIG	Inter-Range Instrumentation Group
IRS	Internal Revenue Service
ISO	International Organizations for Standardization
IT	Information Technology
ITAR	International Traffic in Arms Regulations
kg	Kilogram
KHB	Kennedy Handbook
km	Kilometer

KSC	Kennedy Space Center
L	Launch
LAT	Latitude
LEO	Low Earth Orbit
LHC	Left Hand Circular
LMCM	Launch Management Coordination Meeting
LON	Longitude
LOS	Loss of Signal
LRR	Launch Readiness Review
LSC	Launch Service Contractor
LSP	Launch Service Program
LSTO	Launch Service Task Order
LTM	Load Transformation Matrix
M	Month
MIL-PRF	Military Performance Specification
MIL-STD	Military Standard
MIS	Mission Integration Start
MM	Month
MPCP	Mishap Preparedness and Contingency Plan
MRB	Material Review Board
MSCDR	Mission Specific Critical Design Review
MSDS	Material Safety Data Sheets
MSPDR	Mission Specific Preliminary Design Review
MSPSP	Missile System Prelaunch Safety Package
MSRR	Mission Specific Requirements Review
MSSAR	Mission Specific System Acceptance Review
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NDE	Nondestructive Examination
NEPA	National Environmental Policy Act
NFS	NASA FAR Supplement
NLS	NASA Launch Services
NLT	No Later Than
NNC	NASA and NASA Customer(s)
No.	Number

NPD	NASA Policy Directive
NPR	NASA Procedural Requirements
NSSC	NASA Shared Services Center
NTE	Not To Exceed
NVR	Nonvolatile Residue
OMB	Office of Management and Budget
OMEI	Other Minority Educational Institutions
OR	Operational Requirements
ORCA	Online Representations and Certifications Application
OSHA	Occupational Safety & Health Administration
P/FR	Problem/Failure Reporting
PA	Payload Adapter
PCM	Pulse Code Modulation
PD	Policy Directive
PGAA	Performance and Guidance Accuracy Analysis
PHSF	Payload Handling and Storage Facility
PLF	Payload Fairing
POCC	Payload Operation Control Center
PPC	Procurement Placement Code
PPE	Personal Protection Equipment
PPF	Payload Processing Facility
PRD	Program Requirements Data
PSWG	Payload Safety Working Group
Pub. L.	Public Law
QA	Quality Assurance
QMS	Quality Management System
RAB	Registrar Accreditation Board
RF	Radio Frequency
RFO	Request For Order
RFP	Request For Proposal
RFQ	Request For Quote
RHC	Right Hand Circular
RHU	Radioisotope Heater Unit
RLSP	Request for Launch Services Proposals

RP	Reference Procedure
RTG	Radioisotope Thermal Generator
RX	Receive
SAM	System for Award Management
S/R	Shipping/Receiving
SB	Small Business
S/C	Spacecraft
SDB	Small Disadvantaged Business
SDRL	Supplemental Data Requirements List
SDVOSB	Service-Disabled Small Business
sec	Second
SF	Standard Form
SIC	Standard Industrial Classification
SOW	Statement of Work
SRM	Solid Rocket Motor
SSA	Source Selection Authority
STD	Standard
SW	Southwest
Sync	Synchronous
T	Time (as referenced from launch)
TBD	To Be Determined
TBP	To Be Proposed
TDRSS	Tracking and Data Relay Satellite System
TIC	Target Industry Categories
TIM	Technical Interchange Meeting
TIN	Taxpayer Identification Number
TM	Telemetry
TML	Total Mass Loss
TPA	Test Payload Adapter
TX	Transmit
USAF	United States Air Force
UPN	Unique Project Number
U.S.	United States
U.S.C.	United States Code

VSFB	Vandenberg Space Force Base
VDC	Voltage Direct Current
VETS	Veterans
Vert	Vertical
VOSB	Veteran-Owned Small Business
W	Week
WFF	Wallops Flight Facility
WOB	Woman-Owned Business
WTO	World Trade Organization
X	Number reference, such as CLIN
“X”	Unknown L-period typically in months
YY	Year

ATTACHMENT D11

DEFINITIONS

Anomaly: An unexpected event that can reasonably be considered to pose immediate mission success implications for subsequent NASA missions. An anomaly requires an engineering evaluation and Engineering Review Board disposition prior to a NASA mission flight on an affected launch vehicle configuration.

Authority to Proceed: Notification from the Contracting Officer via task order award to proceed with work on either a Launch Service Task Order (LSTO) or Special Task Assignment.

Baseline ICD: First approved and signed release of ICD.

Certified Common Launch Vehicle Configuration: A common launch vehicle configuration which has met the requirements defined in NPD 8610.7 for a specific launch vehicle risk category.

Common Launch Vehicle Configuration: A distinct combination of core propulsive stages and hardware used to deliver payloads to earth orbit or escape trajectories.

Core Propulsive Stages: All propulsive stages except strap-on motors, final stages (exclusively used for orbit circularization or escape), and trim stages.

Common Launch Vehicle Configuration Upgrades or Modifications: Items that do not substantially affect operating time, total impulse, and/or the thrust profile of one or more propulsive stages are considered upgrades or modifications to the certified common vehicle configuration. Examples of upgrades and modifications include changes in software, payload fairing, payload electrical/mechanical interfaces, incorporation of mission-unique requirements, and the addition or deletion of a final stage (exclusively used for orbit circularization or escape), strap-on motors, and/or trim stage. Upgrades or modifications are not to be interpreted as common vehicle configuration changes.

Common Launch Vehicle Configuration Changes: Items that substantially affect the airframe integrity, operating time, total impulse and/or thrust profile of one or more core propulsive stages are considered changes to the certified common vehicle configuration. Examples of vehicle configuration changes include the replacement of engine types, core propulsive stages, and/or major airframe structures.

Covered Launch Activities: Any and all activities involved in the preparation of a launch vehicle and payload for launch, and conduct of the launch, when those activities take place at a launch site in the United States.

First Flight Item: A first flight item is the planned use of a component for which NASA will be one of the first three flights or uses, and/or which affects qualification margins or subsystem

performance and interfaces. The term “flight” is used generically to include all airborne systems and ground systems used to meet payload requirements and ground systems with direct interfaces to the vehicle. Additionally, any change of component vendor or manufacturing site relocation will be considered a first flight item if a NASA LSP-managed mission is planned to be one of the first three uses of the new component.

Integrated Procedures: Launch vehicle procedures used for various operations that involve the payload and/or the Launch Vehicle/payload interfaces.

Launch: The intentional ignition of the first-stage motor(s) of the launch vehicle that has been integrated with the payload.

Limited Launch Period: Daily launch opportunities constrained by blackout periods associated with scientific requirements (e.g., planetary, lunar, seasonal, and/or lunar exclusions).

Mission Specific: All standard, non-standard, mission unique and first flight services provided to meet the requirements of the payload and mission.

Mission Integration Start: L-30M plus or minus 3M signifies the starting point of Mission Integration Start. All integration work prior to or greater than L-30M plus or minus 3M is considered Early/Extended Integration.

Mission Unique: Services provided that are newly performed or developed specifically to meet the requirements of the payload and mission. Mission unique items are considered first flight items.

NASA LSP Data Center: NASA’s voice, video, communication and data facility located at either CCSFS or VSFB.

NASA Provided Facility: Includes NASA owned facilities as well as NASA procured commercially available facilities for payload processing.

Party or Parties: The Contractor or NASA, or both.

Payload or Spacecraft: All NASA or NASA-sponsored equipment that has been or will be integrated with the launch vehicle for transportation into earth orbit or escape trajectories. Payload and spacecraft are used synonymously throughout the document.

Payload Program Level Requirements: The basis for mission assessments conducted by NASA Headquarters during the development period and provides the baseline for the determination of the science mission success during the operational phase of the payload.

Powered Flight: Any phase of vehicle flight where the vehicle stage main propulsion system is operating, with the sole exception of propellant or pressurant venting phases where no combustion occurs.

Preliminary ICD: Working copy of the ICD released prior to the first signed release.

Primary Payload: The payload which serves to determine the launch day and time, and without which the vehicle would not launch.

Related Party:

- (i) Any of the Parties' directors, officers, agents, employees or customers
- (ii) Any of the Parties' contractors, subcontractors, or suppliers at any tier involved directly or indirectly in the performance of this Contract
- (iii) Any entity having any right, title or interest, whether through sale, lease or service arrangement or otherwise, directly or indirectly, in the payload, the launch vehicle, or the launch service value.

Secondary Payload: Any payload carried by the launch vehicle in space not required by the primary payload. The secondary payload requires the approval of the primary payload customer prior to manifesting. If the secondary payload is not ready to support scheduled launch day, the secondary payload will be demanifested and replaced with a mass simulator.

Successful Flight: A launch which met primary mission requirements.

Third Party: Any person or entity other than NASA, the Contractor, and related parties.

SECTION E

SOLICITATION PROVISIONS

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ATTACHMENT E1

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ATTACHMENT E1

REPRESENTATIONS AND CERTIFICATIONS

Submittal of Representations and Certifications is not required for existing NLS II providers as Representations and Certifications for existing providers were evaluated before award of the current NLS II contracts.

1.0 Reserved

2.0 MINIMUM CONTRACT REQUIREMENT CERTIFICATIONS

2.1 Reserved

2.2 Certification Regarding United States Commercial Provider of Space Transportation Services (Public Law 105-303, Title II, Section 201)

(A) Participation in this procurement is restricted to prime contractors from the United States launch vehicle/services industry meeting the definition of United States commercial provider.

(B) The Offeror certifies, to the best of its knowledge and belief, that it ☐ is ☐ is not a United States commercial provider as defined below--

(i) 'United States commercial provider' means a commercial provider, organized under the laws of the United States or of a State, which is—

(a) more than 50 percent owned by United States nationals; or

(b) a subsidiary of a foreign company and the Secretary of Transportation finds that—

(1) such subsidiary has in the past evidenced a substantial commitment to the United States market through--

(I) investments in the United States in long-term research, development, and manufacturing (including the manufacture of major components and subassemblies); and

(II) significant contributions to employment in the United States; and

(2) the country or countries in which such foreign company is incorporated or organized, and, if appropriate, in which it principally conducts its business, affords reciprocal treatment to companies described in

subparagraph (B)(i)(a) comparable to that afforded to such foreign company's subsidiary in the United States, as evidenced by—

- (I) providing comparable opportunities for companies described in subparagraph (B)(i)(a) to participate in Government sponsored research and development similar to that authorized under this Act;
 - (II) providing no barriers, to companies described in subparagraph (B)(i)(a) with respect to local investment opportunities, that are not provided to foreign companies in the United States; and
 - (III) providing adequate and effective protection for the intellectual property rights of companies described in subparagraph (B)(i)(a).
- (C) "Launch services" means all services required in the performance of this contract, excluding those necessary to produce or manufacture launch vehicle(s), its components and other equipment and facilities required in the performance of the contract. "Controlling interest" means ownership of an amount of equity in such entity sufficient to direct management or to void transactions entered into by management. Ownership of at least fifty-one (51) percent of the equity creates a rebuttable presumption that such interest is controlling.
- (D) The Contractor shall provide in the performance of this contract launch vehicles that are domestic end products. The launch vehicle shall be a domestic end product only if the cost of its components, mined, produced or manufactured in the United States exceeds 50 percent of the cost of all its components. The cost of each component includes transportation costs to the place of incorporation into the launch vehicle and any applicable duty (whether or not a duty-free entry certificate is issued.) "Components," as used in this clause, means those articles, materials and supplies directly incorporated into the end product.
- (E) The Contractor shall provide, in the performance of this contract, domestic launch services. Launch services shall be considered to be domestic if the cost for launch services performed by United States industry sources exceeds 50 percent of the cost of the total required launch services.
- (F) The Offeror certifies that the proposal ☐ is, ☐ is not in compliance with paragraphs (A), (B), (C), (D), and (E) above. Any proposal failing to demonstrate its compliance with these criteria will not be considered for award.

- 2.3 The business entity(ies) actually performing the work specified in this contract shall be AS9100 certified by a registrar that is accredited by a body recognized by either the ANSI-ASQ National Accreditation Board (ANAB) or the International Aerospace Quality Group (IAQG). The Offeror shall comply with this requirement by demonstrating that the corporation(s), corporate division(s), subsidiary(ies), joint venture(s), partner(s), and/or any other business entity actually performing launch vehicle manufacture, payload/launch vehicle integration, testing, and launch activities is/are AS9100 certified. A copy of the AS9100 Certification Certificate shall be provided to NASA.

Corporation Name: _____

AS9100 Certificate Number: _____

Copy of AS9100 Certificate Attached: ☐ Yes

3.0 RESERVED

4.0 RESERVED

5.0 COMPLIANCE WITH VETERANS' EMPLOYMENT REPORTING REQUIREMENTS

By submission of its offer, the Offeror represents that, if it is subject to the reporting requirements of 38 U.S.C. 4212(d) (i.e., the VETS-4212 Report required by Federal Acquisition Regulation clause 52.222-37, Employment Reports on Disabled Veterans and Protected Veterans), it has submitted the most recent report required by 38 U.S.C. 4212(d).

6.0 FAR 52.209-7 INFORMATION REGARDING RESPONSIBILITY MATTERS (DEVIATION) NOV 2025

(a) Definitions. As used in this provision—

Administrative proceeding means a non-judicial process that is adjudicatory in nature in order to make a determination of fault or liability (e.g., Securities and Exchange Commission Administrative Proceedings, Civilian Board of Contract Appeals Proceedings, and Armed Services Board of Contract Appeals Proceedings). This includes administrative proceedings at the Federal and State level but only in connection with performance of a Federal contract or grant. It does not include agency actions such as contract audits, site visits, corrective plans, or inspection of deliverables.

Federal contracts and grants with total value greater than \$10,000,000 means—

- (1) The total value of all current, active contracts and grants, including all priced options; and
- (2) The total value of all current, active orders including all priced options under indefinite-delivery, indefinite-quantity, 8(a), or requirements contracts (including task and delivery and multiple-award Schedules).

Principal means an officer, director, owner, partner, or a person having primary management or supervisory responsibilities within a business entity (e.g., general manager; plant manager; head of a division or business segment; and similar positions).

- (b) The offeror ☐ has ☐ does not have current active Federal contracts and grants with total value greater than \$10,000,000.
- (c) If the offeror checked "has" in paragraph (b) of this provision, the offeror represents, by submission of this offer, that the information it has entered in the Federal Awardee Performance and Integrity Information System (FAPIS) is current, accurate, and complete as of the date of submission of this offer with regard to the following information:
- (1) Whether the offeror, and/or any of its principals, has or has not, within the last five years, in connection with the award to or performance by the offeror of a Federal contract or grant, been the subject of a proceeding, at the Federal or State level that resulted in any of the following dispositions:
- (i) In a criminal proceeding, a conviction.
- (ii) In a civil proceeding, a finding of fault and liability that results in the payment of a monetary fine, penalty, reimbursement, restitution, or damages of \$5,000 or more.
- (iii) In an administrative proceeding, a finding of fault and liability that results in—
- (A) The payment of a monetary fine or penalty of \$5,000 or more; or
- (B) The payment of a reimbursement, restitution, or damages in excess of \$100,000.
- (iv) In a criminal, civil, or administrative proceeding, a disposition of the matter by consent or compromise with an acknowledgment of fault by the Contractor if the proceeding could have led to any of the outcomes specified in paragraphs (c)(1)(i), (c)(1)(ii), or (c)(1)(iii) of this provision.
- (2) If the offeror has been involved in the last five years in any of the occurrences listed in (c)(1) of this provision, whether the offeror has provided the requested information with regard to each occurrence.
- (d) The offeror shall post the information in paragraphs (c)(1)(i) through (c)(1)(iv) of this provision in FAPIS as required through maintaining an active registration in the System for Award Management, which can be accessed via <https://www.sam.gov> (see 52.204-7).

(End of provision)

7.0 FAR 52.211-14 NOTICE OF PRIORITY RATING FOR NATIONAL DEFENSE, EMERGENCY PREPAREDNESS, AND ENERGY PROGRAM USE (APR 2008)

Any contract awarded as a result of this solicitation will be [] DX rated order; [x] DO rated order certified for national defense, emergency preparedness, and energy program use under the Defense Priorities and Allocations System (DPAS) (15 CFR 700), and the Contractor will be required to follow all of the requirements of this regulation.

If the awarded contract contains a DPAS rating (DO-C9), the Contractor shall flow down the rating to all subcontractors and suppliers whose performance is required to meet the delivery requirements of the rated order.

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ATTACHMENT E2

INSTRUCTIONS, CONDITIONS, AND NOTICES TO OFFERORS

1.0 MINIMUM CONTRACT REQUIREMENTS

- 1.1 Provide certification of being a domestic launch service provider pursuant to Section 201 of Public Law 105-303 “Commercial Space Act of 1998”. Foreign sources are not eligible for award. For the purposes of this procurement, a domestic source is defined in Section E, Attachment E1, paragraph 2.2, Certification Regarding United States Commercial Provider of Space Transportation Services (Public Law 105-303, Title II, Section 201).
- 1.2 The business entity(ies) actually performing the work specified in this contract shall be AS9100 certified by a registrar that is accredited by a body recognized by either the ANSI-ASQ National Accreditation Board (ANAB) or the International Aerospace Quality Group (IAQG). The Offeror shall comply with this requirement by demonstrating that the corporation(s), corporate division(s), subsidiary(ies), joint venture(s), partner(s), and/or any other business entity actually performing launch vehicle manufacture, payload/launch vehicle integration, testing, and launch activities is/are AS9100 certified. Provide certification in Attachment E1, paragraph 2.3 and attach a copy of the AS9100 Certification Certificate.
- 1.3 Reserved

2.0 FAR 52.212-1 INSTRUCTIONS TO OFFERORS – COMMERCIAL PRODUCTS AND COMMERCIAL SERVICES DEVIATION(Nov 2025) (TAILORED)

- (a) *North American Industry Classification System (NAICS) code and small business size standard.* The NAICS code and small business size standard for this acquisition appear in Block 10 of the solicitation cover sheet (SF 1449). However, the small business size standard for a concern which submits an offer in its own name, but which proposes to furnish an item which it did not itself manufacture, is 500 employees.
- (b) *Submission of offers.* Submit signed and dated offers to the office specified in this solicitation at or before the exact time specified in this solicitation. Offers may be submitted on the SF 1449, letterhead stationery, or as otherwise specified in the solicitation. As a minimum, offers must show:
 - (1) The solicitation number;
 - (2) The time specified in the solicitation for receipt of offers;
 - (3) The name, address, and telephone number of the offeror;

- (4) A technical description of the items being offered in sufficient detail to evaluate compliance with the requirements in the solicitation. This may include product literature, or other documents, if necessary;
 - (5) Terms of any express warranty;
 - (6) Price and any discount terms;
 - (7) “Remit to” address, if different than mailing address;
 - (8) Reserved
 - (9) Acknowledgment of Solicitation Amendments;
 - (10) Past performance information, when included as an evaluation factor, to include recent and relevant contracts for the same or similar items and other references (including contract numbers, points of contact with telephone numbers and other relevant information); and
 - (11) If the offer is not submitted on the SF 1449, include a statement specifying the extent of agreement with all terms, conditions, and provisions included in the solicitation. Offers that fail to furnish required representations or information, or reject the terms and conditions of the solicitation may be excluded from consideration.
- (c) *Period for acceptance of offers.* The Offeror agrees to hold the prices in its offer firm for 180 calendar days from the date specified for receipt of offers, unless another time period is specified in an addendum to the solicitation.
- (d) *Product samples.* When required by the solicitation, product samples shall be submitted at or prior to the time specified for receipt of offers. Unless otherwise specified in this solicitation, these samples shall be submitted at no expense to the Government, and returned at the sender’s request and expense, unless they are destroyed during preaward testing.
- (e) *On-Ramp Considerations (see Section C Paragraph 2.2).* For the IDIQ portion of this contract, the original solicitation will remain open through June 2026 through the use of an innovative On-Ramp provision. At any time subsequent to the award of the present contract, the Government may award additional contracts for the IDIQ requirements identified in the present contract. Each August, after the award of the initial contract(s), and during the life of this contract, the Government will accept proposals from new launch service providers for IDIQ contracts and proposals from existing IDIQ contractors for additional launch services. If the Government issues a solicitation notice, new launch service providers and current IDIQ Contractors will be allowed to submit proposals, within the notice's stated response time. The minimum contract requirements, the technical acceptability standards, evaluation factors,

solicitation terms and conditions, price reasonableness, and basis for award shall remain in full force and effect for each new proposal. Upon award of each additional contract, the Government shall notify all present Contractors of the award, and the new Contractor shall thenceforth be eligible to compete with all present Contractors for the award of IDIQ task orders.

(f) Late submissions, modifications, revisions, and withdrawals of offers.

(1) Offerors are responsible for submitting offers, and any modifications, revisions, or withdrawals, so as to reach the Government office designated in the solicitation by the time specified in the solicitation. If no time is specified in the solicitation, the time for receipt is 4:30 p.m., local time, for the designated Government office on the date that offers or revisions are due.

(2)(i) Any offer, modification, revision, or withdrawal of an offer received at the Government office designated in the solicitation after the exact time specified for receipt of offers is “late” and will not be considered unless it is received before award is made, the Contracting Officer determines that accepting the late offer would not unduly delay the acquisition; and—

(A) If it was transmitted through an electronic commerce method authorized by the solicitation, it was received at the initial point of entry to the Government infrastructure not later than 5:00 p.m. one working day prior to the date specified for receipt of offers; or

(B) There is acceptable evidence to establish that it was received at the Government installation designated for receipt of offers and was under the Government’s control prior to the time set for receipt of offers; or

(C) If this solicitation is a request for proposals, it was the only proposal received.

(ii) However, a late modification of an otherwise successful offer, that makes its terms more favorable to the Government, will be considered at any time it is received and may be accepted.

(3) Acceptable evidence to establish the time of receipt at the Government installation includes the time/date stamp of that installation on the offer wrapper, other documentary evidence of receipt maintained by the installation, or oral testimony or statements of Government personnel.

(4) If an emergency or unanticipated event interrupts normal Government processes so that offers cannot be received at the Government office designated for receipt of offers by the exact time specified in the solicitation, and urgent Government requirements preclude amendment of the solicitation or other notice of an extension of the closing date, the time specified for receipt of offers will be deemed to be

extended to the same time of day specified in the solicitation on the first work day on which normal Government processes resume.

- (5) Offers may be withdrawn by written notice received at any time before the exact time set for receipt of offers. Oral offers in response to oral solicitations may be withdrawn orally. If the solicitation authorizes facsimile offers, offers may be withdrawn via facsimile received at any time before the exact time set for receipt of offers, subject to the conditions specified in the solicitation concerning facsimile offers. An offer may be withdrawn in person by an offeror or its authorized representative if, before the exact time set for receipt of offers, the identity of the person requesting withdrawal is established and the person signs a receipt for the offer.
- (g) *Contract award (not applicable to Invitation for Bids).* The Government intends to evaluate offers and award a contract without discussions with offerors. Therefore, the offeror's initial offer should contain the offeror's best terms from a price and technical standpoint. However, the Government reserves the right to conduct discussions if later determined by the Contracting Officer to be necessary. The Government may reject any or all offers if such action is in the public interest; accept other than the lowest offer; and waive informalities and minor irregularities in offers received.
- (h) *Multiple awards.* The Government may accept any item or group of items of an offer, unless the Offeror qualifies the offer by specific limitations. Offerors shall respond to the IDIQ portion of the RFP to be considered for contract award.
- (i) *Availability of requirements' documents cited in the solicitation.*
- (j) (1) *Debriefing.* If a post-award debriefing is given to requesting Offerors, the Government shall disclose the following information, if applicable:
 - (1) The agency's evaluation of the significant weak or deficient factors in the debriefed Offeror's offer.
 - (2) The overall evaluated cost or price and technical rating of the successful and the debriefed Offeror and past performance information on the debriefed Offeror.
 - (3) The overall ranking of all Offerors, when any ranking was developed by the agency during source selection.
 - (4) A summary of the rationale for award;
 - (5) For acquisitions of commercial items, the make and model of the item to be delivered by the successful offeror.

- (6) Reasonable responses to relevant questions posed by the debriefed offeror as to whether source-selection procedures set forth in the solicitation, applicable regulations, and other applicable authorities were followed by the agency.

3.0 FAR 52.216-1 TYPE OF CONTRACT (APR 1984)

The Government contemplates award of one or more IDIQ task order contract (FAR 16.5) with NTE prices for launch services, with a catalog of NTE price nonstandard services to be ordered independently or in conjunction with an LSTO, and a provision for firm fixed-price composite labor hours for special task assignments (studies and analyses) performed on a completion basis, resulting from this RFP.

4.0 FAR 52.216-27 SINGLE OR MULTIPLE AWARDS (OCT 1995) *(TAILORED)*

The Government intends to award one or more IDIQ contracts for launch services.

5.0 NFS 1852.223-73 SAFETY AND HEALTH PLAN (JUL 2015)

(a) The offeror must submit a detailed safety and occupational health plan as part of its proposal. The plan must include a detailed discussion of the policies, procedures, and techniques that will be used to ensure the safety and occupational health of contractor employees and to ensure the safety of all working conditions throughout the performance of the contract.

(b) The plan must similarly address subcontractor employee safety and occupational health for those proposed subcontracts or subcontract effort where the work will be conducted completely or partly on a Federally-controlled facility.

(c) This plan, as approved by the contracting officer, will be incorporated into any resulting contract.

(End of provision)

6.0 FAR 52.233-2 SERVICE OF PROTEST (SEP 2006)

- (a) Protests, as defined in section 33.101 of the Federal Acquisition Regulation, that are filed directly with an agency, and copies of any protests that are filed with the General Accounting Office (GAO), shall be served on the Contracting Officer (addressed as follows) by obtaining written and dated acknowledgment of receipt from:

NASA, John F. Kennedy Space Center, Procurement Office
ATTN: Rachel T. Edwards, Contracting Officer
Mail Code: OP-LS
Kennedy Space Center, Florida 32899

- (a) The copy of any protest shall be received in the office designated above within one day of filing a protest with the GAO.

7.0 NFS 1852.233-70 PROTESTS TO NASA (DEC 2015)

(a) In lieu of a protest to the United States Government Accountability Office (GAO), bidders or offerors may submit a protest under 48 CFR Part 33 (FAR Part 33) directly to the Contracting Officer for consideration by the Agency. Alternatively, bidders or offerors may request an independent review by the Assistant Administrator for Procurement, who will serve as or designate the official responsible for conducting an independent review. Such reviews are separate and distinct from the Ombudsman Program described at 1815.7001.

(b) Bidders or offerors shall specify whether they are submitting a protest to the Contracting Officer or requesting an independent review by the Assistant Administrator for Procurement.

(c) Protests to the Contracting Officer shall be submitted to the address or email specified in the solicitation (email is an acceptable means for submitting a protest to the Contracting Officer). Alternatively, requests for independent review by the Assistant Administrator for Procurement shall be addressed to the Assistant Administrator for Procurement, NASA Headquarters, Washington, D.C. 20546-0001.

8.0 FAR 52.252-1 SOLICITATION PROVISIONS INCORPORATED BY REFERENCE (FEB 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The Offeror is cautioned that the listed provisions may include blocks that must be completed by the Offeror and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the Offeror may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at these addresses:

<http://www.acquisition.gov/Far/>

GSA FAR Homepage

<http://www.hq.nasa.gov/office/procurement/regs/nfstoc.htm>

NASA FAR Supplement Homepage

9.0 RESERVED

10.0 GENERAL INSTRUCTIONS FOR PROPOSAL

10.1 Objective of Procurement

Offerors are encouraged to propose more than one launch vehicle configuration to satisfy the requirements of this contract; however, the proposal must demonstrate the technical acceptability of each proposed launch service. If more than one launch vehicle is proposed, the same technical acceptability standard shall be applied to each launch vehicle configuration.

The Government is only soliciting and evaluating proposals for risk category 2 and 3 launch services capable of delivering, at a minimum, a 250 kilogram payload to orbit at an altitude of 200 kilometers and a launch inclination of 28.5°. Proposals for category 2 and 3 launch services shall comply with the certification requirements of Section C, Clause 3.0, Launch Vehicle Certification. Each August during the life of this contract, or any other time established via synopsis, the Government will solicit proposals from new launch service providers for IDIQ contracts and proposals from existing IDIQ contractors for additional launch services.

10.2 General Instructions

Offerors should utilize their existing documentation to the maximum extent possible in providing proposals under this solicitation and in the performance of any resulting contract. The proposal shall:

- (A) Demonstrate an understanding of the overall and specific requirements of the proposed contract
- (B) Describe, in detail, the Offeror's proposed launch service capabilities
- (C) Include prices for each contract line item number (CLIN) and Sub-CLIN, as applicable

It is considered inadequate to merely state that the Offeror understands and complies with this effort, or paraphrase the statements in the RFP or parts thereof. Phrases such as: "Standard procedures will be employed" and "well-known techniques will be used" are considered inadequate. Responses should be complete and thorough in all areas discussed. Any special procedures or specifications to be used shall be thoroughly explained.

10.2.1 Additional Instructions for Existing NLS II Contractors

The following instructions pertain to existing NLS II Contractors:

- (A) A Past Performance Volume does not need to be submitted as part of the existing NLS II Contractor's on-ramp proposal.
- (B) A Technical Volume is required for new common launch vehicle configurations. In addition, significant changes in capability for common launch vehicle configurations currently on the NLS II contract are to be included in Exhibits 1, 2, and 3 of the Model Contract.

- (C) Changes to terms and conditions identified in this solicitation (Sections A through D) are not applicable.
- (D) Changes identified in all Section E Attachments are applicable.
- (E) The following documents are not required to be submitted unless those documents need to be updated as a result of the proposal of new common launch vehicle configuration(s): (1) Attachment D4 – Small Business and Small Disadvantaged Business Subcontracting Plan; (2) Attachment D5 - Safety and Health Plan; (3) Attachment D6 – Reliability Plan; and (4) Attachment D7 – Quality Management Plan.
- (F) Existing NLS II Contractors are not required to submit pricing and other existing contract data for launch vehicle configurations and pricing already on contract.
- (G) Existing NLS II Contractors may submit change pages to the Model Contract in lieu of a full Model Contract. Any changes that take exception to required terms and conditions in the contract may render the proposal as “unacceptable.”
- (H) Existing NLS II Contractors are not required to demonstrate financial capability to provide launch services.
- (I) Page Numbering: If change pages to the model contract increase the content of an existing page to the extent that another page is required, the content of the page that “spills over” to the next page shall be put on a new page and the existing subsequent page numbers shall not be disturbed. Pages added to accommodate a “spill over” shall be numbered using a numeric decimal system (i.e., XX.1, XX.2, etc.). If content is deleted from pages, the page shall be retained in the contract as blank pages with the notation “This page intentionally left blank”. When the content of several sequential pages is removed, a single page can be retained which identifies all the sequentially numbered pages as “Intentionally Blank”. Blank pages shall be added (if necessary) and numbered using the same format specified above to keep existing pages in the original front-to-back orientation. Tables or Figures that are deleted shall be labeled as “Reserved” and the existing content shall be removed. A new number shall be used for Tables and Figures that are added.

10.3 Specific Instructions

- (A) Proposals shall be submitted in 4 volumes: Technical Acceptability, Price, Past Performance, and Contract Volume. Each volume shall be submitted electronically. Only the number of pages shown in Table E2-A will be evaluated. Any pages in excess of the maximum number will be removed from the proposal without being evaluated. No material may be incorporated by reference. A suitable table of contents (excluded from the page limitations) shall be provided with each volume for ready reference to sections, figures, and illustrations. Exclude from the page limitation, any required cross-reference list or chart plus the Small Business and Small Disadvantaged Business Subcontracting, Safety and Health, Reliability, Quality Management, Most Favored Customer Certification, and Work Plans submitted in response to solicitation requirements.

VOLUME	PROPOSAL	PAGE LIMIT
1	Technical Acceptability	
	SOW	250
	Exhibits 1-4, & 7	No Limit
	Attachments D2, D3, & D5 thru D8	No Limit
2	Price	No Limit
3	Past Performance	15
4	Contract	No Limit

Table E2-A: Proposal Volumes and Page Limitation

- (B) The proposal shall be submitted electronically in PDF format via the designated submission portal. Documents shall be formatted for 8 ½" x 11" paper (or the closest metric equivalent) with at least ¾" margins on all sides. Except for the Price Proposal, all volumes shall use a minimum 12-point font; a legible type size shall be used within figures and graphics. All pages must be numbered. Illustrations shall be legible and no larger than 11" x 17" where appropriate for the subject matter. All pages must be numbered. Information which can be construed as belonging in the Technical Acceptability volume will be so construed, added to the end of that volume, and counted against that volume's page limitation.
- (C) The Offeror shall provide a list or chart which relates the page and paragraph numbers of the proposal instructions to the proposal's page and paragraph numbers responding to the instructions (excluded from any page count limitation).
- (D) The Offeror may reference another section in the proposal rather than copy the information in both locations; however, consistency in the logical flow of the subject matter shall be maintained.
- (E) Pages submitted in excess of the page limits enumerated in Table E2-A will not be evaluated, but will be returned to the Offeror. The page limitations apply separately to the initial proposal and, if requested, to the final proposal revision.
- (F) The Offeror's proposal shall be signed by an official authorized to bind the company.
- (G) The proposal is to be dropped into the following Box Link: (<https://nasagov.app.box.com/f/90320d679507450fb47874724c42be34>) and a email sent to the Contracting Officer, Rachel Edwards at Rachel.t.edwards@nasa.gov, to verify submission. Documents should be emailed in MS Word or .PDF format. The Offeror shall provide at least one copy of Volume 4, Model Contract text in MS Word. Price proposal shall be submitted in Excel 2010 or greater format. Pictures, drawings, or figures, not embedded in the body of the document, should be delivered in one of the following formats: .jpg, .gif, or .bmp.

10.4 Proposal Acceptance Period

Offerors shall allow a period of not less than 180 days for acceptance of proposals by the Government. Proposals must be submitted during the open season as specified in Section C, Clause 2.2. The Offeror shall sign the Standard Form 1449 and return it in Volume 4 of the proposal.

11.0 FAR 52.215-20 REQUIREMENTS FOR CERTIFIED COST OR PRICING DATA AND DATA OTHER THAN CERTIFIED COST OR PRICING DATA (Nov 2025) – ALTERNATE IV (Nov 2025)

(a) Submission of certified cost or pricing data is not required.

(b) Provide information described below:

Data other than certified cost or pricing data is required. Data other than certified cost or pricing data means any type of information that is not required to be certified in accordance with FAR 15.406-2 and is necessary to determine price reasonableness. For example, such information may include pricing, sales, or cost information, and includes certified cost or pricing data for which certification is determined inapplicable after submission. The information required is identified in the below Price Proposal Instructions.

12.0 PRICE PROPOSAL INSTRUCTIONS

12.1 General Instructions

The NLS II Offeror shall furnish Launch Services described in this solicitation. Not-to-Exceed Standard Launch Service pricing shall be offered for all of the order years remaining on the NLS II contract, *i.e.*, order years 2025 through 2030 (Not-to-Exceed Non-Standard Launch Service prices and fixed price Special Task Assignment Labor rates through calendar year 2032). Not-to-Exceed pricing is based on a standard integration period of L-30 plus or minus 3 months. The contract will be on a calendar year basis and each launch service begin with authorization to proceed (ATP). The contract structure and periods of performance are summarized in Table E2-B.

	Contract Order Calendar Years						
	2026	2027	2028	2029	2030	2031	2032
IDIQ LSTO Missions Standard Launch Services Non-Standard Services	NTE, TABLE B-9 NTE, TABLE B-10						
Special Task Assignments	FFP, TABLE B-6						

Table E2-B: FFP Labor Hour Rates and NTE Prices for IDIQ Contract Ordering

12.2 Specific Instructions

12.2.1 Offerors are exempt from the requirements of submission of certified cost or pricing data, as defined in FAR 2.101; however, data other than certified cost or pricing data is required for a detailed evaluation of price reasonableness. Presentation of the information in the form regularly maintained by the Offeror as part of its commercial operations is acceptable, provided it contains the information described in the following paragraphs and displayed on the sample formats in Appendix E2-A.

12.2.2 Offerors are requested to submit supporting information which will demonstrate reasonableness of the proposed prices, such as detailed basis of estimates with supporting data for major elements of price and subcontracts, projections, discounts, rough order of magnitude information, commercial catalog or published price lists, market prices, and data from recent sales. If the Contracting Officer cannot determine whether the offered prices are reasonable based on the information provided by the Offeror, the Contracting Officer will require additional information, to include cost data, necessary to determine price reasonableness. The information is to be provided for each Standard Launch Service and Non-Standard Service. A reference column has been inserted on the attached sample formats for Offerors to identify the submitted information. Offerors should ensure that the amounts on the detail pricing forms in the Price Proposal reconcile with the prices submitted in Volume 4, Contract Volume, Section B.

12.2.3 Nothing in this solicitation precludes incumbent IDIQ contract holders from proposing lower LSTO mission NTE prices for standard launch services, associated nonstandard services, or both, already under contract.

12.2.4 Reserved

12.3 Reserved

12.4 Reserved

12.5 Special Task Assignments (Sample Form 3)

The NLS II Offeror may be required to perform special studies and analyses, described in SOW paragraph 5.0. Offerors shall propose fixed-price composite hourly rates, which may be ordered by the Government via Task Orders at any time through 2030. Offerors shall submit a cost element schedule, which displays the labor rate development including skill mix, labor classifications, incidental materials, other direct costs, applicable indirect rates, and anticipated profit. Submit the detailed information in Sample Form 3 for fixed price labor hours in CYs 2026 through 2030.

12.6 Launch Service Task Orders (Sample Form 4)

IDIQ contracts shall contain Not-to-Exceed (NTE) prices for the standard launch services which may be ordered at any time through June 30, 2030 (Non-Standard services through

December 31, 2032). Offerors shall state the proposed NTE price of the standard launch service and all non-standard services for each vehicle configuration proposed and launch site in Sample Form 4. The offeror should provide enough pricing details for the Government to complete a full evaluation for price reasonableness. Mission unique pricing is not required for IDIQ launch services.

12.7 Reserved

12.8 Financial Capability

Offerors shall demonstrate their financial capability to provide launch services and properly execute multiple LSTOs under a contract of this type and magnitude.

In the proposal, submit one copy of financial statements and accompanying notes for the two most recently completed fiscal years. In addition, provide data which show the amount of established and/or available lines of credit, the financial institution extending the line and the dollar amount (if any) presently in use.

The Offeror shall disclose the limits of its general liability insurance coverage.

13.0 VOLUME 4, CONTRACT - INSTRUCTIONS FOR CONTRACT TERMS AND CONDITIONS

13.1 The Offeror shall complete Sections A, B, C, D, and E of the model contract and return signed with its proposal. Throughout Sections A through E of the contract, additional instructions to Offerors appear in *italics* where information is required to be proposed and inserted by Offerors. Offerors shall replace all **TBP**s with the applicable proposal information. Offerors may reformat the size and number of rows and tailor these sections to accommodate the amount of data required by tables, clauses, and exhibits. There is no page limitation on Volume 4.

13.2 Section A – Standard Form 1449

The Offeror shall complete blocks 17a, 30a, 30b, and 30c of SF 1449.

13.3 Section B – Continuation of SF 1449/Schedule of Supplies and Services

Standard launch service pricing shall be offered for all of the order years remaining on the NLS II contract, *i.e.*, order years 2026 through 2030 (Non-Standard and Special Task Assignment Labor rates through December 31, 2032) for the tables identified below.

Tables B-1, B-2, B-3, and B-4 are reserved. Table B-5 will be used to order individual special task assignments for CY 2026-2032 in accordance with Table B-6. As required in Table B-6, Offerors shall insert a firm fixed-price composite hourly labor rate for task orders used for special task assignments.

Table B-8 will be used to order IDIQ launch services between CY 2026-2030. In Table B-9, Offerors shall insert NTE prices for all available launch vehicle configurations. The Offerors shall only include those launch vehicle configurations capable of delivering a 250 kilogram payload or greater to orbit. These launch services shall represent the Offerors entire range of performance capability above 250 kilograms that could be made available to the Government in support of IDIQ requirements between CY 2026-2030. In Table B-10, Offerors shall insert NTE prices for all available non-standard services associated with those launch vehicle configurations listed in Table B-9. The Offeror shall add as many tables and additional rows (in Table B-10 format) necessary to identify all non-standard services with associated NTE prices for each launch vehicle configuration proposed.

The Offeror shall propose a plan for validating compliance with and enforcing the certification found in Section B, Clause 4.0, Most Favored Customer after contract award. This plan shall be included as Attachment D9 to the contract and is not included in the page count.

13.4 Section C - Contract Terms and Conditions

13.4.1 Clause 5.0, Launch Service Payments, Milestone Events and Completion Criteria

- (A) Prices for all proposed launch services shall be based on contract Section C, Clause 5.0, Table C-1, Launch Service Payment Schedule. The percentages specified in Table C-1 shall be based on the total launch service CLIN price, consisting of three SubCLINs, i.e., standard launch service, non-standard services, and mission unique services. Under future Requests for Launch Service Proposals only, Offerors shall be required to insert two additional columns into Table C-1. One column shall contain invoice submission dates and the other column shall contain the dollar amount of each payment.
- (B) Offerors shall adhere to the Contract Work Plan, Attachment D8. The “Program Event” and “Accomplishment Criteria” descriptions in the Work Plan may be tailored to align with the contractor’s standard launch service process as long as the “Program Event” and “Accomplishment Criteria” descriptions remain substantially the same as is currently stated in the Work Plan. In accordance with FAR 32.204, launch service payment schedules are specified in this solicitation and satisfy the following criteria:
 - (i) Begin no earlier than L-“X”/L-30M months before launch
 - (ii) Meet the definition of Commercial Interim Payments^a
 - (iii) Specify an amount to be paid coincident with launch mission success payments (10%) and an additional amount returned (15%) to the Government in the event of a mission failure^b.

- (iv) Based on milestones or substantial progress^c
- (v) Individual payments through launch shall not exceed 10%
- (vi) Individual payments occur no more often than quarterly, unless noted in the work plan
 - a “Commercial interim payment” means any payment that is not a commercial advance payment or a delivery payment. These payments are contract, financing payments for prompt payment purposes (*i.e.*, not subject to the interest penalty provisions of the Prompt Payment Act in accordance with FAR Subpart 32.9). A commercial interim payment is given to the Offeror after some work has been done, whereas a commercial advance payment is given to the Offeror when no work has been done.
 - b The combined amount for mission failure shall be 25% of the total launch service price.
 - c Commercial interim payments are contract financing payments that are not payment for accepted items. Commercial interim payments shall be made on the basis of achievement or occurrence of specified milestone events.

13.4.1.1 Reserved

13.4.1.2 Reserved

13.4.2 Clause 12.0, Order Limitations

Offerors shall insert and complete the information on order limitations in paragraph (b) of Section C, Clause 12.0, FAR 52.216-19, Order Limitations. In addition to the awarded initial mission set of requirements on June 16, 2000, the Government anticipates ordering a maximum of seventy launch services and associated non-standard services via the IDIQ task ordering process during the twenty-year life of the NLS I & II contracts. The minimum order for a stand-alone IDIQ contract is one launch service, a special task assignment, or a non-standard service equal to \$20,000.00. Offerors shall identify any time or quantity limitations with respect to ordering these services within the twenty-year contract period of performance due to launch rates, production capacities and any other constraints.

13.4.3 Clause 19.0, Adjustments to Launch Schedule

- (A) Offerors shall propose, in Table C-5, Liquidated Damages paid to NASA for each day of launch delay, which would result from a Contractor delay of launch. The maximum amount paid for delays by the Contractor/NASA shall be the lesser amount of either a 5% ceiling cap based on the highest priced launch vehicle configuration at year of contract award or a total of \$9,400,000. Offerors shall not adjust the Time Periods or Grace Days in Table C-5. The Offeror’s proposed schedule of launch delay postponement fees shall be based on the number of months

before launch stated in the solicitation and are subject to the following ground rules and assumptions:

- (i) Cumulative launch schedule adjustments up to eighteen months (excluding grace period) for both parties shall be subject to postponement fees with a cap or ceiling for a maximum postponement of eighteen months
 - (ii) Launch schedule adjustments greater than eighteen months shall be subject to equitable adjustment (i.e., Changes clause)
- (B) Offerors may propose dollar amounts in Table C-6 lower than the ones proposed in Table C-5. Offerors may also propose grace days in Table C-6 higher than the ones proposed in Table C-5. However, the Time Period in Table C-6 shall not be adjusted. Offerors are cautioned, under no circumstance shall offerors propose higher NASA Postponement Fees than Contractor Liquidated Damages, or lower NASA Grace Days than Contractor Grace Days.

13.4.4 Reserved

13.4.5 Termination for Government's Convenience

As indicated in contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement under FAR 212-4(l), it is the Government's intent to incorporate an advance agreement on maximum termination liability in the event the Government decides to exercise its right to terminate for the Government's convenience all or part of the contract under FAR 52.212-4(l). Offerors shall not take exception to the specified maximum termination liability for any standard launch service and the percentage of cumulative performance based milestone payments to be returned to the Government in the event of a termination for convenience, as shown in the table in contract Section C, Clause 28.0, Advance Understanding Regarding Termination Settlement under FAR 52.212-4(l).

For extended integration periods greater than L-30M plus 3 months, the Contractor shall propose a percentage TBD (50 to 100) that represents the percentage of cumulative payments made to date to be returned to the Government in the event of a Termination for Convenience of the Government (Table C-1).

Table E2-C: Reserved

Table E2-D: Reserved

13.5 Small Business Subcontracting Plan

- (A) In keeping with Congressionally-mandated goals, NASA seeks to encourage the placement of a fair portion of its contract dollars, where feasible, with Small, Small Disadvantaged, and Women-Owned Business concerns as defined in FAR 52.219-8, Utilization of Small Business Concerns. For this procurement, NASA has determined the following goals as reasonable at contract award:

Small Business (SB)*	5%
Small Disadvantaged Businesses(SDB)**	1%
Woman-Owned Businesses (WOB)***	1%
Historically Black Colleges and Universities (HBCU)	
And Other Minority Educational Institutions (OMEI)	.25%
HUBZone Small Business (HUBZone)	.25%
Veteran-Owned Small Business (VOSB)	.25%
Service-Disabled Small Business (SDVOSB)	.25%

* includes SB, SDB, WOB and others

- (B) The goals are stated as a percentage of total contract value, not total planned subcontracting dollars. These goals shall be improved upon each year and increased until after a period of four (4) years the goals shall reflect the following four year average:

Small Business (SB)*	5.5%
Small Disadvantaged Businesses(SDB)**	1.5%
Woman-Owned Businesses (WOB)***	1.5%
Historically Black Colleges and Universities (HBCU)	
And Other Minority Educational Institutions (OMEI)	.25%
HUBZone Small Business (HUBZone)	.25%
Veteran-Owned Small Business (VOSB)	.25%
Service-Disabled Small Business (SDVOSB)	.25%

* Includes SB, SDB, WOB and others

- (C) NASA prime Offerors may count first-tier subcontracting dollars toward achievement of the contract specific goals under their commercial plan. This solicitation contains FAR 52.219-9, Small Business Subcontracting Plan, Alternate II. The Subcontracting Plan required by the clause including the associated subcontracting goals must be submitted with your proposal. The plan will be included in any resultant contract as an attachment.
- (D) A commercial plan is the preferred type of subcontracting plan for Offerors furnishing commercial items. The commercial plan shall relate to the Offeror's planned subcontracting generally, for both commercial and Government business, rather than solely to the Government contract. Commercial plans are also preferred for subcontractors that provide commercial items under a prime contract, whether

or not the prime Offeror is supplying a commercial item. The subcontracting plan percentage goals shall be based on total contract dollars.

13.6 Section D - Documents, Exhibits, and Attachments

Offerors shall follow instructions and fully comply with the technical proposal instructions to provide data and information for all SOW tables, exhibits and attachments. The Offerors shall include all Section D exhibits, and attachments, listed in Table E2-A for Volume 1 and place in Volume 1. A complete description of all launch services proposed shall be contained in the SOW Exhibits 1 through 3.

13.7 Reserved

14.0 LAUNCH VEHICLE CERTIFICATION CRITERIA

The Offeror shall provide the following data and plan(s) in Exhibit 7.

14.1 The Offeror shall describe how each proposed common launch vehicle configuration will be certified to category 2 or 3 as defined in NPD 8610.7, ELV Risk Mitigation Policy for NASA-Owned and NASA-Sponsored Payloads. Nothing shall preclude or restrict selecting the utilization of a lower risk launch vehicle for any payload.

14.2 For a common launch vehicle configuration being proposed, the Offeror shall provide the data required in either (A) or (B):

(A) For common launch vehicle configurations previously certified by the LSP, the Offeror shall provide a copy of certification letter from the LSP (provide the vehicle configuration and date of certification by the LSP).

(B) For common launch vehicle configurations that have not been previously certified, the Offeror shall provide the data required in either (i) or (ii):

(i) For launch vehicle configurations that have flown 14 or more consecutive successful flights, provide mission names, launch dates, launch vehicle configuration, associated customers, and post-flight data reports for the most recent string of 14 consecutive successful flights of the launch vehicle being proposed.

(ii) For common launch vehicle configurations that are to be certified prior to the launch of the NASA payload, the Offeror shall provide a launch vehicle certification plan describing the Offeror's ability to meet the certification requirements in NPD 8610.7 and LSP-PLN-324.01. This plan shall include:

(a) Mission names, launch dates, common launch vehicle configuration, associated customers, and post flight data reports for the previous launches of the common launch vehicle configuration being proposed.

- (b) A description of the launch vehicle design, acceptance test, qualification test, and quality processes used in the development of the proposed common launch vehicle configuration, along with an example of a qualification rationale for a single hardware element (design and qualification requirements, support analyses, qualification test report including any test anomalies/resolutions, and test-like-you-fly exceptions). Also include a description of the launch vehicle analysis processes and standards, along with an example of a coupled loads analysis (methodology, ground rules and results). Provide a description of the similarities and differences between the common launch vehicle configuration being proposed and any certified common launch vehicle configuration.
- (c) A manifest showing the planned flight experience with the proposed configuration and any scheduled launches within the required time frame to satisfy the certification requirements.
- (d) An approach and schedule for completing the necessary certification requirements and supporting the NASA review of certification data with sufficient margin allowing for clarification of technical data and completion of outstanding actions.
- (e) A plan for support of AS9100 2nd party audits performed by NASA/KSC in accordance with AS9101 requirements. The audit will be performed as part of the certification process. To facilitate this requirement, the Offeror shall identify a point-of-contact for coordination of NASA's on-site audit and locations where on-site audits should be conducted.
- (f) A description of the risk mitigation program which addresses outstanding hardware and software qualification risks, outstanding risks associated with launch vehicle analyses, quantification of the risk level, and plans to mitigate these risks.

14.3 Reserved

15.0 TECHNICAL PROPOSAL INSTRUCTIONS

15.1 Introduction

- (A) The Technical Proposal shall include a discussion of the Offeror's approach to meeting the requirements of the entire SOW. The Offeror shall discuss its approach to meeting the requirements of the SOW for each launch service proposed.

- (B) The Offeror shall provide all data/information labeled **TBP** in the SOW, its exhibits, and attachments and show it as **BOLDED** right in the SOW section where it goes as well as reference it in the Technical Proposal area.
- (C) The technical approach shall be specific, detailed, and complete to a level, which clearly and fully demonstrates the Offeror's understanding of the requirements and the inherent risks associated with launch services. The Technical Proposal must adequately describe how the Offeror proposes to comply with the applicable requirements, including a full explanation of the techniques and procedures the Offeror intends to follow. The Offeror shall provide representative schedules for the services being proposed. Information submitted prior to receipt of proposals, if any, will be considered only to the extent it is resubmitted. It should not be incorporated by reference.

15.2 Standard Launch Service

The Offeror shall describe all launch service tasks necessary to deliver payloads to defined orbital parameters in compliance with mission requirements. The Offeror shall describe all arrangements required for the launch Range authorization and support for vehicle processing; integrated payload/vehicle processing; launch; and launch site maintenance and modifications.

The Offeror shall describe all services, equipment, and infrastructure necessary to perform program management, mission integration, launch site support, ground and flight system safety, and flight assurance. The Offeror shall describe the facilities and services at Offeror facilities, which will be utilized by NASA personnel performing insight and approval functions.

The Offeror shall describe all necessary services, test hardware and software, and mission specific elements required to integrate the payload(s) to the launch vehicle systems. All capabilities and conditions stated in Exhibit 1 must be consistent and compatible with all other capabilities and conditions stated in the exhibit and in response to the requirements of the SOW. For example, any stated condition for a particular payload environment characteristic must be applicable for all stated performance capabilities. Any exceptions to the stated capabilities or conditions must be specifically noted.

The Offeror shall describe how they will coordinate and assist NASA Public Affairs Office with press releases, launch commentary, news conferences, and press kit documentation development. The Offeror shall identify and describe all services included in the proposed standard launch service(s) necessary to meet the requirements of this SOW. Describe any limitations associated with these services. This information shall be provided in Exhibit 2, Standard Services.

15.2.1 Launch Vehicle

The Offeror shall describe the approach to meeting each standard launch vehicle requirement identified in paragraph 2.1 of the SOW.

15.2.2 Program Management

Describe the overall program management approach.

15.2.2.1 Formal Reviews

15.2.2.1.1 Program Reviews

Describe the program reviews that will be conducted by the Offeror. Address the review structure, content, participants, and frequency or timing of these reviews.

15.2.2.1.2 Design Reviews - Mission Specific

Describe the mission specific design reviews (MSRR, MSPDR, MSCDR, and MSSAR) that will be conducted by the Offeror. Address the review structure, content, participants, and frequency or timing of these reviews.

15.2.2.1.3 Readiness Reviews

Describe the pre-mate readiness review that will be conducted by the Offeror. Address the review structure, content, and participants of this review. Describe how the Offeror will participate in the Launch Vehicle Systems Readiness Review, Pre-Mate Readiness Review, LMCM, FRR, and LRR.

15.2.2.2 Risk Management

Describe risk mitigation techniques utilized by the Offeror to identify, quantify and mitigate risk to mission success and launch schedules. Describe the criteria, methods, and procedures used for identifying critical items and obtaining the proper level of management visibility into these items.

15.2.2.3 Configuration Management

Describe the methods of configuration management to be used for launch vehicle production, assembly, testing and processing for launch vehicle components/subsystems, hardware, and software.

15.2.2.4 Manifest Policy

Provide the Program Master Schedule, Launch Vehicle Planning Manifest and Manifest Policy addressing the Offeror's overall approach to ensuring timely launch of NASA, Department of Defense (DOD) and commercial payloads identified in paragraph 2.2.4 of the SOW.

15.2.2.5 New Launch Vehicle Orientation

For a new launch service, describe the process for supporting an orientation briefing at KSC and address the specific requirements identified in paragraph 2.2.5 of the SOW.

15.2.3 Mission Integration Services

15.2.3.1 Mission Integration Management

- (A) Describe the mission integration activities performed to develop, document, implement and verify the interfaces between the payload flight and ground systems payload with the launch vehicle.
- (B) Describe all kick-off meetings, working-group, and technical interchange meetings conducted by the Offeror. Address the structure, content, participants, location, and frequency or timing of these meetings.
- (C) Describe how the Offeror will track development status of and resolve issues associated with mission specific hardware and software.
- (D) Describe the methods to be used to plan, coordinate, and implement mission specific requirements.
- (E) Describe methods to be used to manage the design, development, qualification, testing and integration of mission specific requirements.
- (F) Describe integration activities performed at the launch site.
- (G) Describe the method to be used to evaluate the capability of a launch vehicle to accommodate NASA secondary payloads.
- (H) Describe methods to be used to provide mission status to NASA and NASA customers throughout the launch campaign from Authority to Proceed (ATP) through mission success determination.
- (I) If the Offeror proposes to co-manifest an NLS II mission, describe any hardware and analyses required and any mission unique integration activities. If necessary, describe any design and development work required along with the plan/schedule to meet the launch date.

15.2.3.2 Mission Analyses

Describe how the Offeror will prepare and submit the mission analyses identified in Table D1-A of the SOW.

15.2.3.3 Mission Success Determination

Describe how the Offeror will comply with the mission success determination requirements identified in paragraph 2.3.3 of the SOW.

15.2.4 Launch Site Support

15.2.4.1 Pre-Launch Checkout and Launch Support

15.2.4.1.1 Launch Vehicle Preparation and Launch

Describe all launch vehicle preparations and launch site operations necessary to safely and successfully prepare the launch vehicle and payload and associated hardware and software for launch.

15.2.4.1.2 Telemetry Data and Launch Countdown Support

15.2.4.1.2.1 Vehicle and Ground Support Equipment

Describe the methods/processes for providing telemetry as identified in Section 2.4.1.2.1.

15.2.4.1.2.2 Network Requirements

Describe the methods/processes for providing defined signals as identified in Section 2.4.1.2.2.

15.2.4.1.2.3 Operational Voice Requirements

Describe the methods/processes for providing defined signals as identified in Section 2.4.1.2.3.

15.2.4.1.2.4 Video Requirements

Describe the methods/processes for providing video as identified in Section 2.4.1.2.4.

15.2.4.1.2.5 Timing Requirements

Describe the methods/processes for providing countdown clock signal as identified in Section 2.4.1.2.5.

15.2.4.1.3 NASA Launch Flight Operations

Describe the methods/processes for providing defined signals as identified in Section 2.4.1.3.

15.2.4.1.4 NASA Launch Countdown Support

Describe the methods/processes for providing pre-launch and launch support requirements as identified in Section 2.4.4.1.4.

15.2.4.2 Launch Site Payload Support

15.2.4.2.1 Payload Processing Facility

Describe the physical attributes and the capabilities of any payload processing facilities to be available as standard service for any launch service being proposed at a location other than KSC, CCSFS, WFF, or VSFB. The description shall include the items listed below:

- (A) Reserved
- (B) Control Room Space and Capabilities
- (C) Crane Size and Hook Height
- (D) Electrical Service
- (E) Environment, i.e., cleanliness level, relative humidity
- (F) Environment Monitoring Equipment
- (G) Temperature
- (H) Hazardous Capability
- (I) Power & Grounding (i.e. UPS, backup generator)
- (J) Processing Space Floor Capability
- (K) Processing Space Size
- (L) Office Space
- (M) Security/Access
- (N) Fire Protection
- (O) Compressed gasses
- (P) Communications and data handling (i.e. voice communications, video, command/control and telemetry)

15.2.4.2.2 Launch Site Payload Integration Support

- (A) Describe the services, equipment, and support provided at the launch site for the integration and launch of each payload.
- (B) Describe the standard services and support for the integration and launch of payloads, in particular as it relates to the items listed in 2.4.2.2 of the SOW.

15.2.4.2.3 Contamination and Environmental Control

The Offeror shall describe the approach to meeting the contamination and environmental control requirements specified in paragraph 2.4.2.3 of the SOW.

Where stated in the SOW, identify any periods or configurations where these requirements cannot be met.

15.2.4.2.4 Operational Support Services

Describe how the Offeror will provide the operational support services identified in paragraph 2.4.2.4 of the SOW.

15.2.4.3 Range Support and Services

Describe the approach for meeting Range launch support including the specific items listed in paragraph 2.4.3 of the SOW. Identify all services that will be provided by the Range in conjunction with the standard launch services.

15.2.4.5 Lightning Monitoring

Describe how the Offeror will provide the Lightning Monitoring identified in paragraph 2.4.5 of the SOW.

15.2.5 Safety, Reliability and Quality Assurance

15.2.5.1 Safety and Health Plan

The Offeror shall submit a detailed Safety and Health Plan including the specific items listed in paragraph 2.5.1 of the SOW. Describe the approach for implementing and/or managing a ground and flight system safety program. This plan is excluded from the page count limitations set in Table E2-A.

15.2.5.2 Reliability Plan

The Offeror shall submit a detailed Reliability Program Plan describing techniques and methods of maintaining vehicle design reliability greater than 95% at an 80% confidence level. This plan is excluded from the page count limitations set in Table E2-A.

15.2.5.3 Quality Assurance

The Offeror shall submit a detailed Quality Management Plan. This plan must address all the elements of AS9100, describe techniques to be used to link and incorporate quality standards and practices into specific procedures, and describe the approach and standards used to evaluate the quality systems of sub-contractors that are not AS9100 certified. This plan is excluded from the page count limitations set in Table E2-A.

Describe the approach for implementing and maintaining a flight and ground software assurance program. Define the degree and methods of checkout, certification and control of software.

15.2.6 NASA Insight and Approval

Describe the methodology to be employed by the Offeror to support NASA's insight and approval requirements in Section C, Clause 25.0.

Describe the methodology used by the Offeror to notify NASA of qualification or test anomalies involving similar launch vehicles, systems, subassemblies and components.

Describe the methodology used by the offeror to support a resident office if NASA elected to have representation as a resident office at the Contractor's major manufacturing and engineering facilities for the life of the contract.

15.3 Non-Standard Services

The Offeror shall construct Exhibit 3 by defining individual non-standard services and providing a description of each.

15.4 Mission Unique Requirements

15.4.1 Mission Unique Hardware

Describe the process for designing, manufacturing, testing, and qualifying for flight the launch vehicle mission unique hardware that is required to support the payload and mission.

15.4.2 Mission Unique Software

Describe how the Offeror will prepare and submit the mission unique software deliverables identified in paragraph 4.2 of the SOW.

15.4.3 Unanticipated Mission Unique Services

Describe the method for complying with the unanticipated mission unique requirements identified in paragraph 4.3 of the SOW.

15.5 Special Task Assignments

Describe the method for complying with the special task assignment requirements identified in Section 5.0 of the SOW.

15.6 Information Technology

(A) Reserved

(B) The offeror shall discuss its approach to providing an IT security compliant system or application (including hardware and software), schedule for development and implementation, anticipated user base, anticipated benefits, and risks.

(C) Reserved

15.7 Capabilities, Specifications and Environments (Exhibit 1)

This exhibit will be used similar to a mission planner's guide. The Offeror shall provide all the data necessary to complete the associated tables and diagrams.

Describe the technical performance capabilities of the proposed standard launch and non-standard services. Describe how the launch service(s) to be provided meet each of the requirements specified in Exhibit 1 of the SOW. Provide a revised SOW Exhibit 1 describing the performance capability of the Offeror's launch services to be incorporated into the resulting contract(s). Adhere to the format of Exhibit 1 to the maximum extent possible.

If more than one vehicle configuration is proposed and the data for each configuration are different, the data provided shall be presented separately. For example, if the instruction is to provide performance capabilities for payload mass and orbit in Table D1-D, and the data to be provided vary with configuration, the data shall be provided separately in Table D1-D1, D1-D2, etc. for each configuration.

The Offeror shall describe each proposed launch vehicle configuration with the specific information requested in the paragraphs herein.

The Offeror shall include a listing of all assumptions used in which vehicle performance tables were constructed. Such items as performance reserve, fairing blanket thickness, shall be included within Exhibit 1.

15.7.1 Vehicle Configuration

Provide vehicle configuration data in a quick look table. This table is intended to present an inclusive list of the major components such as solid rocket motors, upper stages, payload adapters, and available payload fairings. All standard subsystems such as electrical, mechanical and propulsion are assumed to be included and are not required as part of this table. Table D1-C in Exhibit 1 is provided as an example only. Multiple tables are acceptable.

15.7.2 Payload Orbit and Mass Requirements

Using the formats provided in paragraph 2.0 of Exhibit 1, provide the data required to complete Tables D1-D through D1-G3 for each proposed vehicle configuration(s). Only

applicable configurations need to be addressed for the different types of mission profiles (e.g., upper stage configurations for escape missions). For the LEO and GTO data, a separate table shall be provided for each applicable vehicle configuration. List and describe all pertinent mission design assumptions, as applicable per table, including, but not limited to: inclination, park orbit, and DLA.

15.7.3 Insertion Accuracies

Provide an inclusive list of the insertion accuracies for each proposed vehicle configuration(s) as defined in paragraph 1.0 for each of the maximum and minimum mission profiles as defined in paragraph 2.0 of Exhibit 1. Table D1-H of Exhibit 1 lists the orbital parameters that shall be considered.

15.7.4 Payload Separation Attitude Accuracy and Rates

Using the format provided in Table D1-I of Exhibit 1, provide an inclusive list of the payload deployment attitudes and rates. Errors shall be defined for each vehicle configuration(s) and consist of all known dispersion sources (i.e., guidance, navigation, and attitude control system limitations).

15.7.5 Payload Fairing (PLF)

For each PLF and PA combination specified in paragraph 1.0 of Exhibit 1, provide a separate diagram showing the maximum payload static envelope and the associated dynamic envelope for a rigid payload hard mounted to the PA.

15.7.6 Mechanical Interface

Provide a separate detailed drawing for each PA, which includes dimensions, mounting provisions, and interfaces. Describe the manner in which the payload is protected from separation system debris and re-contact with the launch vehicle.

15.7.7 Electrical Interface

Describe, in detail, the payload electrical interfaces for the proposed vehicle configuration(s). Provide the data required to complete Table D1-J of Exhibit 1. Describe the provisions for the prevention of static charging on umbilical connectors.

15.7.8 Payload Environment

- (A) Provide the electromagnetic radiation levels encountered by the payload as proposed in Figure D1-3 of Exhibit 1.
- (B) Provide the pressure profiles and depressurization rates for each proposed PLF configuration as proposed in Figure D1-4 of Exhibit 1.
- (C) Describe the maximum expected flight environments due to acoustics (50% fill), random vibration, and shock as proposed in Tables D1-K, D1-L, and D1-M, of Exhibit 1, respectively.
- (D) Describe the maximum expected total acceleration (limit load factors), including uncertainty factors (key dynamic events may be broken out as appropriate) as proposed in Table D1-N of Exhibit 1.
- (E) Describe the maximum expected sinusoidal vibrations at the base of the PA as proposed in Table D1-O of Exhibit 1.
- (F) Provide the maximum quasi-steady acceleration as a function of payload separated mass in as proposed in Figure D1-5 of Exhibit 1.
- (G) Provide the maximum thermal environments during ascent in Figure D1-5A for every launch vehicle component (i.e., upper stage, unblanketed skin, thermal/acoustic blankets, etc.) inside the fairing with a view factor to the payload. Provide as many figures as necessary to encompass the entire launch service.

15.7.9 Payload Environment Instrumentation

Propose the specific content of Table D1-P in Exhibit 1 to include sensor type, number of locations, range, sensitivity, and bandwidth.

15.7.10 Generic Secondary Payload Characteristics/Interface

Describe the generic secondary payload characteristics and interfaces for each proposed vehicle configuration(s). Include a graphical description using the format as proposed in Figures D1-6 and D1-7 of Exhibit 1 for separating and non-separating payloads. Complete Table D1-Q of Exhibit 1 and state any assumptions or conditions that apply to the secondary payload accommodations.

15.8 Standard Launch Services List (Exhibit 2)

The Offeror shall identify all vehicle configurations to be provided as part of their standard launch service(s). The description and performance capabilities for these vehicles may be included as reference to Exhibit 1.

The Offeror shall describe all services provided to meet the requirements of standard launch service section of the SOW. Any, and all, services proposed by the Offeror, as part of the standard launch service, shall be listed in Exhibit 2.

The Offeror shall identify all additional services they have included as part of their standard launch service.

15.9 Non-Standard Services List (Exhibit 3)

The Government has provided a list of items to which the Offeror shall respond whether the non-standard service is available. The Offeror shall identify whether any items on the Government's list of items are included in the standard launch service. The Offeror shall identify any restrictions or launch vehicle specific requirements, which apply to the non-standard services listed in Exhibit 3. Any impacts to the capabilities and/or conditions defined in Exhibit 1 introduced by the non-standard service shall be explained in detail.

The Offeror shall list and describe any additional non-standard services offered.

The non-standard services delineated in Exhibit 3 shall be used by the Offeror to complete the matrix in contract Section B, Table B-10.

For payload processing facilities available as a non-standard service at KSC, CCSFS, WFF, or VSFB, describe the physical attributes and capabilities of the proposed processing facilities including those listed below:

- (A) Reserved
- (B) Control Room Space and Capabilities
- (C) Crane Size and Hook Height
- (D) Electrical Service
- (E) Environment, i.e., cleanliness level, relative humidity
- (F) Environment Monitoring Equipment

- (G) Temperature
- (H) Hazardous Capability
- (I) Power & Grounding (i.e. UPS, backup generator)
- (J) Processing Space Floor Capability
- (K) Processing Space Size
- (L) Office Space
- (M) Security / Access
- (N) Fire Protection
- (O) Compressed gasses
- (P) Communications and data handling (i.e. voice communications, video, command/control and telemetry)

For LSC provided Telemetry available as a non-standard service, describe the service using the following information:

- (A) Range shall be the geographical location of the launch site
- (B) Orbit shall be defined as: earth orbiting, geosynchronous, solar, lunar, planetary

The LSC shall consider the following telemetry types for this NSS: main vehicle and GSE telemetry, special instrumentation/wideband telemetry system and video. The LSC shall also consider the following assets for this NSS: aircraft staged locally, aircraft deployed to a remote location, sea-based staged locally, sea-based deployed to remote location, ground team deployed to remote location, ground fixed asset, and TDRSS. The LSC should also consider that special instrumentation/wideband telemetry may not be able to be transmitted via TDRSS.

- 15.10 Reserved
- 15.11 Reserved
- 15.12 Launch Vehicle Certification Data (Exhibit 7)

The Offeror shall provide all data, as identified in Attachment E2, paragraph 14.0, Launch Vehicle Certification Criteria, to build Exhibit 7.

- 15.13 Data Requirements List (Attachment D2)

The Offeror shall provide all **TBPs** listed in Attachment D2.

The Offeror shall identify any new, existing, or commercial products proposed to satisfy the description/data requirements of each CDRL. The Offeror shall describe, in detail, the information/data contained in these documents and shall state how this information will satisfy the description/data requirements of each CDRL.

15.14 Supplemental Data Requirements List (Attachment D3)

The Offeror shall provide all **TBP**s listed in Attachment D3.

16.0 PAST PERFORMANCE PROPOSAL INSTRUCTIONS

16.1 Contract Past Performance, History, and Experience (New Offerors Only)

- (A) The Offeror shall provide a narrative description of relevant past experience as a Launch Service Provider. The Offeror shall provide a listing of five (5) or more previous or existing commercial and governmental launch service contracts, including contract number, contract type, type of vehicle and mission, and total original and present or final contract value. A listing of customers' names, addresses and phone numbers should be provided.
- (B) The Offeror shall discuss its ability to manage price increases and contract changes as experienced in recent launch service contracts. In a separate table, identify the number of contract modifications by type: administrative, exercise of option, change order, and supplemental agreement (bilateral) for each mission. Identify the change and the contract value related to any change order or supplemental agreement for each mission. Also in table format, identify all task orders and task assignments by task description, period of performance, total number of hours and total value at completion for each mission. These tables will not count against the page limitation.
- (C) The Offeror shall discuss its ability to meet requirements and to receive milestone or performance based payments as experienced in recent launch service contracts. Any examples of delayed payments because of missed milestones or inadequate performance should be discussed along with remedial action taken.
- (D) The Offeror shall discuss its experience and history of design, fabrication, test and launch of launch vehicles and its ability to perform tasks associated with vehicle and payload integration. History of meeting technical requirements and performance standards should be discussed as well as how processes have evolved to correct problems and incorporated lessons learned. The Offeror should provide examples of how significant technical problems were identified and resolved and its history of requirement deviations and waivers. Discussions should include how processes may have been updated in order to reduce requirement deviations and waivers.

16.1.1 Contract Past Performance, History, and Experience (Existing Offerors)

- (A) Existing Offerors do not need to provide the past performance information as required above. The Government will evaluate the past 3 years of the contractor's past performance evaluations under the NLS, NLS II, and other contracts.

16.2 Launch Manifest History

The Offeror shall provide a manifest of the previous three (3) years launch history of the proposed launch services. This history should include the original launch dates at contract turn-on and the actual launch dates. The Offeror shall discuss its ability to meet the planned launch dates, the reasons for launch slips and the processes in place which were used to mitigate the effect of technical, operational and programmatic problems on launch dates.

16.3 Quality of Launch Service and Customer Support

The Offeror shall discuss the quality of its launch service and its ability to provide a high level of satisfaction to its customers as it pertains to its history as a Launch Service Provider. Examples of how the Offeror used customer feedback in its Lessons Learned process should be cited. If data exists on customer satisfaction, these data should be cited.

16.4 Mission Failure History

The Offeror shall furnish information for any launches performed during the past five (5) years, which were less than fully successful. The Offeror shall provide a brief explanation of the condition that caused the failure and a summary of any corrective action taken as a result. The Offerors approach to determining probable root cause should be discussed as well as how impacts to subsequent missions and customers were mitigated.

17.0 PRE-AWARD SURVEY

The Government may elect to perform a pre-award survey of any Offeror in order to determine the Offeror's responsibility in accordance with FAR Subpart 9.1. If performed, the evaluation team will utilize the pre-award survey forms provided in FAR 9.1. If a pre-award survey is performed, the Government anticipates the survey will occur between 30-45 days after receipt of proposal. The Government will give the Offeror fifteen (15) days notice of the site visit.

As prescribed by the FAR, the Government may review the Offeror's facilities, production capabilities, financial capabilities, accounting system, management systems, safety program, and quality assurance systems during the pre-award survey. Any unsatisfactory results from the survey will be communicated to the Offeror via the Contracting Officer for reconciliation.

18.0 RESERVED

APPENDIX E2-A

SAMPLE PRICING FORMS

Sample 1	Reserved.....E2-A2
Sample 2	Reserved.....E2-A3
Sample 3	Special Task Assignments.....E2-A4
Sample 4	Indefinite Delivery/Indefinite Quantity.....E2-A5

SAMPLE 1 – Reserved

SAMPLE 2 – Reserved

SAMPLE 3 – SPECIAL TASK ASSIGNMENTS

DESCRIPTION								
(List each service)	2026	2027	2028	2029	2030	2031	2032	REF.
DIRECT LABOR								
(Identify Skill Mix)								
BURDENS								
-								
- (Overhead)								
-								
INCIDENTAL MATERIALS								
OTHER DIRECT COSTS								
G&A								
TOTAL COST								
ANTICIPATED PROFIT								
COMPOSITE HOURLY RATE								

Note: Pricing shall be offered for all of the calendar years remaining on the NLS II contract, *i.e.*, calendar years 2026 through 2032.

☐ ☐	STANDARD LAUNCH SERVICES	LAUNCH SITE	VEHICLE CONFIGURATION
	NON-STANDARD SERVICES	(Identify)	(Identify)

[illegible]

E2-A5

APPENDIX E2-B

RESERVED

ATTACHMENT E3
EVALUATION CRITERIA
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ATTACHMENT E3

EVALUATION CRITERIA

Paragraph 4.0 applies to existing NLS II contractors seeking to add new launch and related services to their existing NLS II contracts. All remaining paragraphs of this Attachment apply to all offerors to the extent and in the manner specified herein.

1.0 MINIMUM CONTRACT REQUIREMENTS

The Offeror must meet minimum contract requirements at the time of proposal submission to be considered for NLS II contract award. Proposed launch services not meeting the following minimum requirements will be determined non-compliant and excluded from NLS II evaluation:

- (A) Reserved
- (B) Offeror shall be a domestic launch service provider pursuant to 51 U.S.C. § 50131(a).
- (C) The business entity(ies) actually performing the work specified in this contract shall be AS9100 certified by a registrar that is accredited by a body recognized by either the ANSI-ASQ National Accreditation Board (ANAB) or the International Aerospace Quality Group (IAQG). The Contractor shall comply with this requirement by demonstrating that the corporation(s), corporate division(s), subsidiary(ies), joint venture(s), partner(s), and/or any other business entity actually performing launch vehicle manufacture, payload/launch vehicle integration, testing, and launch activities is/ are AS9100 certified.

2.0 FAR 52.212-2 EVALUATION-COMMERCIAL PRODUCTS AND COMMERCIAL SERVICES DEVIATION (Nov 2025) *(TAILORED)*

- (a) (i) The Government may award one or more contracts for Indefinite Delivery/Indefinite Quantity (IDIQ) launch services resulting from this solicitation to the responsible Offeror whose proposal, conforming to the solicitation, provides fair and reasonable not to exceed (NTE) prices and Firm Fixed Price Labor Rates meeting or exceeding the acceptability standards for non-price factors. The following evaluation factors and significant sub factors establish the requirements of acceptability and shall be used to evaluate proposals for new providers:
 - (1) NTE Price and Firm Fixed Price Labor Rate
 - (2) Technical Acceptability
 - (i) Technical Approach
 - (ii) Management Approach

(3) Past Performance

- (ii) Launch service task orders will be awarded on the basis of 'best value'. Evaluation criteria will be established and documented in the request for launch service proposal.

(b) Reserved

- (c) A written notice of award or acceptance of an offer furnished to the successful Offeror(s) within the time for acceptance specified in the offer, shall result in a binding contract without further action by either party. Before the offer's specified expiration time, the Government may accept an offer (or part of an offer), whether or not there are negotiations after its receipt, unless a written notice of withdrawal is received before award.

3.0 GENERAL CRITERIA

- 3.1 Source evaluation will be conducted IAW FAR 15.305, Proposal Evaluation. The Government may elect to award more than one contract. The ordering period of the IDIQ contract(s) will extend through June 30, 2032. IDIQ missions will be competed among those Offerors under IDIQ contract via Launch Service Task Orders (LSTO).
- 3.2 The proposal submission instructions, Section E, Attachment E2, Instructions, Conditions, and Notices to Offerors, provide guidance to the Offeror concerning the type of documentation used in proposal evaluation. The proposal information submitted in response to Attachment E2 will be evaluated under the appropriate evaluation factor. Oral or written discussions, or both, may be conducted with some or all Offerors, which meet the minimum contract requirements of Attachment E3, paragraph 1.0, in accordance with applicable FAR and NASA regulations. Findings associated with the evaluation factors will be presented to the Source Selection Authority (SSA) who is responsible for making the source selection decision as to the contract(s) awarded.
- 3.3 The information contained in proposals may be supplemented by information obtained during audits of or discussions with organizations such as: the Defense Contract Audit Agency, commercial sources (i.e., previous customers or Dun and Bradstreet), or the Offerors themselves.
- 3.4 The Government may elect to perform a pre-award survey of the Offeror's facilities, production capabilities, financial capabilities, accounting system, management systems, safety program, and quality assurance systems. Findings from any pre-award survey will be considered in determining the Offeror's responsibility in accordance with FAR Subpart 9.1.
- 3.5 It is mandatory proposals be prepared as prescribed in this RFP. Accordingly, the Government reserves the right to reject any proposal failing to include any material

information requested, failing to provide complete information in significant areas, or otherwise failing to conform to the requirements of this solicitation.

4.0 EVALUATION INFORMATION SPECIFIC TO EXISTING NLS II CONTRACTORS

The following pertains to existing NLS II Contractors:

- (A) Existing NLS II contractors were deemed to meet the minimum contract requirements upon their on-ramp onto the NLS II contract. Therefore, existing NLS II contractors are deemed to meet the minimum contract requirements unless they submit data as part of their proposals that state otherwise or other information known to the evaluators establishes otherwise.
- (B) A Past Performance Volume does not need to be resubmitted as part of an existing NLS II contractor's proposal. The Government will evaluate the past 3 years of the contractor's past performance evaluations under the NLS II, and other contracts.
- (C) For existing NLS II contractors, the original evaluation of those portions of their prior proposals, not required to be resubmitted in accordance with Attachments E-2 and E-3 (Revision T), will remain valid and be considered for purposes of evaluating their current proposal(s) and for selecting and adding new capabilities to their existing contracts.
- (D) While all applicable evaluation criteria will be considered, proposal evaluation and all subsequent activities leading to modifications of existing NLS II contracts (under the authority of Section C, Clause 2.0, On-Ramp and Technology Insertion), will be conducted in accordance with applicable contract administration regulations, and center policies and procedures.

5.0 PRICE FACTOR

The price evaluation factor is used to assess what each Offeror's proposal will cost the Government should it be selected for award. The proposal will be evaluated using appropriate price analysis techniques IAW FAR Subpart 15.4 to determine if the Offeror's proposed prices are reasonable and consistent with the types, quantities, qualities and performance/delivery schedules of all products and/or services described in the technical proposal.

6.0 COMPLIANCE WITH SOLICITATION REQUIREMENTS

The Offeror's proposal will be evaluated for compliance with solicitation instructions, representations and certifications, contract terms and conditions, financial capability, equal employment opportunities, the small and small disadvantaged business subcontracting plan, the work plan, and the most favored customer certification plan.

7.0 TECHNICAL ACCEPTABILITY FACTOR

7.1 The Technical Acceptability factor evaluates whether the launch service meets or exceeds all technical requirements. Evaluation will focus on the Offeror's technical ability and approach to management of the launch service. The Technical Approach and Management Approach subfactors must be deemed Acceptable in order for the offeror to receive an Acceptable rating on the Technical Acceptability Factor. The Technical Acceptability factor for new offerors will be rated either Acceptable or Unacceptable based on the following definitions:

- (A) Acceptable is defined as a proposal, which meets or exceeds the minimum requirements of the RFP. The proposal may include individual "weaknesses" as long as these weaknesses do not increase the risk of unsuccessful contract performance to an unacceptable level. In addition, the proposal does not contain any deficiencies.
- (B) Unacceptable is defined as a proposal, which fails to meet the requirements of the RFP. A proposal that contains a deficiency will be determined to be unacceptable. A proposal will be determined to be unacceptable if it takes exceptions to the RFP to such a degree that it increases the risk of unsuccessful contract performance to an unacceptable level.

Acceptability for existing NLS II contractors shall be determined in accordance with applicable contract administration regulations, and center policies and procedures.

7.2 The Government will evaluate the Offeror's proposal for Technical Acceptability based upon the two sub factors of Technical Approach and Management Approach. Findings generated from any LSP launch vehicle certification process, NASA Space Act Agreement or other NASA contracts may be used in determining the Offeror's Technical Acceptability.

7.2.1 Technical Approach

The evaluation will address the Offeror's response to each of the technical requirements as either Acceptable or Unacceptable. The evaluation will be based on the Offeror's demonstrated:

- (A) Ability to meet or exceed the requirements of the SOW
- (B) Ability to satisfy the requirements of Section C, Clause 3.0, Launch Vehicle Certification. NASA will evaluate launch vehicle certification status. For launch services not certified at proposal submission, NASA will evaluate the risk associated with the Offeror's plan and schedule to meet or exceed the certification requirements prior to mission launch.
- (C) Reserved

- (D) Ability to successfully launch and deliver a payload to orbit using a launch service capable of placing at a minimum a 250kg payload into a 200 km circular orbit at an inclination of 28.5°.

7.2.2 Management Approach

The Offeror's Management approach will be evaluated on its ability to ensure launch service and mission success. The evaluation will address the Offeror's response to each of the management approach requirements as either Acceptable or Unacceptable. The evaluation will be based on the Offeror's:

- (A) Overall, program management approach to ensuring launch success.
- (B) Approach to managing its subcontractors and the complexity of the work to be subcontracted.
- (C) Plan for ensuring the safety and health of personnel and facilities.
- (D) Quality Management Plan using the AS9100 standard as a guideline. The software assurance approach will also be evaluated using AS9115 as a guideline.
- (E) Configuration management approach for ensuring hardware and software design implementation, change process control, time/age-sensitive component control, and as-built drawing control.
- (F) Reliability plan using NASA-STD-8729.1, Planning, Developing and Managing an Effective Reliability and Maintainability (R&M) Program, as a guideline.
- (G) Risk mitigation program, including the risk associated with certifying new launch services prior to mission launch IAW LSP-PLN-353.01.
- (H) Manifest policy, with regard to the national priorities of NASA missions.
- (I) Approach to keep NASA and the payload customer informed on launch campaign progress, activities, and anomalies from contract award through mission success determination.
- (J) Approach to ensuring Government insight throughout a mission flow, as well as Offeror's approach to Government cognizance of mission unique issues.
- (K) Approach for meeting IT Security Requirement IAW NPR 2810.1A.

8.0 PAST PERFORMANCE

- 8.1 This factor will be used to evaluate how the experience of the Offeror relates to and will benefit Government activities. The Past Performance factor will be evaluated in accordance with FAR 15.305 and NFS 1815.305.

Past Performance shall be evaluated for each offeror using the following levels of confidence ratings as defined in NFS 1815.305.

- (A) Very High Level of Confidence
- (B) High Level of Confidence
- (C) Moderate Level of Confidence
- (D) Low Level of Confidence
- (E) Very Low Level of Confidence

In the case of an offeror without a record of relevant past performance or for whom information on past performance is not available, the offeror may not be evaluated favorably or unfavorably on past performance and a “Neutral” rating will be assigned. Past Performance will be deemed acceptable unless the evaluation results in a Very Low Level of Confidence rating.

- 8.2 The relevance of the Offeror’s experience and the quality of performance will be evaluated on the most recent and relevant experience. Past Performance from the last three years will be considered more relevant than Past Performance older than three years. To determine relevancy of experience, the Government will consider the demonstrated accomplishment of work, which is comparable in nature and magnitude to the work required by this acquisition.

- 8.3 In evaluating Past Performance, the Government will consider the Offeror’s previously demonstrated:

- (A) Contract past performance, history and experience.
- (B) Ability to control price increases and its processes for minimizing contract changes as experienced in recent Launch Service contracts.
- (C) Ability to accomplish requirements to receive milestone or performance based payments. This includes activities such as interim deliverables, delivery of CDRLs or other technical and business reports, and completion of customer direction through task assignments or technical directions.
- (D) Ability to perform as a launch service provider with responsibilities, including design, fabrication, vehicle and payload integration.
- (E) Ability to meet technical requirements and performance standards for previous work.
- (F) Ability to meet launch dates.

- (G) Approach in determining probable root cause for less than fully successful missions and resultant actions taken to improve reliability of launch services.
- (H) Launch vehicle history.

9.0 EVALUATION AND AWARD

No firm missions are included in this on-ramp solicitation. Therefore, evaluation of timely proposals against the RFP's acceptability and reasonableness criteria may proceed at different rates. Upon completion of evaluation of one or more acceptable proposals, the Government may award an IDIQ contract(s) and/or modify an existing NLS II IDIQ contract(s) to on-ramp additional launch service capabilities, while still evaluating other proposals received by the proposal due date.